TM Forum Specification

Open Gateway Operate API Onboarding and Ordering User Guide

TMF931

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Introduction

The following document is the user guide of the REST API for TMF931 Open Gateway Onboarding and Ordering Operate API. The GSMA Open Gateway initiative aims to establish standardized Service APIs (under CAMARA standard) but also homogeneous mechanisms for the integration of Channel Partners, enabling seamless integration and interoperability across various platforms and telecommunications service providers.

This Operate API component suite covers the operations required between the Channel Partners and the Operators as described in the GSMA Playbook (https://www.gsma.com/solutions-and-impact/gsma-open-gateway/wp-content/uploads/2024/02/Channel-Partner-Onboarding-Guide-WA.101-v1.0.pdf).

This Operate API component suite implements a subset of the proposed scenarios:

- ApplicationOwner Management (acceptance, onboarding, lifecycle management)
- Application Management (acceptance, onboarding, lifecycle management)
- Service API Order Management (ordering, retrieving orders)
- Product Management (retrieving products)

In the user guide and OAS :

- Aggregator is modelled as a Channel Partner
- Developer or ASP is modelled as an ApplicationOwner

A Technical Playbook is work in progress under GSMA in order to restrict the possibilities of implementation of TMF931 by Operators and Channel Partners, for improved interoperability. At the time of release of this API version, this document is not yet validated and published. Further evolution of this API specification might implement decisions approved in this Technical Playbook.

Domain Context Specialization

This API is a composite DCS (Domain Context Specialization) of 4 TMF API for Open Gateway context.

- TMF622 Product Ordering Management v5.0
- TMF637 Product Inventory Management v5.0
- TMF669 Party Role Management v5.0
- TMF639 Resource Inventory v5.0

This DCS API doesn't yet fully follow the TMF DCS guidelines. Future evolutions of this API may apply those guidelines.

The internal translation of TMF931 operations and resources should be as follows :

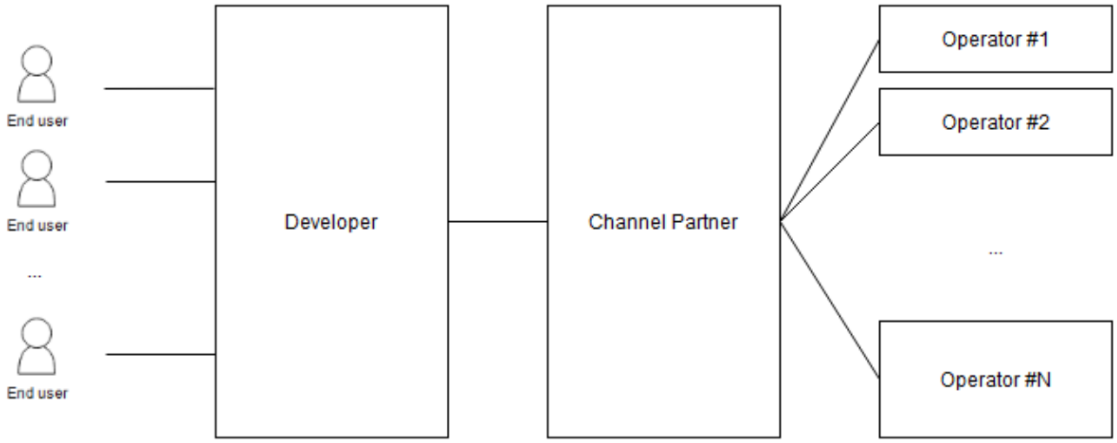
- TMF931 POST /ApplicationOwner corresponds to TMF669 POST /partyRole. This TMF931 operation contains the Party (engagedParty) by value that is a difference vs the TMF669 standard.

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- TMF931 POST /Application corresponds to TMF639 POST /logicalResource. This TMF931 operation returns the resourceRole of the Application as consumer of API products.
- TMF931 POST /ApiProductOrder corresponds to TMF622 POST /productOrder. This TMF931 operation includes a productAction that contains the configurable characteristics (replaces the product attribute in TMF622). Each orderItem is done for one application and one offer, but it allows have several orderItem in the same productOrder. During the processing of the “POST /apiProductOrder” to order a set of API offers, there is creation of “apiProduct” instances in the product inventory.
- TMF931 GET /ApiProduct corresponds to TMF637 GET /product. There is one “apiProduct” instance for each API offer being ordered for a specific Application. 2 Applications can’t therefore consume the same “apiProduct” instance.

Scenarios

The actors involved in this Open Gateway API can be seen in the figure:



Open Gateway Channel Partner model is composed of different players
(WA.101 Open Gateway Channel Partner OnboardingProcess, July 2023)

- Operators are CSP (Communications Service Provider) that provide value to Channel Partner through the use of Operate APIs.
- The Channel Partner (CP) are providers/aggregators who offer its services to one or more ApplicationOwners.
- ApplicationOwners create Applications that are acquired and utilized by end users
- End users are the ApplicationOwner customers at the end of the value chain and are the ultimate beneficiaries of the value offered by the Operate APIs.

This Operate API focuses on the operations between Channel Partners and Operators. The operations allow to subscribe to Service APIs, manage subscriptions; create and manage ApplicationOwners; and create and manage Applications for ApplicationOwners.

The ordering (API subscription/unsubscription) is done on offers exposed to ApplicationOwners via the TMF936 Product Catalog API DCS for Open Gateway.

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The GSMA Open Gateway initiative takes into account two different business models:

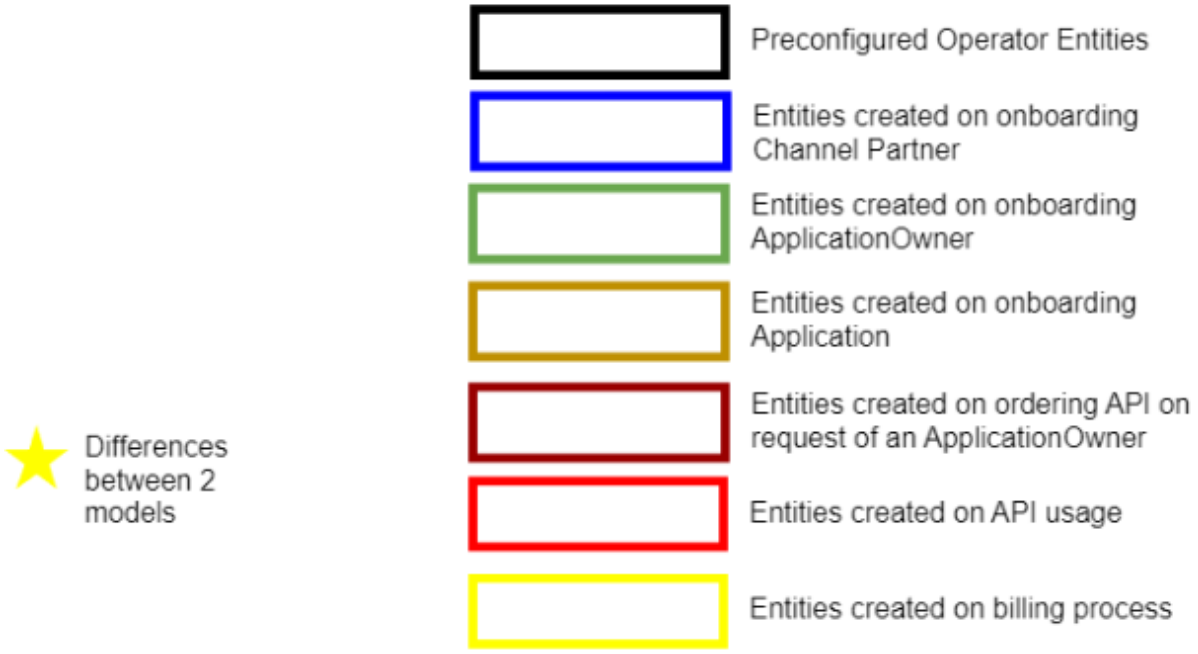
- In the Wholesale Reseller model, the service is fully managed by the Channel Partner.
- In the Aggregated Marketplace model, the Channel Partner leverages its Marketplace to individually expose the APIs. To ensure a coherent user experience, a UX layer is applied, creating a unified interface for the ApplicationOwner.

This Operate API considers the two models defined in GSMA Open Gateway and both can be implemented by using it.

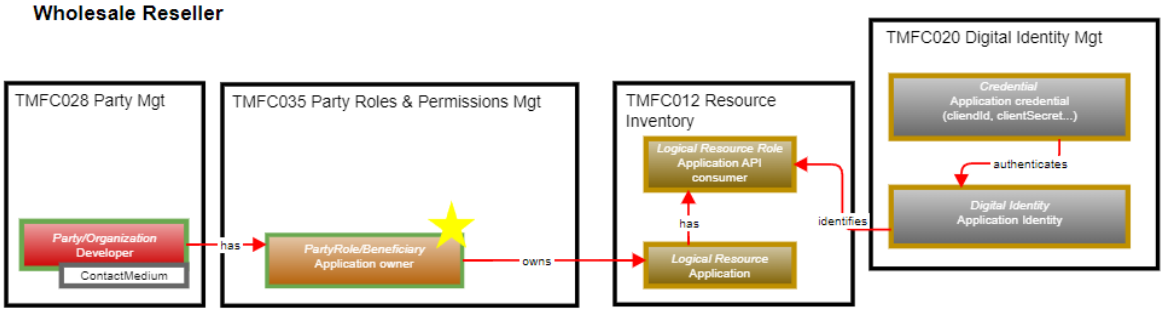
Entities modelling

This modelling is illustrative. It permits to understand how, inside the Operator Information System, are modelled the main entities, their relationships and the target ODA components that are responsible of these entities. Depending on the agreement model between Channel Partner and Operator, the modelling may differ.

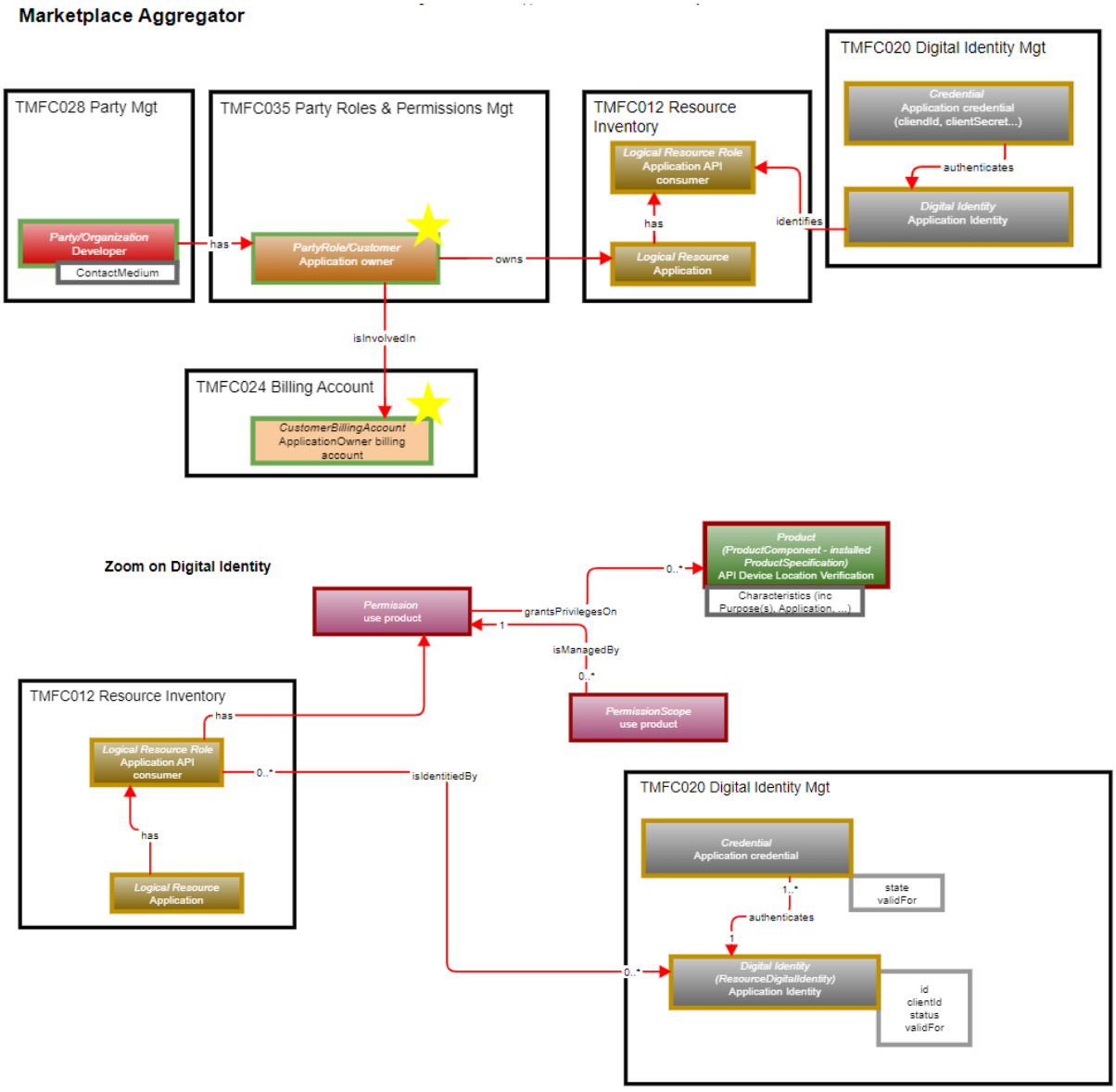
The main steps are represented, with yellow stars highlighting the differences between both models.



ApplicationOwner and Application onboarding

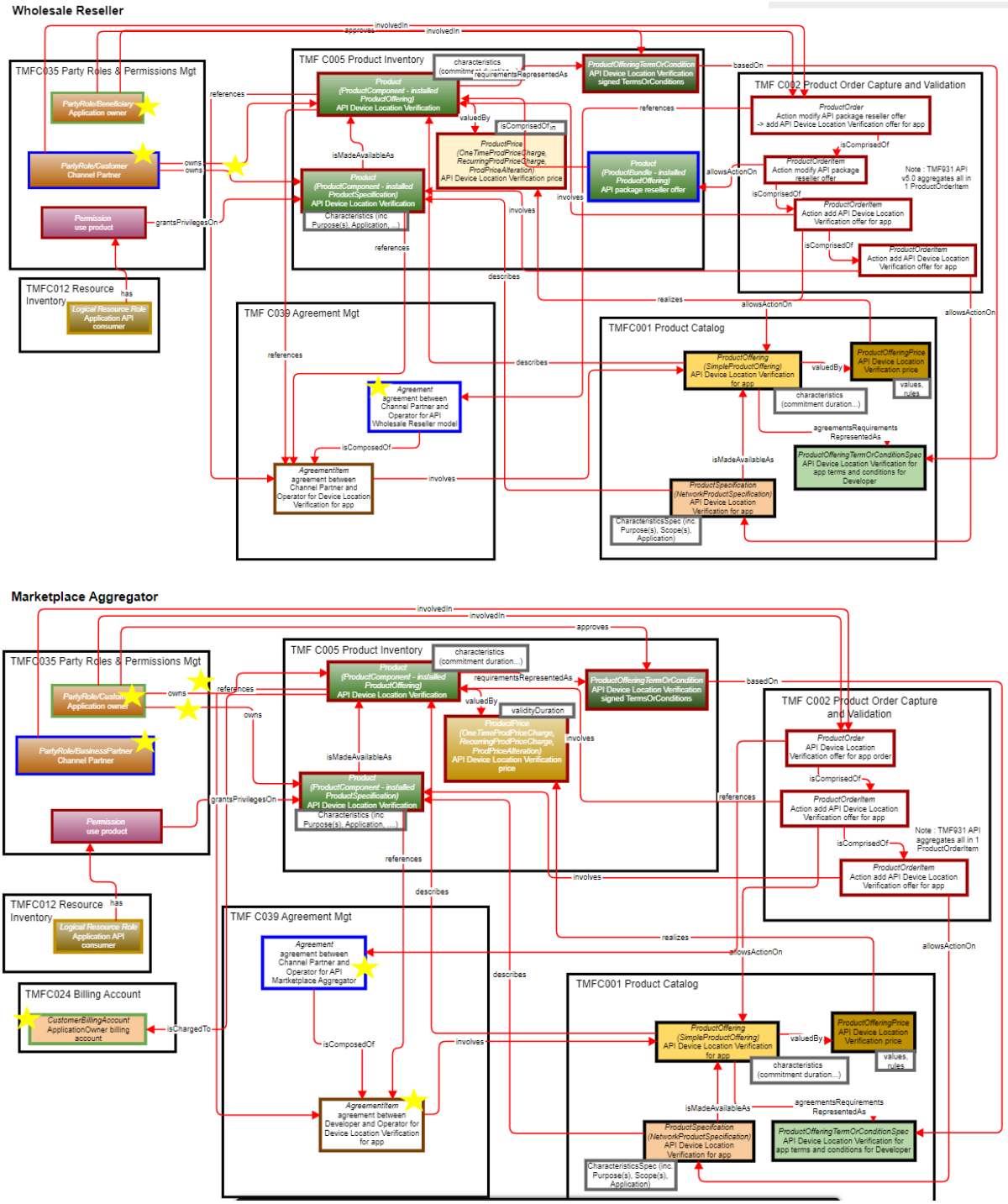


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Ordering

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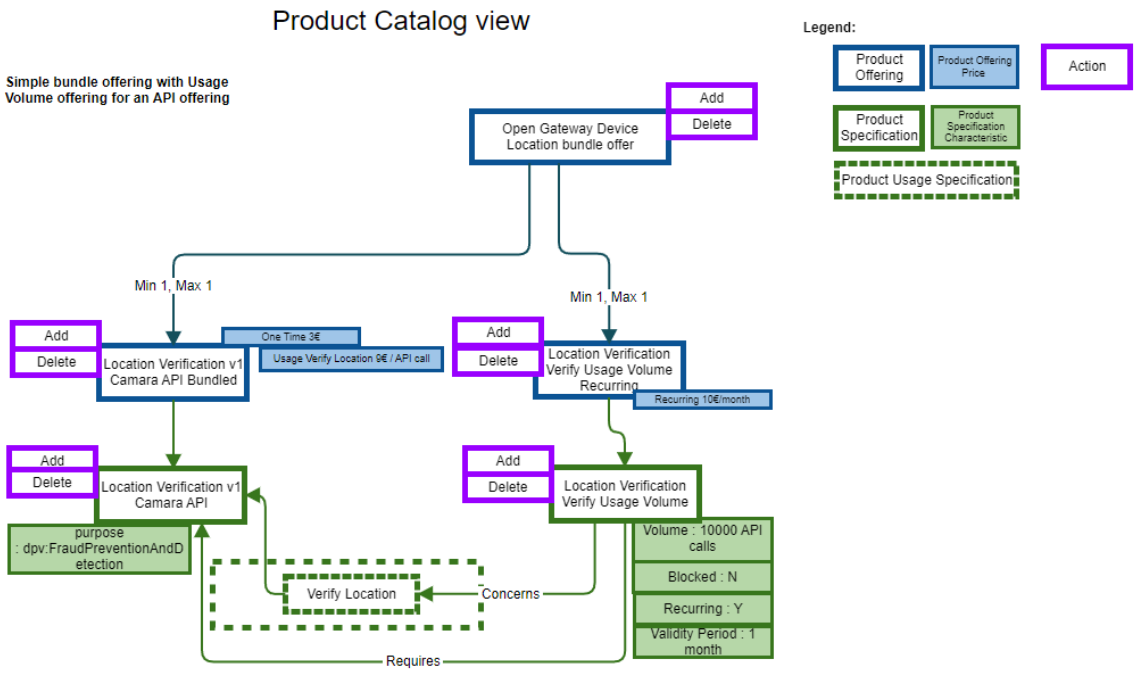
Catalog modelling (from TMF936 User Guide)

The examples in the OAS and User Guide are based on this ficticious catalog model.

Disclaimer: All prices examples appearing in this specification are fictitious. Any resemblance to real offers is purely coincidental.

Bundle offer (Operator internal view) :

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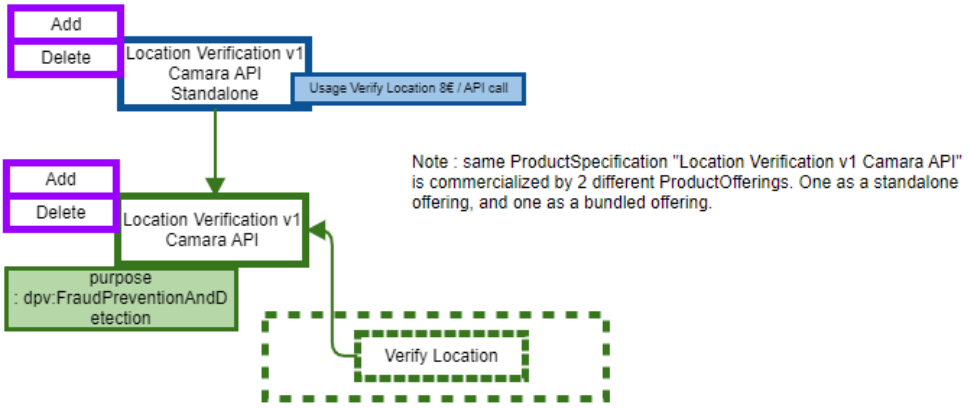


The product catalog offers for Open Gateway Location Verification a flexible model based on subscription and usage volume for an API verification service. The main offering is the Open Gateway bundle offer, which allows users to add or remove components based on their specific needs. This structure provides an adaptable solution for different use cases.

Within this offering, there are two main location verification included offerings. The first one, Location Verification v1 CAMARA API Bundled, includes the Camera API as part of a package and is specifically marketed for fraud prevention and detection.

The second included offering, Location Verification Verify Usage Volume Recurring, follows a usage-based model with recurring cost, enabling continuous access to the service. Both included offerings can be configured dynamically based on user requirements.

Standalone offer (Operator internal view) :



The concepts of ProductSpecification and ProductOffering permits Channel Partner to know which are the offerings made upon a same unique ProductSpecification.

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An Operator can design as many atomic ProductOffering as he needs based on a same ProductSpecification. An atomic ProductOffering can be based on maxiumum 1 ProductSpecification. A composite ProductOffering (bundle, package, …) is composed of atomic ProductOfferings or composite ProductOfferings.

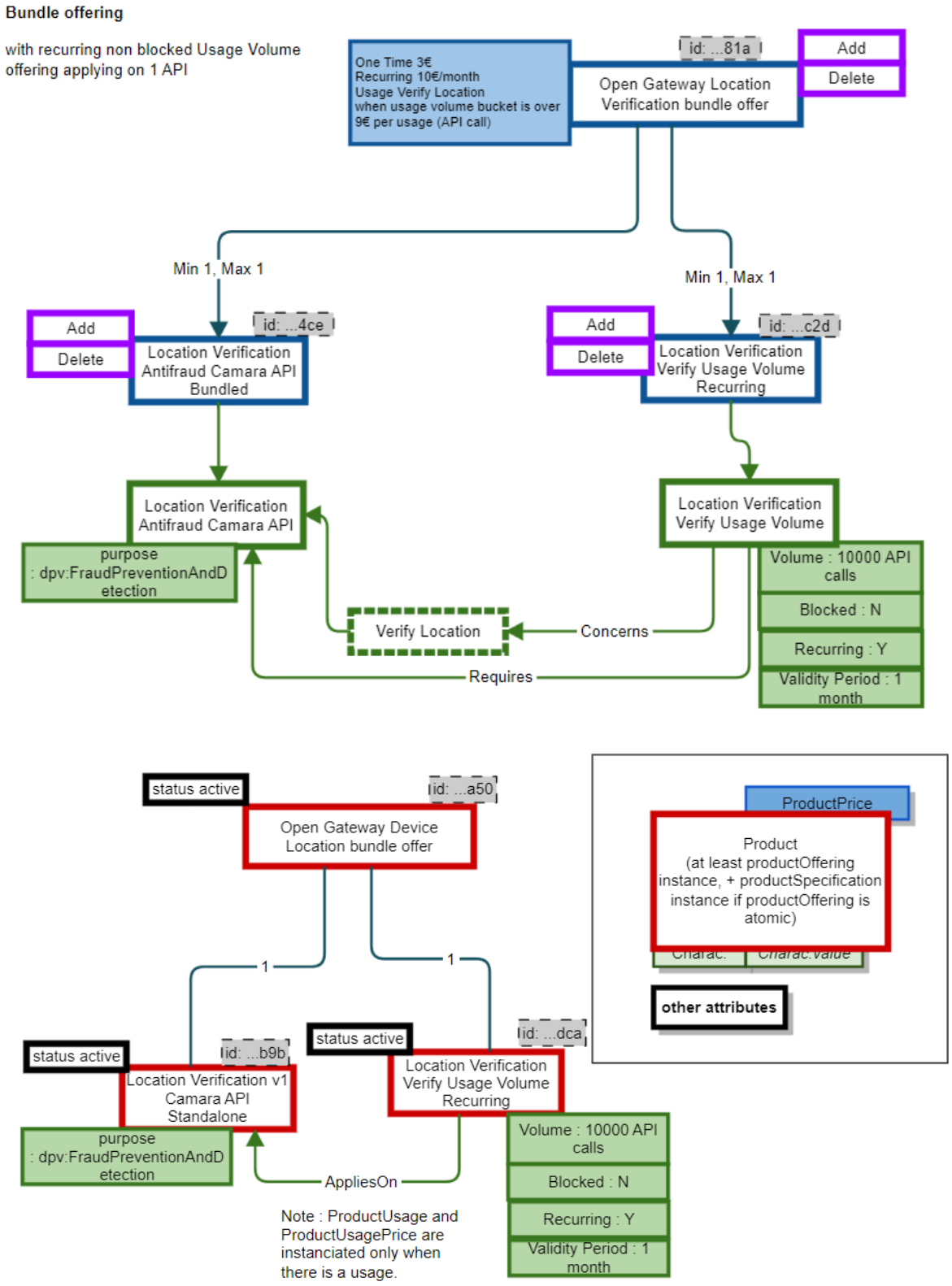
Important notice : it has been decided to simplifiy this view for the Channel Partner. We consider that the action (add, delete, modify) done at the atomic productOffering level applies same at its related productSpecification level. This implies that the product instantiated from an atomic productOffering order (action add) aggregates the instance of the atomic productOffering and the instance of the productSpecification. Inside the Operator IT it could be 2 separated products, but the view given to Channel Partner is only one product. For a composite productOffering, each productOffering that is part of the composition (including the main composite offer itself) has an instantiated product created for each productOffering that has an action add on the order. It also implies that the product order contains only the id of the productOfferings (not the id of the productSpecifications) but their productAction contain productSpecification characteristics, so apply to the ProductSpecification.

In the examples below, the id are the last 3 characters of the full id present in the OAS examples of TMF936 and TMF931. This aims to provide comprehensive understanding of the OAS examples based on these diagrams. Please note that the ProductOrder examples in OAS are not represented in these diagrams. In TMF931 there are :

- 1 example of order with action add for the standalone ProductOffering/ProductSpecification
- 1 example of order with action delete for the standalone Product (instance of the standalone ProductOffering/ProductSpecification)
- 1 example of order with 3 order items : 1 item with action add for the bundle ProductOffering/ProductSpecification, 1 item with action add for the atomic API ProductOffering/ProductSpecification and 1 item with action add for the atomic usage volume ProductOffering/ProductSpecification. These 3 order items are in relationship, keeping same relationships as in the diagram. The 3 order items are mandatory to fulfil the order. This permits to pass the necessary characteristics (in ProductAction) and other attributes that are specific for each ProductOffering/ProductSpecification.

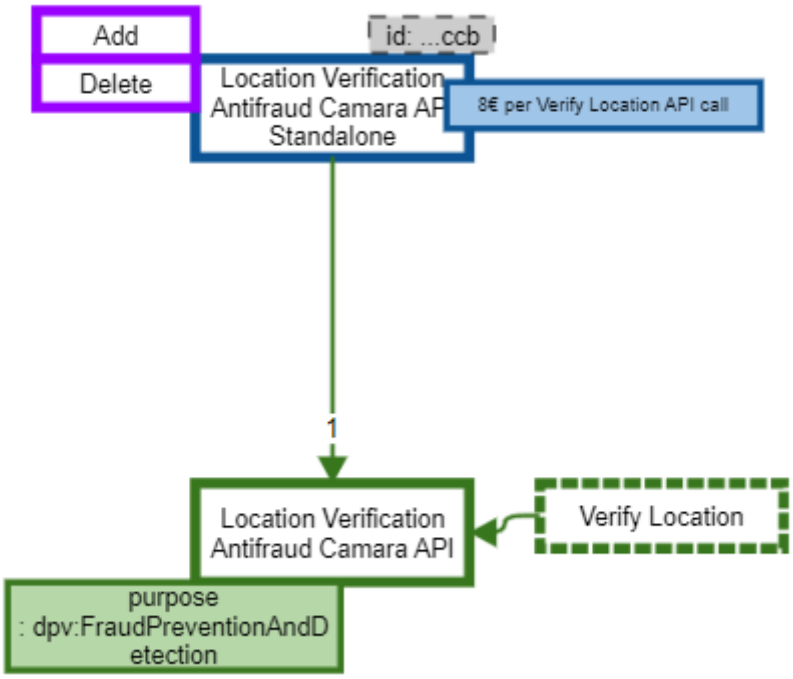
Bundle offer (Channel Partner view of catalog and inventory) :

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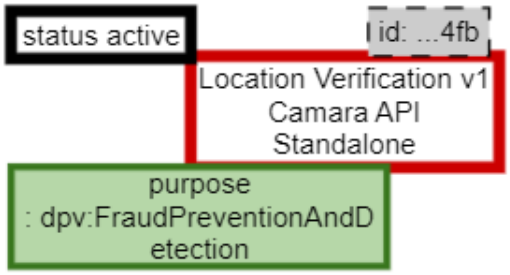


Standalone offer (Channel Partner view of catalog and inventory) :

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Standalone API offering



Note : as per choice to not expose Product Order Action Item at ProductSpecification level, the instanciation of Atomic ProductOffering and Atomic ProductSpecification are aggregated into 1 single Product

Limitations of this specification version

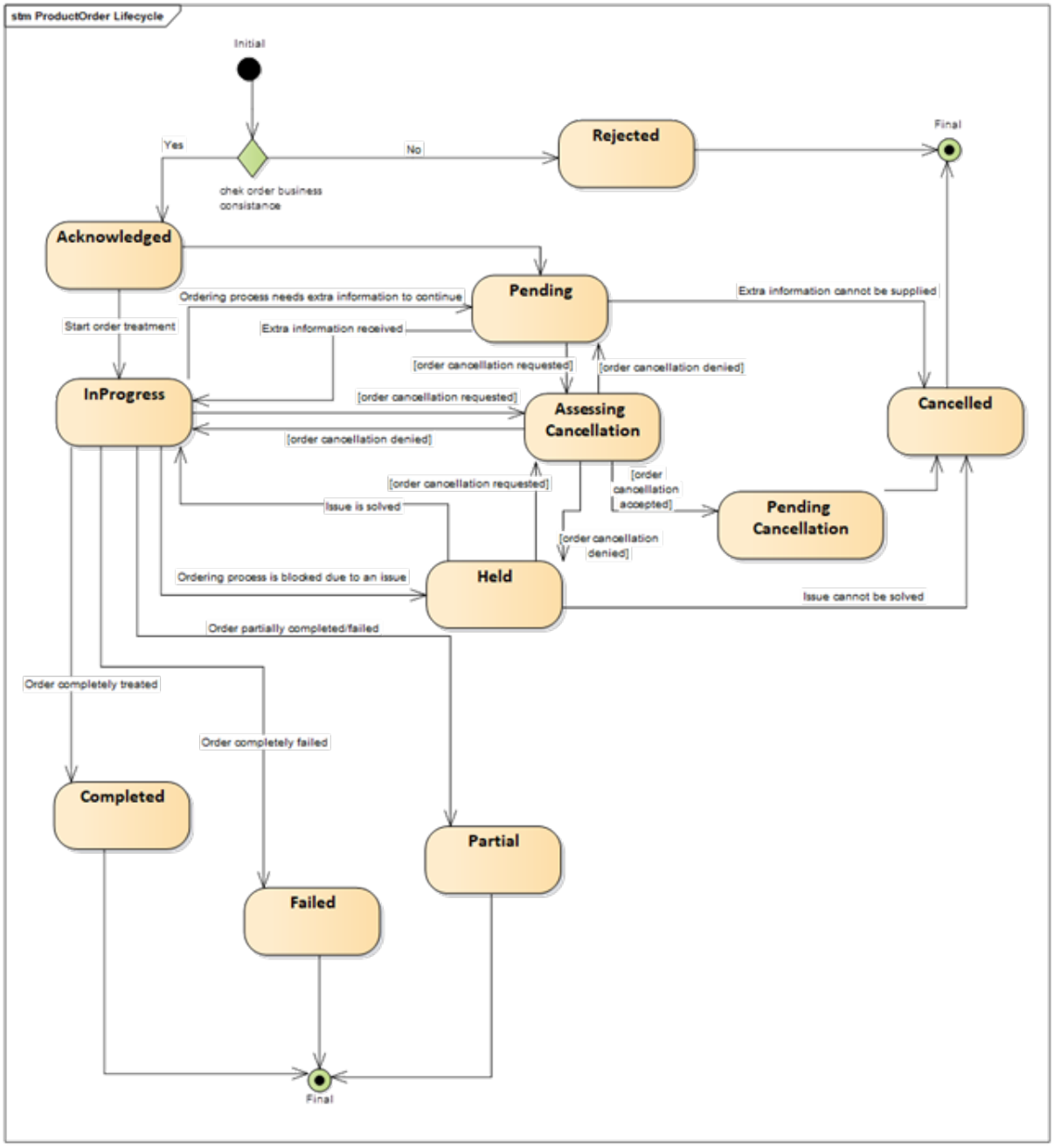
This version of this Operate API permits to implement the ordering and retrieval of products that are not only API products, so permits to manage bundle (composite product) and usage volume product. But to maintain backward compatibility it has been decided to keep the ApiProduct and ApiProductOrder schemas that fit for API products, but not so much for composite bundle product or for usage volume product. A further version of this API might propose a better fitted set of schemas for other kind of products that are not API.

State lifecycle

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Product Order

Note : ongoing GSMA Technical Playbook proposes to implement only a subset of this lifecycle.



State	Definition
acknowledged	The Acknowledged state is where an order/item has been received and has passed message and basic business validations.

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State	Definition
rejected	The Rejected state is where: An order failed the Order Feasibility check / Invalid information is provided through the order request / The order request fails to meet business rules for ordering Same rules applied for order item.
pending	The Pending state is used when an order/item is currently in a waiting stage for an action/activity to be completed before the order/item can progress further, pending order amend or cancel assessment. In situations where Access Seeker action is required, an “information required” notification will be issued on transition into this state. A pending stage can lead into auto cancellation of an order/item, if no action is taken within the defined timeframes to be described under the Agreement.
inProgress	The In Progress state is where an order Item has passed the Order Feasibility check successfully and product delivery has started. For the order at least one order item is inProgress
held	The Held state is used when an order/item cannot be progressed due to an issue. SP has temporarily delayed completing an order/item to resolve an infrastructure shortfall to facilitate supply of order. Upon resolution of the issue, the order/item will continue to progress.
assessingCancellation	Following a cancel request, the SP is assessing if cancel can be done for the order/item (or if the PO has reached PONR). If cancellation request is not accepted after assessment the order will return in Held or Pending or InProgress state.
pendingCancellation	Once a cancel order has been accepted by SP, it could in some UC take time to effectively cancel the order/item. During this time when accepted cancellation is effectively processed we use Pending Cancellation.
cancelled	The Cancelled state is where an In-Flight Order/item has been successfully cancelled.
completed	The Completed state is where an item has complete provision and the service is now active. For an order all order item are completed
failed	Order item as not a successful delivery completion. The product is not delivered and the order item failed. All Order items have failed which results in the entire Order has Failed.
partial	Some Order items have failed and some have succeeded so the entire Order is in a Partial state. This provides support for partial Failure of an Order. This state is not available at item level

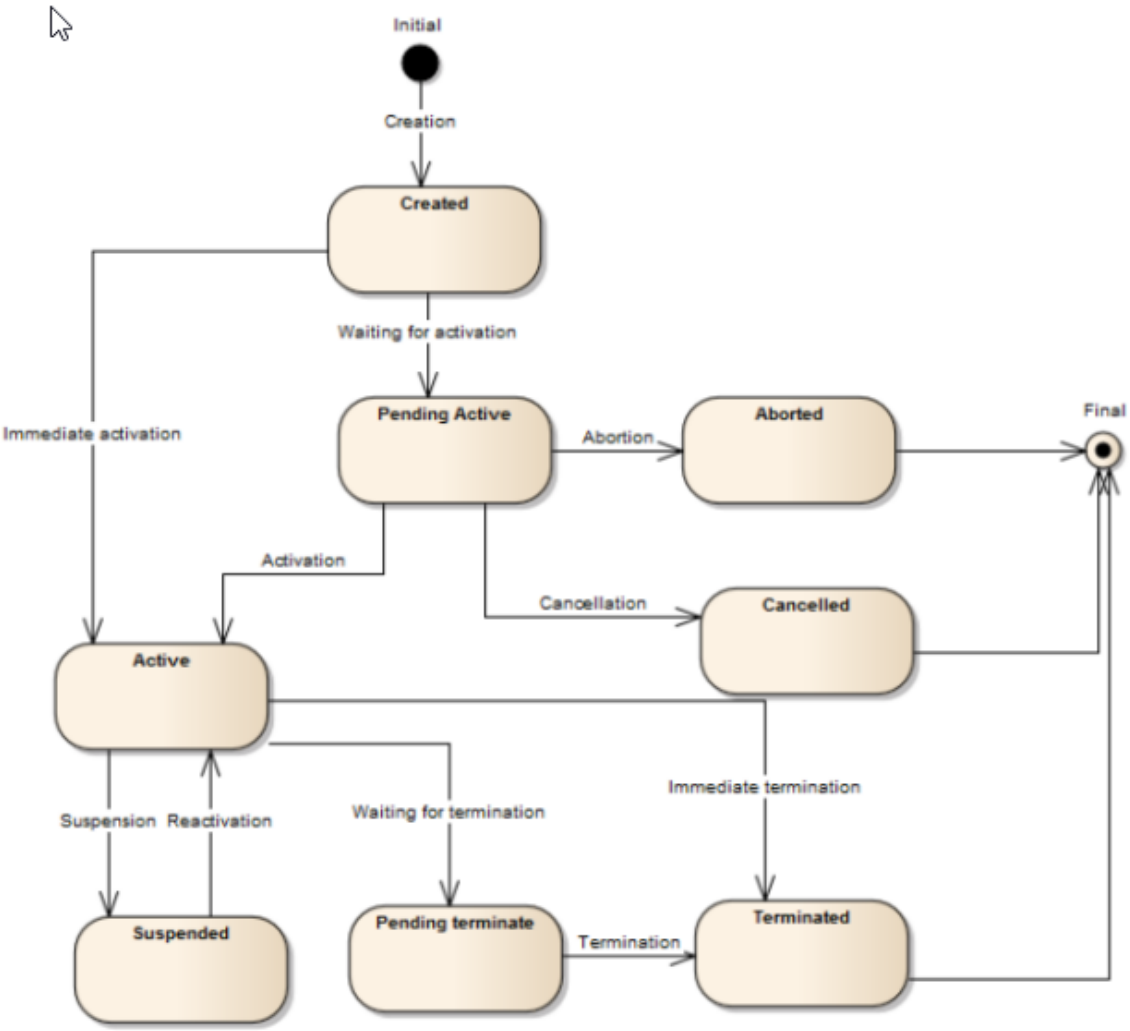
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State engine comments:

- The choice to accept order is done for the complete order and we cannot have a partial acceptance with some item in rejected and some other in acknowledged.
- Consistence rules must be defined between order state & order item state (example: an order has a failed state if it has all order item state with Completed or Failed state with at least one Completed and one Failed)

Product

Note : ongoing GSMA Technical Playbook proposes to implement only a subset of this lifecycle.



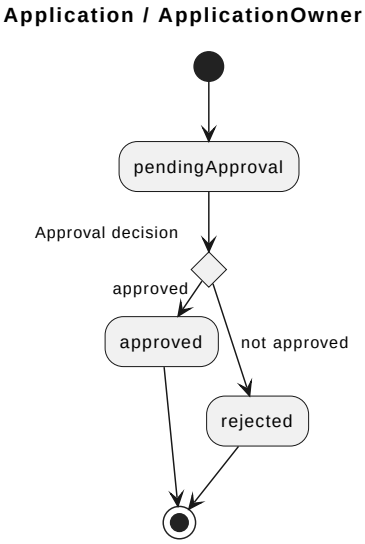
State	Definition
created	The product instance has been created but not yet activated.
pendingActive	The product is in activation phase. Not yes active but it is in progress

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State	Definition
aborted	The product activation has been stopped by abnormal condition. There is probably an unexpected delivery issue.
cancelled	The product activation has been cancelled – it could come from customer or provider.
active	The product is live and running
suspended	The product is suspended – it could be an outcome from a customer request or provider decision.
pendingTerminate	The product is still active but a termination process is in progress. It will soon be terminated
terminated	The product is not anymore active.

Application / ApplicationOwner

The creation of these 2 entities is controlled by the Operator via a manual or automatic process.



State	Definition
pendingApproval	The entity is pending for approval.
approved	The entity is approved.
rejected	The entity hasn't been approved.

Error Management

In error handling different cases must be considered, even at the functional level that it is possible to modify the error message returned to the API consumer. For this error handling there are two possible alternatives listed below:

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- Error handling done with custom policies in the API admin tool.
- Error management performed in a service queried by API.

The essential requirements to consider would be:

- Error handling should be centralized in a single place, regardless of whether they are APIs that are proxies to end services or APIs that invoke an integrator service on the ESB.
- Customization of the generated error based on the error content returned by the final core service should be contemplated.
- Latency should be minimized in its management.

In the following, we elaborate on the existing client errors. In particular, we identify the different error codes and cluster them into separate tables, depending on their nature: i) syntax exceptions, ii) service exceptions, and iii) server errors.

Client Standard Code and Reason Errors

Syntax Exceptions

HTTP Code	Error Code	Error Reason	Description
400	SVC1000	INVALID_ARGUMENT	Invalid request made
409	SVC1001	CONFLICT	The request could not be processed because of conflict in the request.
400	SVC1011	OUT_OF_RANGE	Client specified an invalid range.
403	SVC1013	PERMISSION_DENIED	Client does not have sufficient permissions to perform this action.
409	SVC1002	ABORTED	Concurrency conflict.
409	SVC0005	ALREADY_EXISTS	The resource that a client tried to create already exists.

Service Exceptions

HTTP Code	Error Code	Error Reason	Description
401	SEC1004	UNAUTHENTICATED	Request not authenticated due to missing, invalid, or expired credentials.
404	SVC1006	NOT_FOUND	The specified resource is not found.

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HTTP Code	Error Code	Error Reason	Description
429	SEC1003	TOO_MANY_REQUESTS	Either out of resource quota or reaching rate limiting.
401	SEC1004	AUTHENTICATION_REQUIRED	New authentication is required.

Server Exceptions

HTTP Code	Error Code	Error Reason	Description
400	SVR1001	FAILED_PRECONDITION	Request cannot be executed in the current system state.
422	SVR1010	{UNPROCESSABLE_ENTITY}	Request content type cannot be understood in the current system.
500	SVR1002	DATA_LOSS	Unrecoverable data loss or data corruption.
500	SVR1000	INTERNAL	Unknown server error. Typically a server bug.
502	SVR1004	BAD_GATEWAY	An upstream internal service cannot be reached.
503	SVR1006	UNAVAILABLE	Request timeout exceeded.
504	SVR1008	TIMEOUT	Request timeout exceeded.
501	SVR1003	NOT_IMPLEMENTED	This functionality is not implemented yet
405	SVR1005	METHOD_NOT_ALLOWED	The requested method is not allowed/supported on the target resource.
406	SVR1007	NOT_ACCEPTABLE	The server cannot produce a response matching the content requested by the client through Accept-* headers.
415	SVR1009	UNSUPPORTED_MEDIA_TYPE	The server refuses to accept the request because the payload format is in an unsupported format.

Note: When no login has been performed or no authentication has been assigned, a non-

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descriptive generic error will always be returned in all cases, a UNAUTHENTICATED 401 “Request not authenticated due to missing, invalid, or expired credentials.” is returned, whatever the reason.

Then we can use next regex to restrict the code field: *"(SVC)|(SVR)|(SEC[0-9]{4})$"*

Example of returned body:

```
{
    "code": "SVC1000",
    "reason": "INVALID_ARGUMENT",
    "message": "A human readable extra description of what the event represent",
    "referenceError": "http://example.com",
    "@baseType": "string",
    "@schemaLocation": "http://example.com",
    "@type": "string"
}
```

Strongly typed schemas for characteristics

Introduction

A Product Ordering business process requires the execution of a series of previous actions (including API calls) in a defined order so that the process can be carried out correctly. Like the more generic product ordering process managed with TMF Product Ordering, the specialized API Product Ordering is also catalog-oriented, so in order to be carried out it needs to know, through the execution of previous steps, the specifications of the products to be delivered in the order, as well as the actions required by the requester on these products. There is a clear alignment between the way a product is modelled in the product catalog and the way this product can be ordered via an API Product Order call. The modelling of the productSpecification or ProductOffering in the Product Catalog has also an impact on the way the Product is finally reflected in the product inventory.

With the ApiProductOrder resource definition as a specialization of a ProductOrder resource, a new way to ensure such alignment has been introduced: the use of strongly typed referenced schemas for ProductOrderItem and ProductAction. The underlying idea can be summarized as follows: each ProductSpecification managed in an ApiProductOrder requires a set of optional or mandatory attributes to be valued by the API client to ensure its successful delivery. The attributes to be in input depend on the productSpecification definition and the action to be performed over it (add, modify or delete), as each product may require different information. A schema-based model is mandatory : there are for each ProductSpecification definition a reference to N ProductAction schemas, one for each possible action available over that ProductSpecification (via a ProductOffering). Each schema includes the corresponding set of attributes necessary for that ProductOffering/ProductSpecification to be delivered via an ApiProductOrder. Same applies for the ApiProduct : for each ProductSpecification there is a reference to an ApiProduct schema that includes the corresponding set of attributes necessary for that ApiProduct description.

ProductAction replaces the product passed in the TMF622 orderItem. The idea is that productAction contains only productSpecification configured characteristic during the order

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(purpose, …) + reference of the application. This permits to avoid to use the standard product attribute that contains the full Product schema that would not be useful because needed only for characteristics.

Examples

Warning: these examples may not be 100% alinged with the OAS schemas, this is illustrative to understand the use of strongly typed schemas and the @schemaLocation to retrieve dynamically the schemas that permit to validate the payload.

Let's take the example of a Device Location Verification API that is offerd through an atomic ProductOffering based on a ProductSpecification. Let's assume that this ProductSpecification describes a specific characteristic for this API (coming from the Customer Facing Service definition) : bestAccuracy. This characteristic is not in the common schemas ApiProduct and ProductSpecification, so it's not described in the OAS, because the OAS can't contain all specific schemas from all product specifications that will be exposed in all Operator catalogs. This rule permits to not having to upgrade the Operate API OAS each time a new ProductSpecification is added into an Operator catalog.

We need to use specialized schemas that will allow this specific characteristic to be valued and read by the Operate API client. So it has to be described in a JSON schema that the API client can discover. The way TMF decided to do is :

- @type : name of the schema. It can be the standard name of schema from OAS, or can be a specific name because the schema needs to be specialized
- @schemaLocation : URL of the json schema describing the schema used in the payload. This attribute is not mandatory when @type value is the standard OAS schema; but becomes mandatory when the @type is the name of a schema not in the OAS.
- @baseType : can be used when the schema is specialized in order to indicate the parent schema

ApiProductDeviceLocationVerification.schema.json

This is the JSON schema that describes the specialized ApiProduct schema for Device Location Verification. The API client needs it when it will receive for API server a payload with an @type:ApiProductDeviceLocationVerification.

```
{
    "$schema": "http://json-schema.org/draft-07/schema#",
    "$id": "ApiProductDeviceLocationVerification.schema.json",
    "title": "ApiProductDeviceLocationVerification",
    "definitions": {
        "ApiProductDeviceLocationVerification": {
            "type": "object",
            "properties": {
                "bestAccuracy": {
                    "type": "integer"
                }
            },
            "allOf": [
            {
                "$ref": "https://.../ApiProduct.schema.json#/definitions/ApiProduct"
            }
```

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```
            ]
        }
    }
}
```

ApiProduct.schema.json

This schema, while already in OAS, is needed because the specialized ApiProductDeviceLocationVerification schema does an allOf this one. It could be possible also to put all ApiProduct description in the ApiProductDeviceLocationVerification schema in order to avoid having 2 different schemas.

```
{
    "$schema": "http://json-schema.org/draft-07/schema#",
    "$id": "ApiProduct.schema.json",
    "title": "ApiProduct",
    "definitions": {
        "ApiProduct": {
            "type": "object",
            "properties": {
                "@type": {
                    "type": "string"
                },
                "@baseType": {
                    "type": "string"
                },
                "@schemaLocation": {
                    "type": "string"
                },
                "href": {
                    "type": "string"
                },
                "id": {
                    "type": "string"
                },
                "name": {
                    "type": "string"
                },
                "description": {
                    "type": "string"
                },
                "productOffering": {
                    "$ref": "#/definitions/ProductOfferingRef"
                },
                "productSpecification": {
                    "$ref": "#/definitions/ProductSpecificationRef"
                },
                "apiName": {
                    "type": "string"
                },
                "apiVersionDetails": {
                    "type": "array",
                    "items" : {
                        "$ref": "#/definitions/ApiVersionDetails"
                    }
                },
                "productOrderItem": {
                    "type": "array",
                    "items" : {
```

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```
                        "$ref": "#/definitions/RelatedOrderItem"
                    }
                },
                "approvedProductOfferingTermOrCondition": {
                    "$ref": "#/definitions/ProductOfferingTermOrCondition"
                },
                "state": {
                    "enum" :["created", "pendingActive", "cancelled", "active",
"pendingTerminate", "terminated", "suspended", "aborted"]
                },
                "creationDate": {
                    "type": "string",
                    "format": "date-time"
                },
                "startDate": {
                    "type": "string",
                    "format": "date-time"
                },
                "orderDate": {
                    "type": "string",
                    "format": "date-time"
                },
                "terminationDate": {
                    "type": "string",
                    "format": "date-time"
                },
                "channelPartner": {
                    "$ref": "#/definitions/PartyRoleRef"
                },
                "agreementItem": {
                    "$ref": "#/definitions/AgreementItemRef"
                }
            },
            "required": [
                "@type"
            ]
        },
        "ProductOfferingRef": {
            "type": "object",
            "properties": {
                "@type": {
                    "type": "string"
                },
                "@baseType": {
                    "type": "string"
                },
                "@schemaLocation": {
                    "type": "string"
                },
                "href": {
                    "type": "string"
                },
                "id": {
                    "type": "string"
                },
                "name": {
                    "type": "string"
                },
                "@referredType": {
                    "type": "string"
                },
                "version": {
                    "type": "string"
```


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```
                }
            }
        },
        "ProductSpecificationRef": {
            "type": "object",
            "properties": {
                "@type": {
                    "type": "string"
                },
                "@baseType": {
                    "type": "string"
                },
                "@schemaLocation": {
                    "type": "string"
                },
                "href": {
                    "type": "string"
                },
                "id": {
                    "type": "string"
                },
                "name": {
                    "type": "string"
                },
                "@referredType": {
                    "type": "string"
                },
                "version": {
                    "type": "string"
                },
                "targetProductSchema": {
                    "$ref": "#/definitions/TargetProductSchema"
                }
            }
        },
        "ApiVersionDetails": {
            "type": "object",
            "properties": {
                "@type": {
                    "type": "string"
                },
                "@baseType": {
                    "type": "string"
                },
                "@schemaLocation": {
                    "type": "string"
                },
                "apiMajorVersion": {
                    "type": "integer"
                },
                "basePath": {
                    "type": "string"
                },
                "countryCode": {
                    "$ref": "#/definitions/ISO31661Alpha2StandardIdentifier"
                },
                "apiGrantInformation": {
                    "$ref": "#/definitions/ApiGrantInformation"
                }
            }
        },
        "ProductOfferingTermOrCondition": {
            "type": "object",
```


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```
            "properties": {
                To Be completed...
            }
        },
        "PartyRoleRef": {
            "type": "object",
            "properties": {
                To Be completed...
            }
        },
        "AgreementItemRef": {
            "type": "object",
            "properties": {
                To Be completed...
            }
        }
    }
}
```

ApiProductActionAddDeviceLocationOptimisation.schema.json

The API client needs it when it have to fill the body of the POST apiProductOrder for an order that will add a product (subscription). It has to put @type:ApiProducActionAddtDeviceLocationVerification and use this json schema structure for the body. The future Operate API Product catalog will permit the API Client to know that for this specific ProductSpecification the @type value must be ApiProducActionAddtDeviceLocationVerification the URL of the ApiProductActionAddDeviceLocationOptimisation.schema.json.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "$id": "ApiProductActionAddDeviceLocationOptimisation.schema.json",
  "title": "ApiProductActionAddDeviceDeviceLocationOptimisation",
  "definitions": {
    "ApiProductActionAddDeviceDeviceLocationOptimisation": {
      "$id": "#ApiProductActionAddDeviceDeviceLocationOptimisation",
      "description": "Schema for product action on a product order item for Device
Location Optimisation API.",
      "type": "object",
      "properties": {
        "bestAccuracy": {
          "type": "integer"
        }
      },
      "allOf": [
            {
                "$ref":
"https://.../ApiProductActionAdd.schema.json#/definitions/ApiProductActionAdd"
            }
            ]
      }
  }
}
```

ApiProductActionAdd.schema.json

This schema, while already in OAS, is needed because the specialized

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ApiProductActionAddDeviceLocationVerification schema does an allOf this one. It could be possible also to put all ApiProductActionAdd description in the ApiProductActionAddDeviceLocationVerification schema in order to avoid having 2 different schemas.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "$id": "ApiProductActionAdd.schema.json",
  "title": "ApiProductActionAdd",
  "definitions": {
    "ApiProductActionAdd": {
      "$id": "#ApiProductActionAdd",
      "description": "Schema for product action on a product order item for an
API.",
      "type": "object",
      "properties": {
        "@type": {
          "const": "ApiProductActionAddDeviceDeviceLocationOptimisation",
        },
        "@baseType": {
          "type": "ApiProductActionAdd",
        },
        "@schemaLocation": {
          "type": "string",
          "description": "The URI of this JSON-Schema file that defines additional
attributes and relationships"
        },
        "purpose": {
          "type": "string"
        },
        "countryCode": {
          "type": "array",
          "items": {
             "type": "string"
          }
        },
        "targetApplication": {
          "type": "object",
          "properties": {
            "@type": {
              "const": "ApplicationRef"
            },
            "id": {
              "type": "string"
            }
        }
      }
    }
  }
}
}
```

Call flows

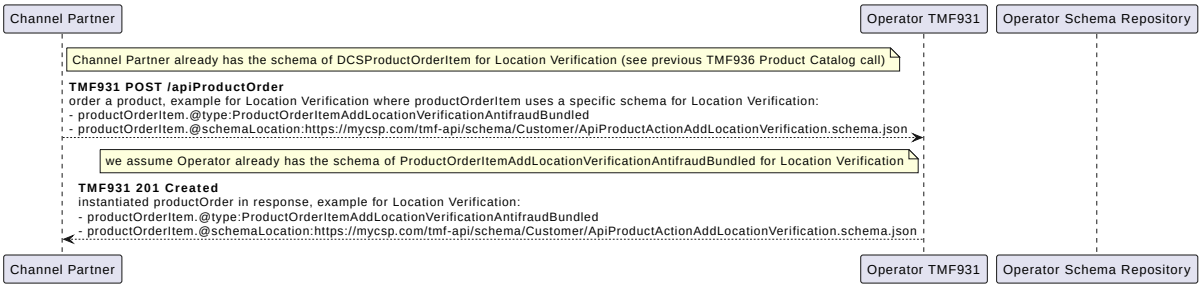
In order to better understand how to manage strongly typed schemas, this call flow is an illustrative way of doing it.

Getting product offering and product specification

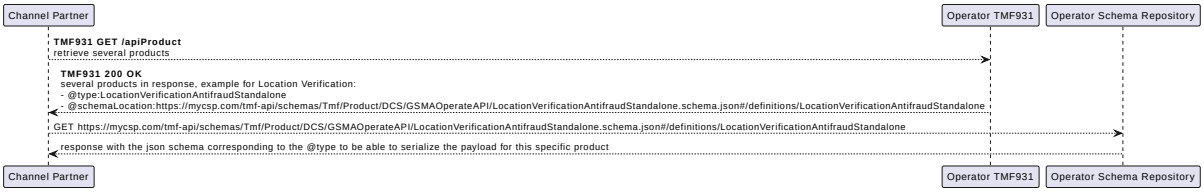
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Posting product order use case



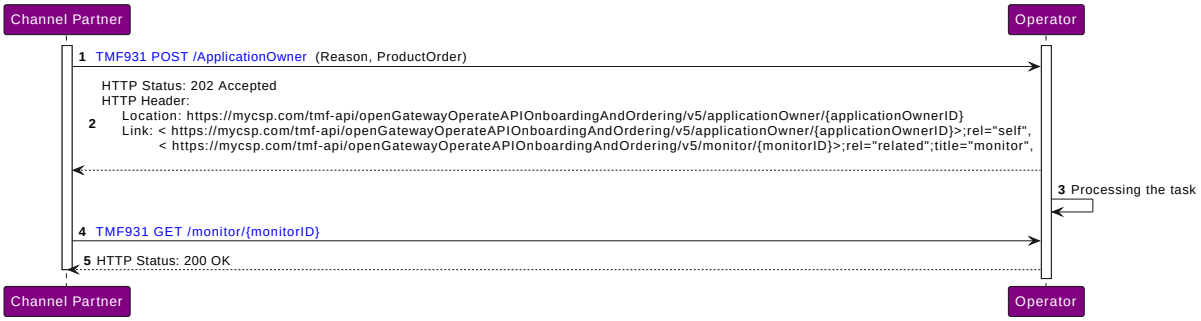
Reading product inventory use case



Monitor pattern

Monitor pattern is described in TMF630 REST guidelines.

A classic interaction with Asynchronous response and monitor pattern looks like below:



In the above example we can notice the following:

The client does not send any header to indicate to the server if it would like a synchronous answer or an asynchronous answer (no Expect Header used).

According to the API definition, the server can return either 201 Created or 202 Accepted - meaning that it can support both patterns (synchronous and asynchronous) and is up for the server implementation to decide or if there is support for the client to decide via the Expect

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header.

The server is:

- returning a 202 Accepted HTTP Status Code - representing the asynchronous behavior
- returning a Location header to indicate to the consumer the path to the applicationOwner resource. In this case, the server is able to provide as part of the response the applicationOwnerID.

The location path can look something like this:

Location: https://mycsp.com/tmf-api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-40d9-8d16-112275a28385 (where 31 is the applicationOwner identifier)

Note If the server is unable to provide a resource identifier, then it must provide the monitor resource. The link to the actual resource (self)

- Link: https://mycsp.com/tmf-api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-40d9-8d16-112275a28385;rel=self,

Due to the fact that the server is implementing a "monitor" pattern it will also return the relevant "hypermedia links"

The link to the monitor resource, where the consumer can check regularly in order to check the status of the operation

- Link: https://mycsp.com/tmf-api/openGatewayOperateAPIOnboardingAndOrdering/v5/monitor/987654;rel=related;title=monitor, (where the 987654 is the monitor resource identifier assigned for this resource)

Note : This pattern can be done for PATCH operations also.

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Sample Use Cases

This section explains how the functions are implemented using the Onboarding Operate API. As mentioned above, there are three groups of operations in this API:

- Application Service Provider (ApplicationOwner) Management This functional area covers the management of a ApplicationOwner as a Party Role. This includes the creation of the ApplicationOwner PartyRole in this context. It also covers the retrieval of the ApplicationOwner.

*Application Management This functional area covers the management of the LogicalResource representing the Application in the Channel Partner Resource Inventory. This includes the creation of the Application as a LogicalResource + its LogicalResourceRole in this context. This Application, created in the Channel Partner resource inventory has to be known by the Operator. The Operator will then have to create this representation of this Application in a dedicated resource inventory.

An Application is modeled as a subclass of a LogicalResource.

It covers also the deletion of the Application from the Channel Partner side.

- Products Management This functional area covers the Channel Partner ordering the Operator for an API to be consumed by an Application. This requirement is mandatory as it's the core need for this API business requirements. It permits the Channel Partner to order a Product to the Operator when it is mandated in the Order Delivery process of the Channel Partner. This happens when ApplicationOwner orders an API to Channel Partner.

A Service API is modeled as a Product (ProductOffering/ProductSpecification instance).

The order can be for subscribing or unsubscribing a Service API.

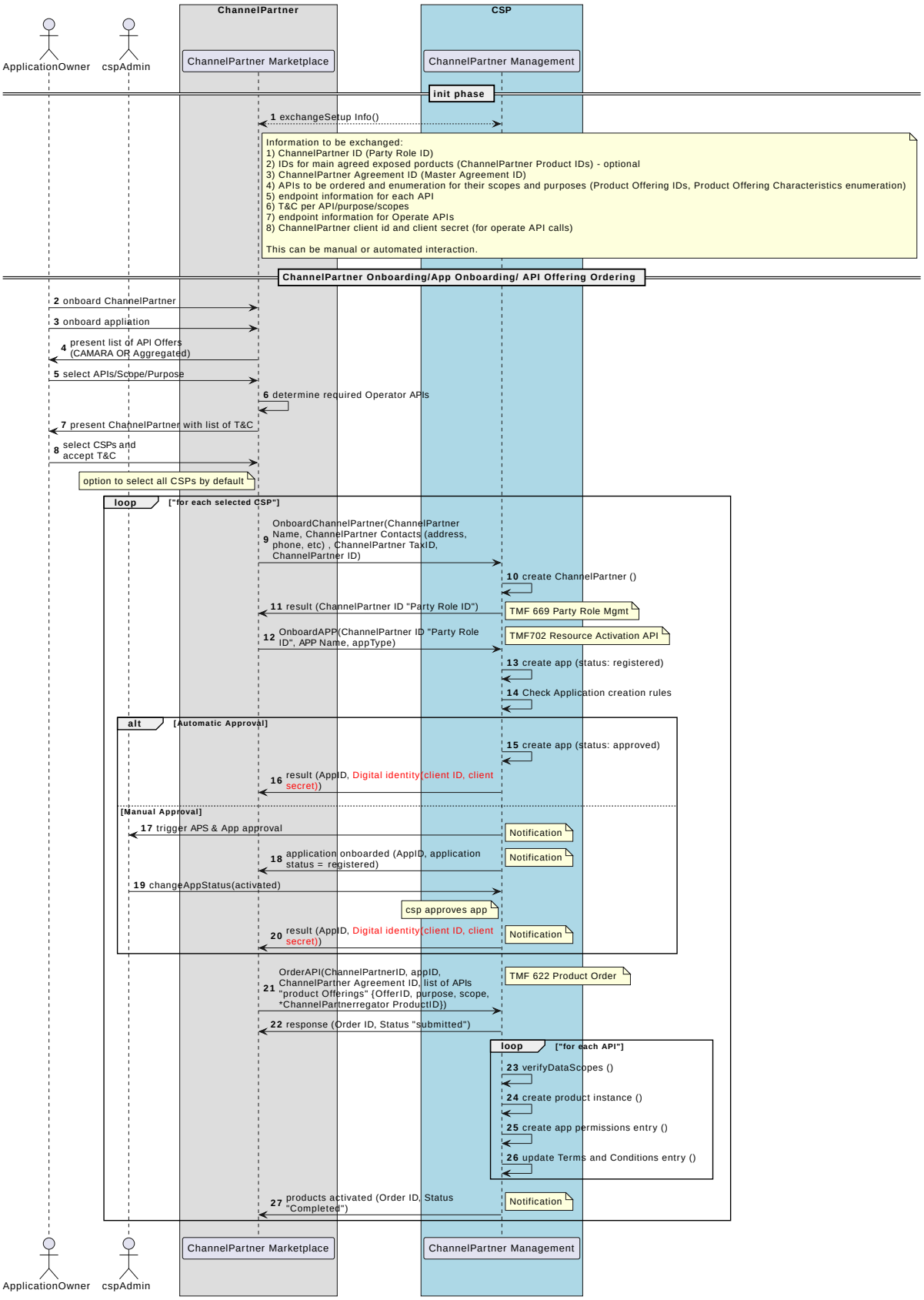
It also covers the retrieval to the Operator of the Channel Partner Products that result from these orders.

UC1: ApplicationOwner Onboarding / Application Onboarding / API Offering Ordering

The use case covers fully onboarding of the full hierarchy of ApplicationOwner, Application and providing access to Service APIs.

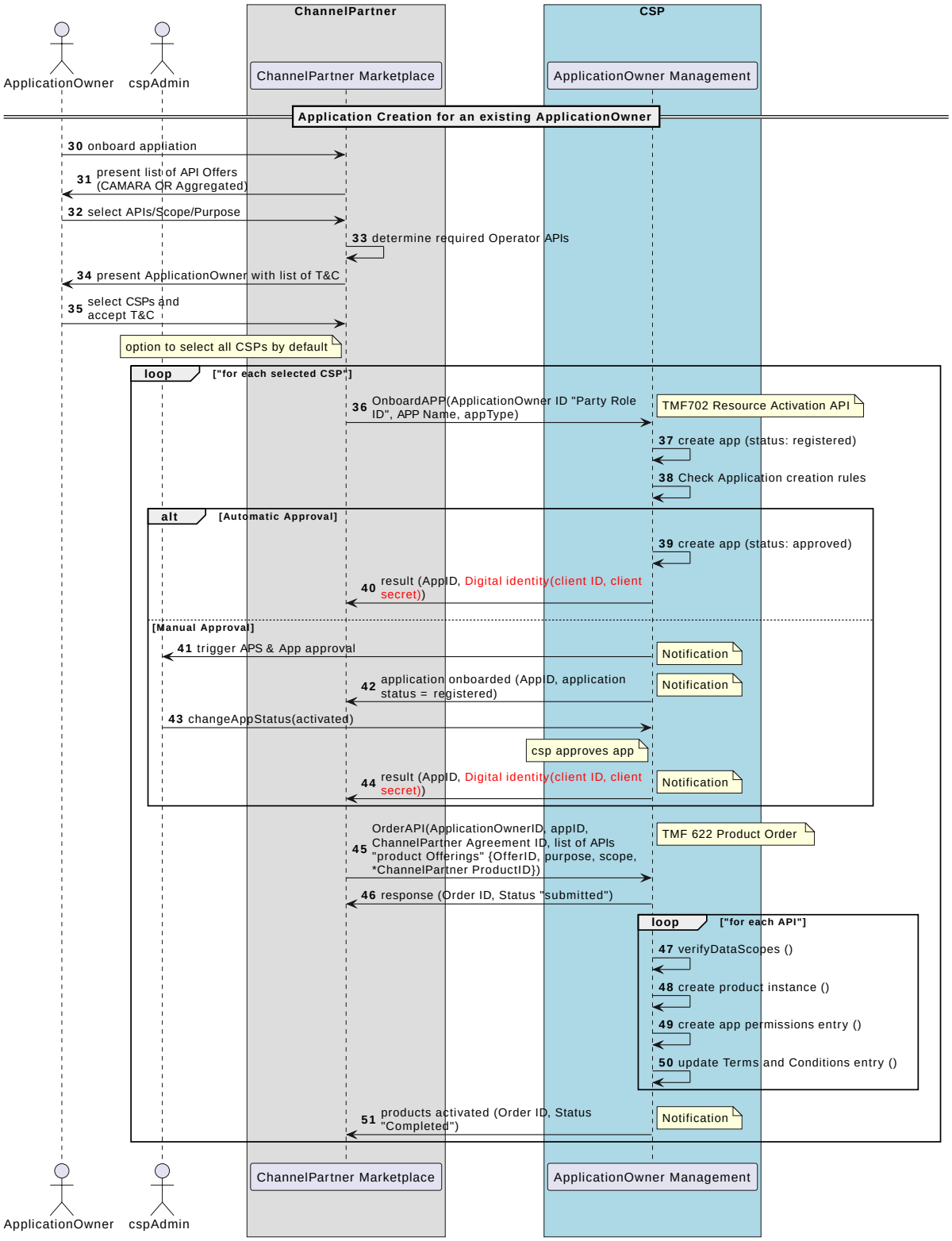
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Onboard ApplicationOwner and application in ChannelPartner/Wholesale Scenario (*Part One*)



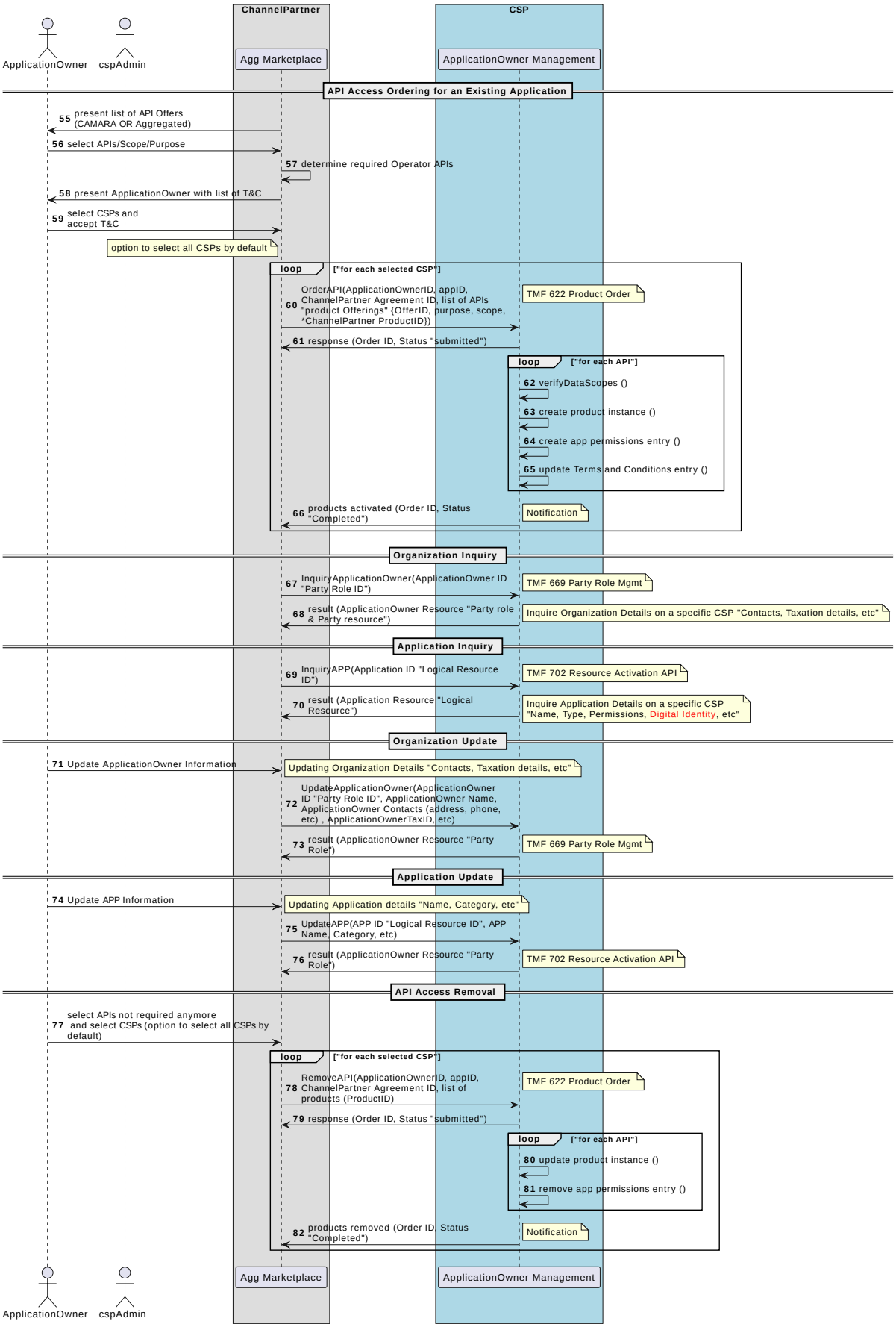
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Onboard ApplicationOwner and application in ChannelPartner/Wholesale Scenario (*Part Two*)

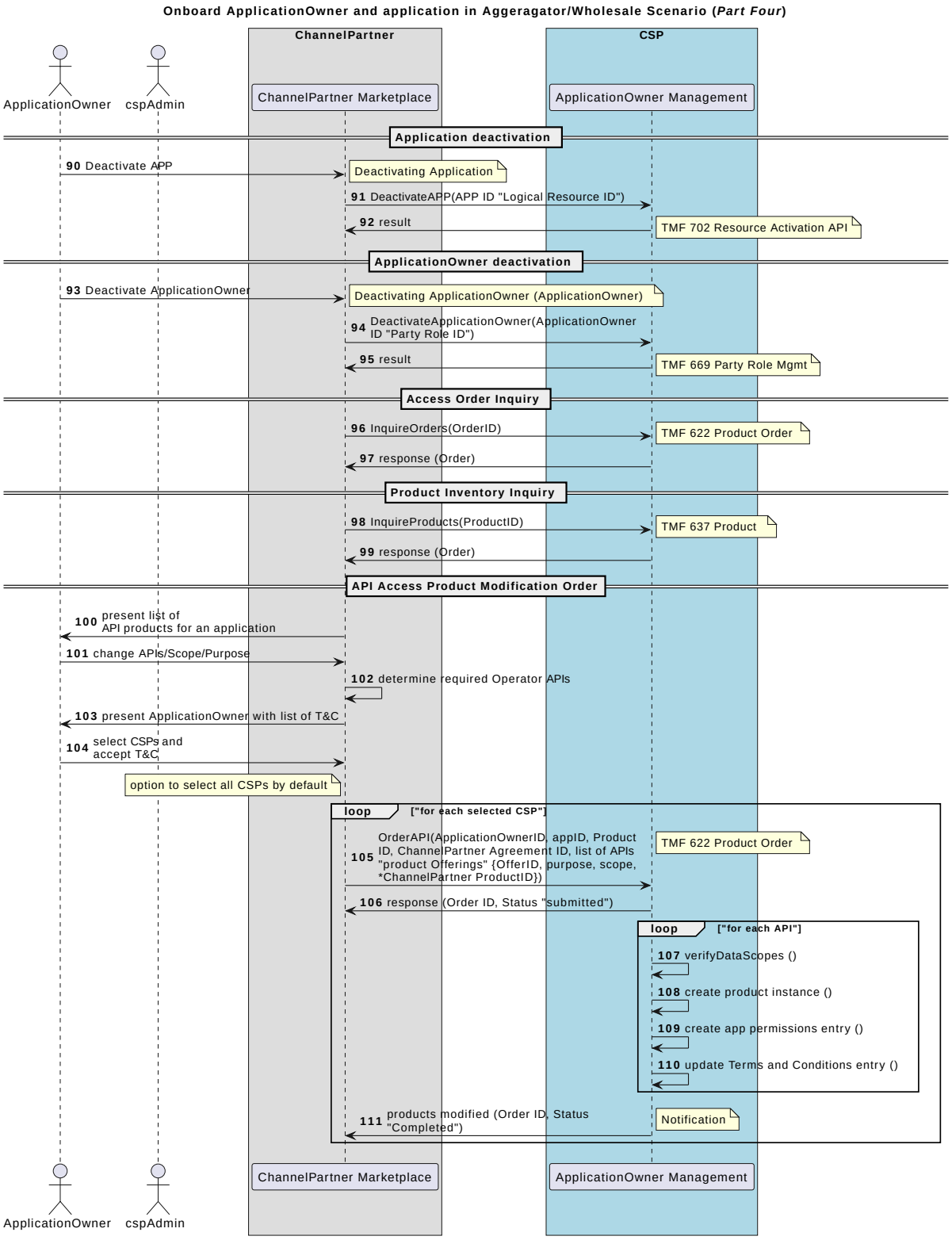


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Onboard ApplicationOwner and application in Aggeragator/Wholesale Scenario (*Part Three*)



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In the first part, the ApplicationOwner (Application Service Provider) registers with the Channel Partner. As a consequence, the Channel Partner presents the list of available Service APIs to the ApplicationOwner. Those APIs can be provided by different Operators, since the Channel Partner can have previous business agreements with several Operator.

The ApplicationOwner then can select the list of APIs that fit its needs. It follows a sequence where the Channel Partner sends the Terms and Conditions for the ApplicationOwner to

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accept. This sequence can be a loop where the ApplicationOwner registers on several Operators and accepts the Terms and Conditions of each of them.

Once the ApplicationOwner is registered and the Terms & Conditions have been accepted, the Channel Partner starts working directly with the Operator. For each Operator selected by the ApplicationOwner, the Channel Partner proceeds to onboard the ApplicationOwner in the Operator and then to onboard each of the Applications that the ApplicationOwner wants to register for the use of the Operator APIs.

In a third sequence, for each Application, one or more APIs are ordered to the Operator, who does internal work to create the new Products instances, together wih permissions and Terms & Conditions checking. When this internal processing is finished, the Operator informs the Channel Partner of the positive result of the request, that is, the activation of the APIs for the ApplicationOwner to use in its Application.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
8	Onboarding ApplicationOwner	POST /applicationOwner
11	Onboard Application	POST /application
16,17,19	Application Approval Status Change Event	POST /client/listener
20	Order API	POST/APIProductOrder

The assumption is that manual checks can be taken on application level, and for initial ApplicationOwner creation both the ApplicationOwner itself as well as the first application can be assessed at the same time.

These operations may require a manual authorization, so has to be designed to be compatible with synchronous or asynchronous functional response.

This use case is preceded by the configuration of its products by the Operator and the initialization of the Channel Partner - Operator agreement. The information below is expected to be exchanged prior this use-case:

- Channel Partner ID
- Identifiers for the main agreed exposed Product Offering (Channel Partner Product Offering IDs) - optional
- Channel Partner Agreement Identification (Master Agreement)
- APIs to be ordered and enumeration for their scopes and purposes (Product Offering IDs, Product Offering Characteristics enumeration)
- Endpoint information for each API
- Terms and Conditiones per API/purpose/scopes
- Endpoint information for Operate APIs
- Channel Partner client id and client secret (for operate API calls)

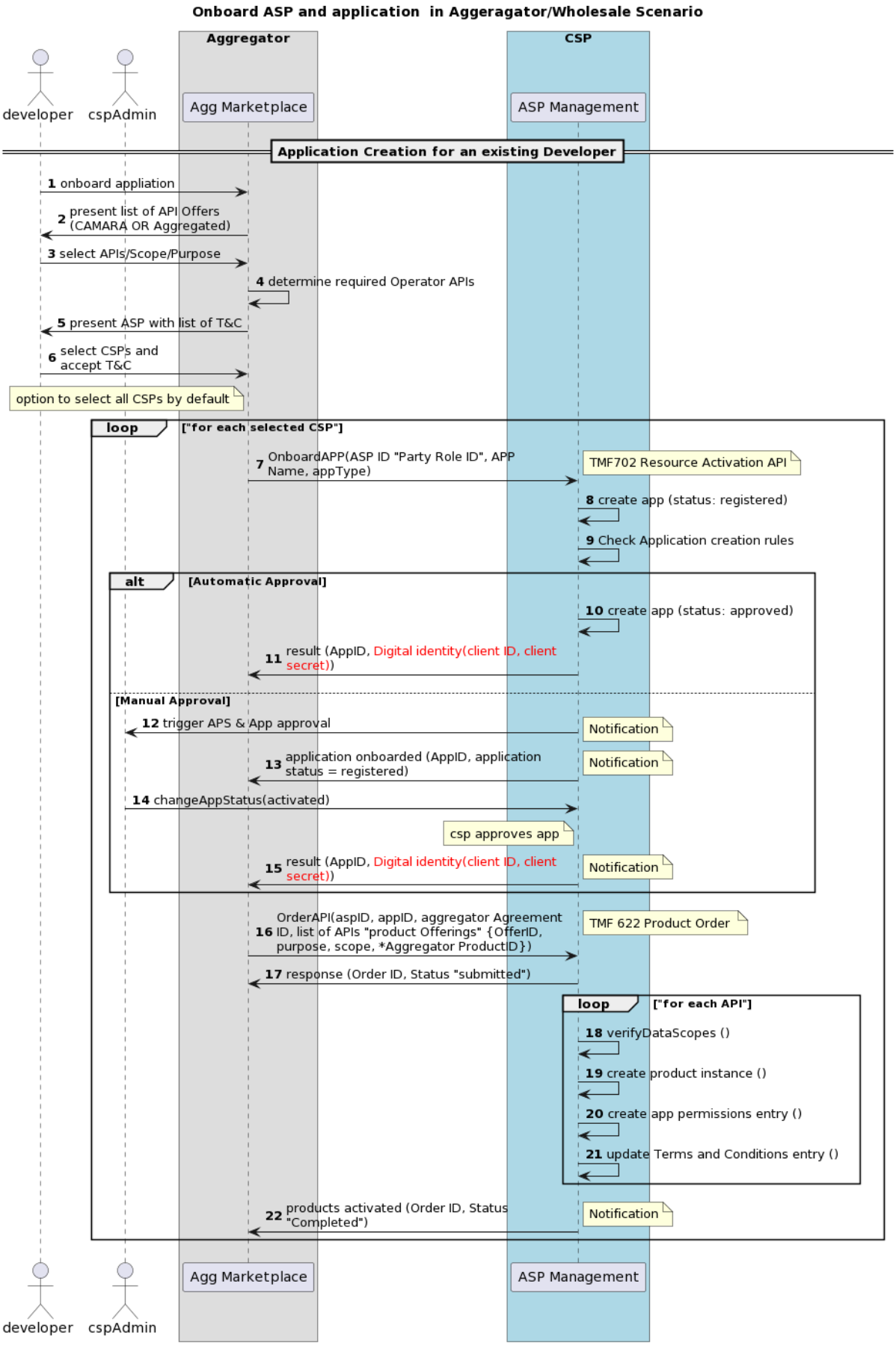
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UC2: Application Creation for an existing ApplicationOwner

This use case covers onboarding a new Application for an ApplicationOwner which was already registered in the Channel Partner (and hence in the Operator) and providing access to Service APIs.

The use case is preceded by the Operator configuration of their products initialization of the Channel Partner Operator agreement, and creation of the ApplicationOwner.

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The scenario starts in a similar way to Use Case 1. Now the process of ApplicationOwner onboarding is not needed, since the ApplicationOwner is already registered in the Channel Partner and the Operator. A new Application is registered on the Channel Partner, who then proceed to onboard the Application on the Operator. The onboarding includes the registration of the list of Service APIs needed for the Application, together to the internal Operator steps to create new Product and validate permissions and Terms & Conditions.

This function may require a manual authorization, so has to be designed to be compatible with synchronous or asynchronous functional response.

The use case allows for 2 variations of managing application digital identity: Digital Identity is created during application creation : Operate API will orchestrate the Operator security server returning back the client ID and client secret (Digital Identity is returned in application creation response) Digital identity not created during application creation : Seperate call need to be orchestrated by the Channel Partner directly to the Operator security server (Digital Identity not returned in application creation response)

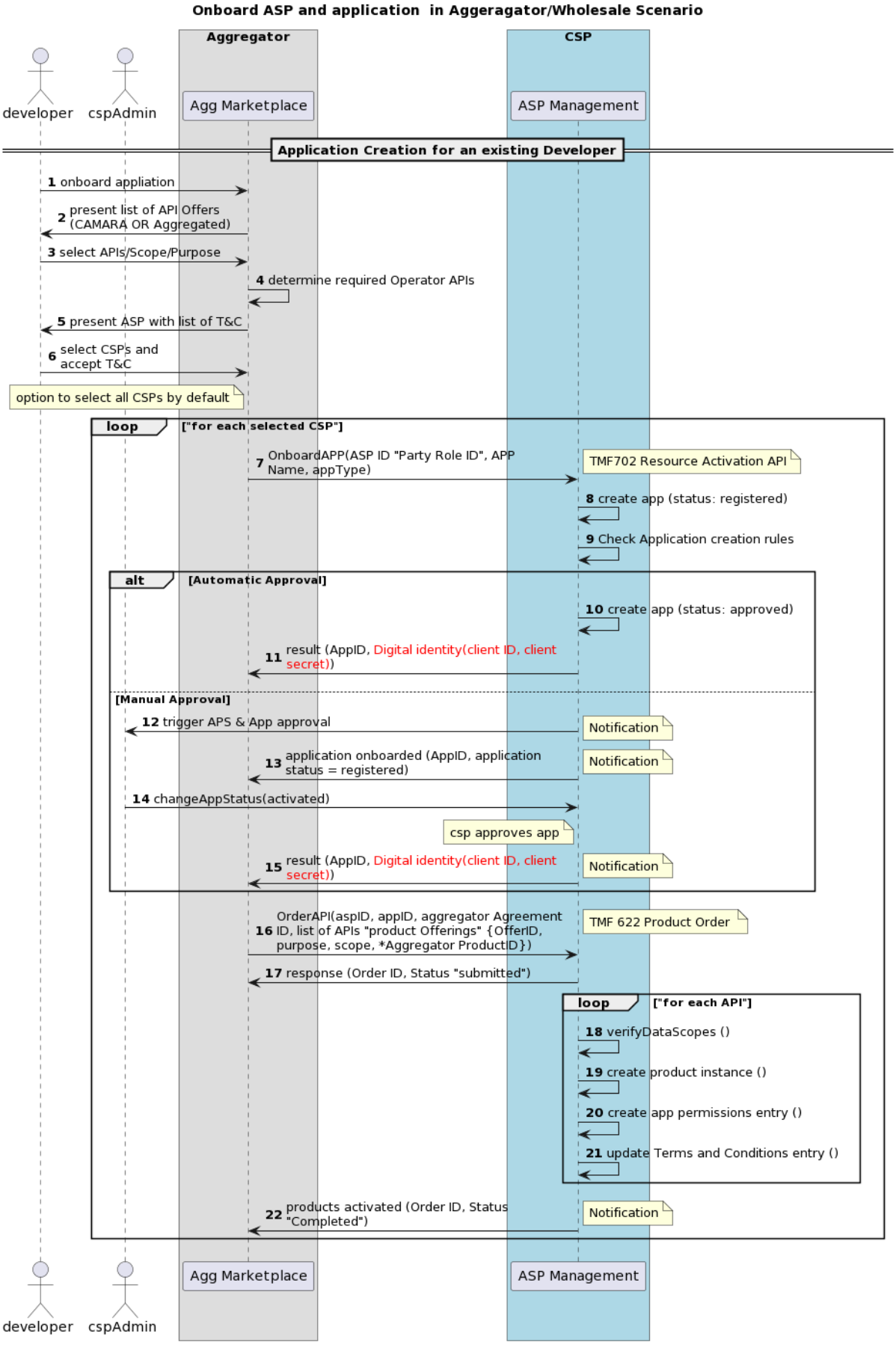
The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
7	Onboard Application	POST /application
12/13/15	Application Approval State Change Event	POST /client/listener/applicationApprovalStatusChangeEvent
16	Order API	POST /apiProductOrder

UC3: API Access Ordering for an Existing Application

In this use case an Application get access to Service APIs. This use case is preceded by the Operator configuration of their products, the initialization of the Channel Partner Operator agreement, the creation of the ApplicationOwner as well granting application access to the API.

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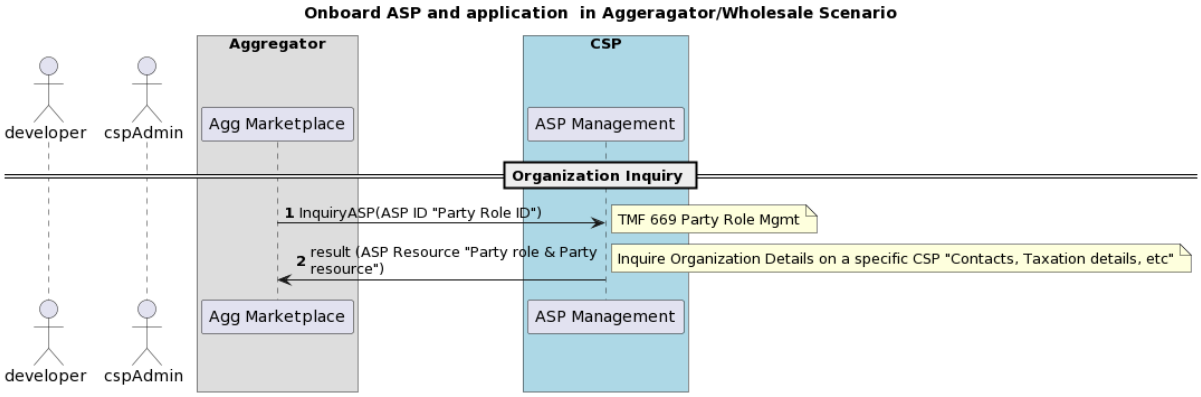
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The ApplicationOwner selects one or several APIs/Scopes/Purposes already offered by the Channel Partner. The APIs could be provided by one or several Operators. For each of the Operators involved, the ApplicationOwner has to accept the Terms & Conditions. Once this is accomplished, the Channel Partner request onboarding for the APIs to the corresponding Operator. Each of the Operators does the internal work required to validate permissions and do Terms and Conditions checking. Activation of each of the APIs is notified to the Channel Partner.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
7	Order API	POST/ apiProductOrder

UC4: Organization Inquiry



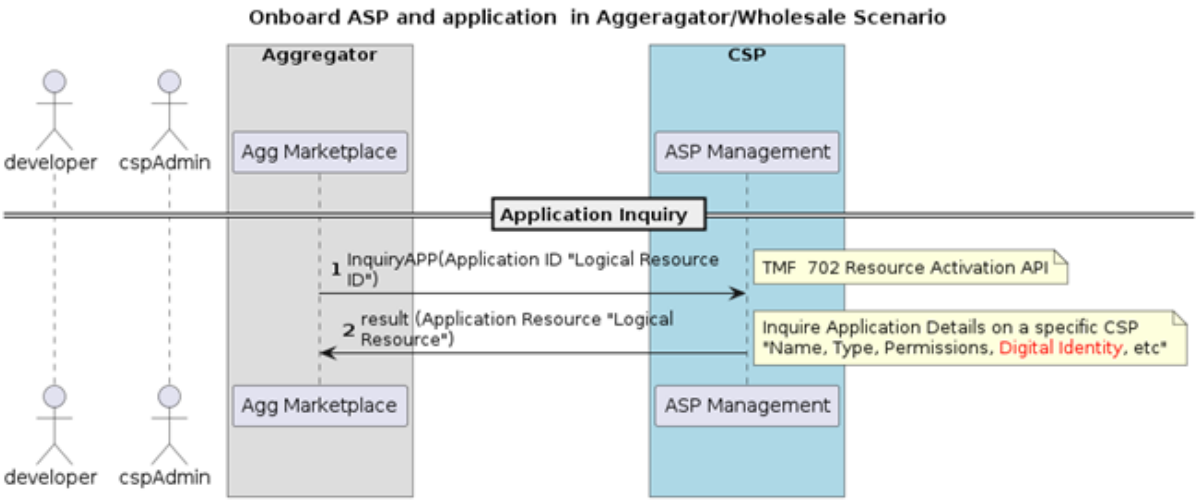
In this use case, the Channel Partner asks the Operator about the information it holds regarding an ApplicationOwner that was previously registered.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
1	Retrieve ApplicationOwner	GET /applicationOwner

UC5: Application Inquiry

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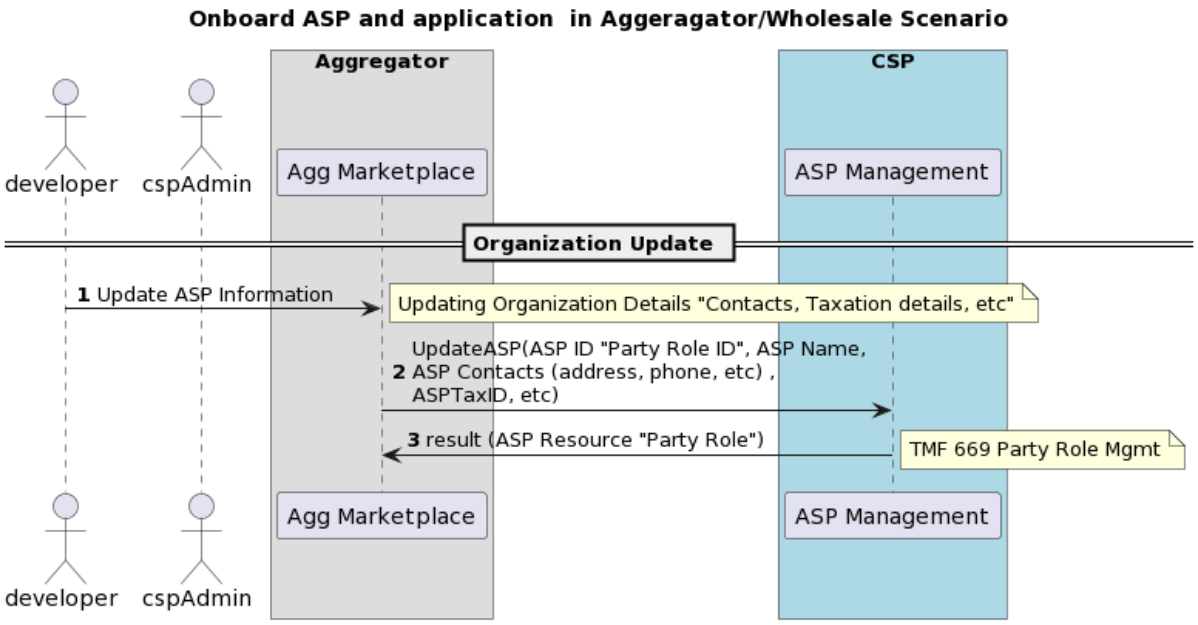


In this use case, the Channel Partner asks the Operator about the information it holds regarding an Application pertaining to an ApplicationOwner. Both ApplicationOwner and its Application should have been previously registered.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
1	Retrieve Application	GET /application

UC6: Organization Update



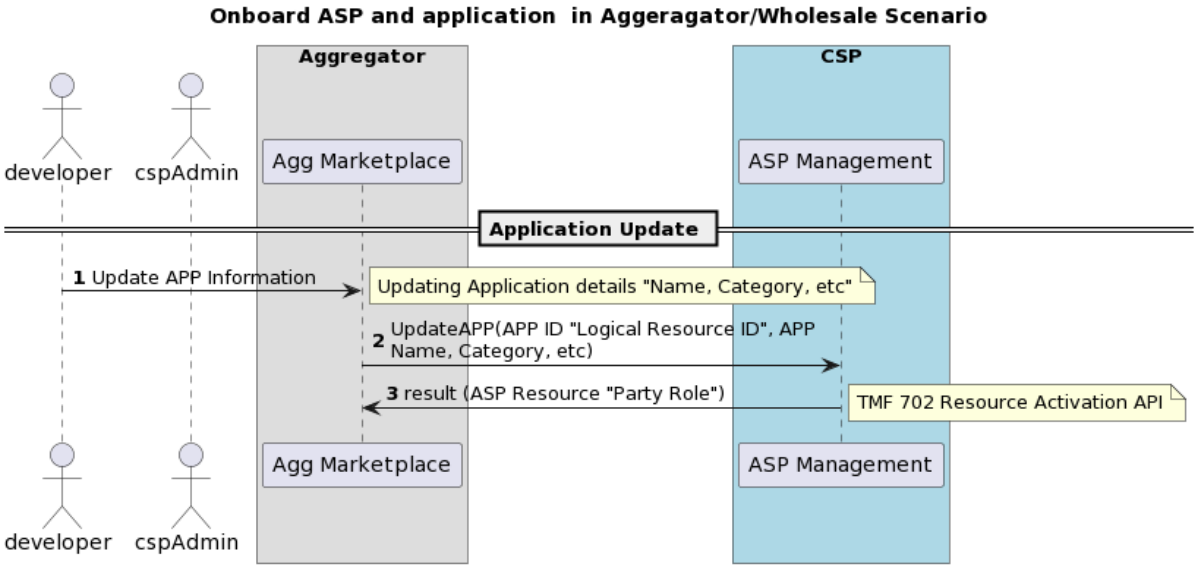
In this use case, the ApplicationOwner requires the Channel Partner to update some details about itself, such as the contact information, tax information, etc. The Channel Partner in turn sends an update request to the Operator to synchronize the changes required. This operation implies that the ApplicationOwner has been previously registered on the Channel Partner and the Operator.

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The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
2	Update ApplicationOwner	PATCH /applicationOwner

UC7: Application Update



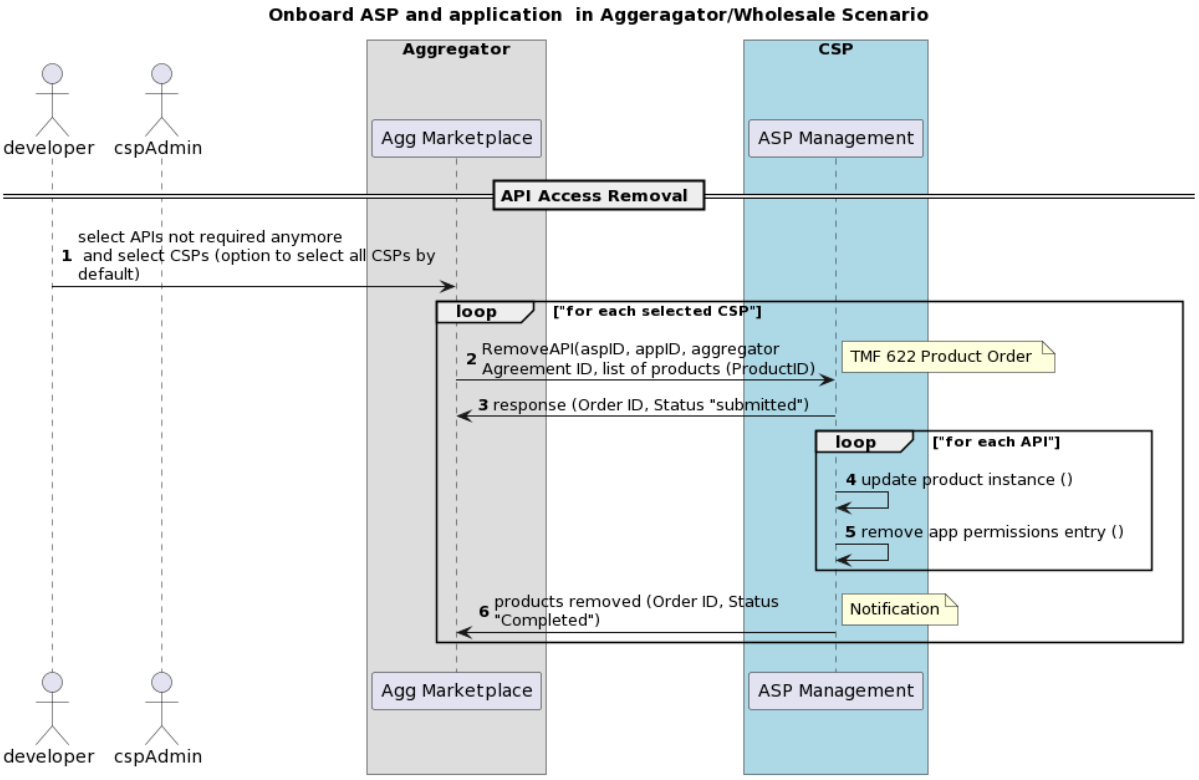
In this use case, the ApplicationOwner requires the Channel Partner to update some details about one of the Applications that were previously registered. The updates involve attributes such as the name or the category. The Channel Partner in turn sends an update request to the Operator to synchronize the changes required. This operation implies that the ApplicationOwner and the Application have been previously registered on the Channel Partner and the Operator.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
2	Update Application	PATCH /application

UC8: API Access Removal

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In this use case, the ApplicationOwner requires the Channel Partner to remove some APIs that are not needed any more. Operator providing the APIs and Applications that were using them should be specified. The Channel Partner in turn sends a removal request to the Operator. If there are more than one Operator or more than one API, the Channel Partner sends the removal requests in a loop until all removals have been sent. Each Operator confirms removal for all their APIs.

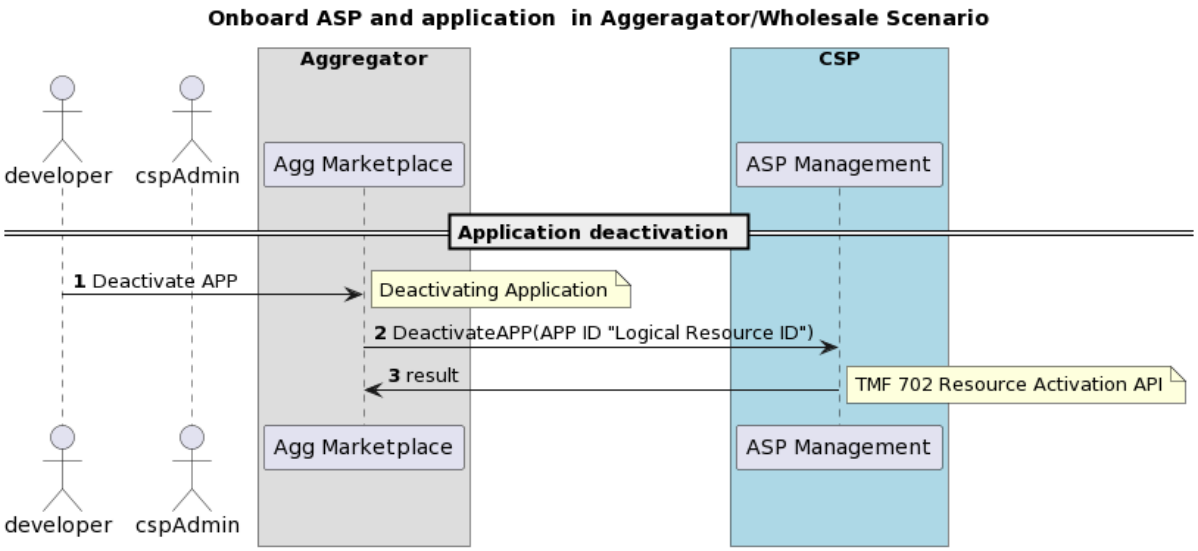
This operation implies that the ApplicationOwner, the Applications and the APIs have been previously registered on the Channel Partner and the Operator.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
2	Order API product for unsubscription	POST /apiProductOrder

UC9: Application deactivation

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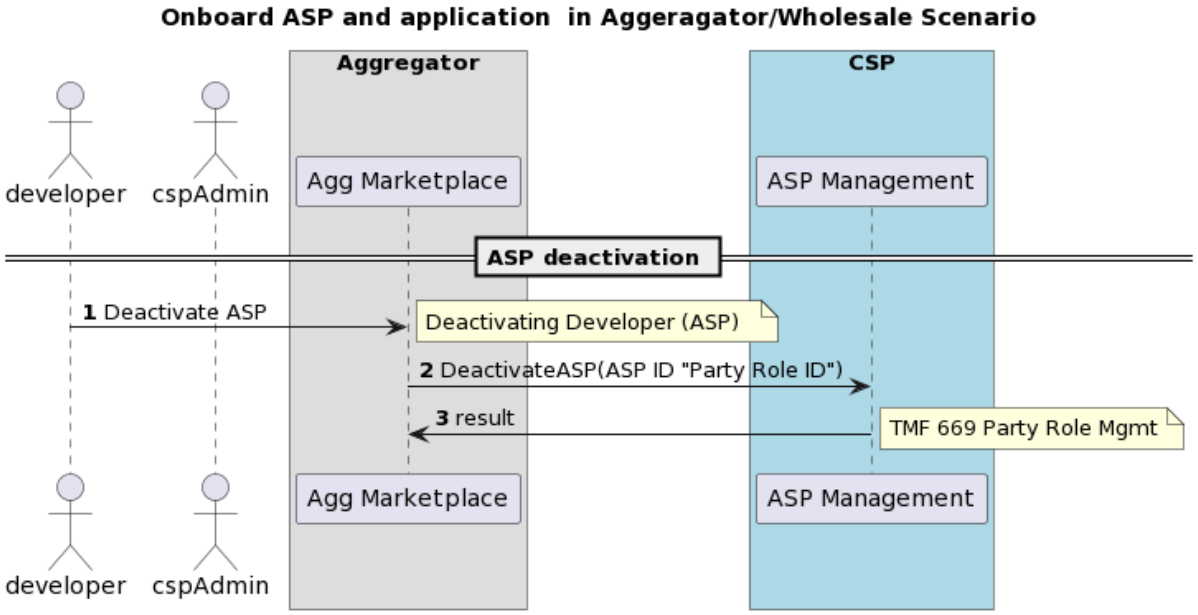
In this use case, the ApplicationOwner requires the Channel Partner to deactivate one of its applications. The Channel Partner, in turn, sends a deactivation request to the Operator and receives a confirmation from it.

This operation implies that the ApplicationOwner and the Application have been previously registered on the Channel Partner and the Operator.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
2	Deactivate Application	PATCH /application

UC10: ApplicationOwner deactivation



In this use case, the ApplicationOwner requires the Channel Partner to deactivate itself from

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the Channel Partner. The Channel Partner, in turn, sends a deactivation request to the Operator and receives a confirmation from it.

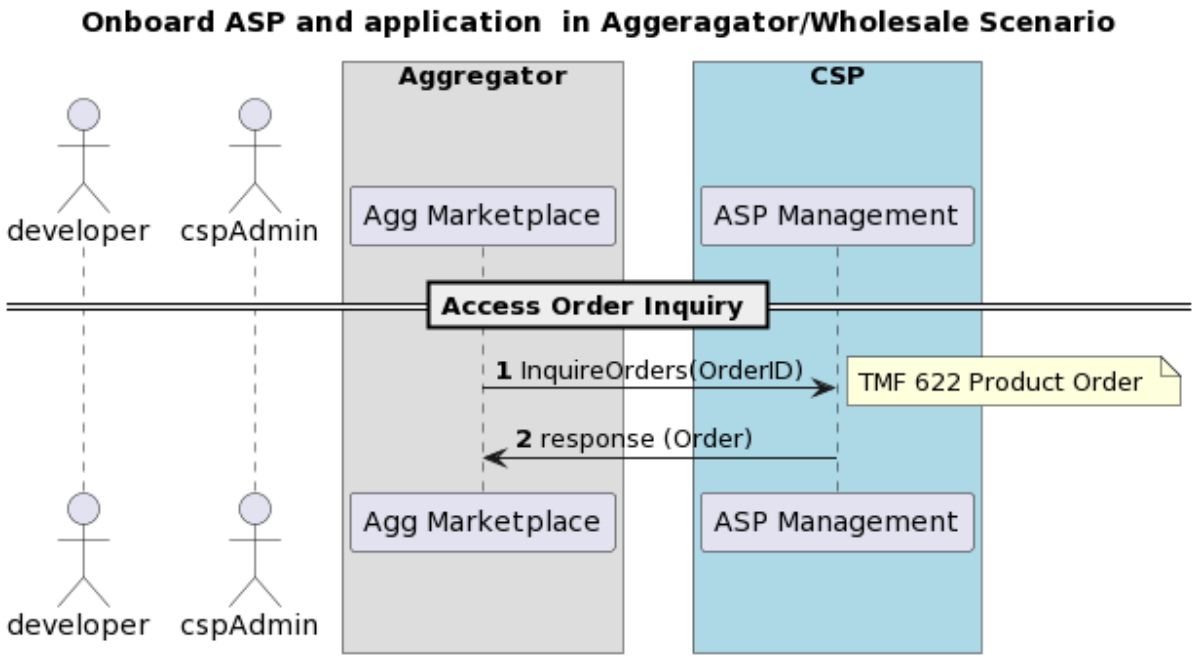
This operation implies that the ApplicationOwner has been previously registered on the Channel Partner and the Operator.

For the operation to succeed no associated applications should be in an active state. Deactivation of any active applications need to precede a successful deactivation action.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
2	Deactivate ApplicationOwner	PATCH /applicationOwner

UC11: Product Order Inquiry



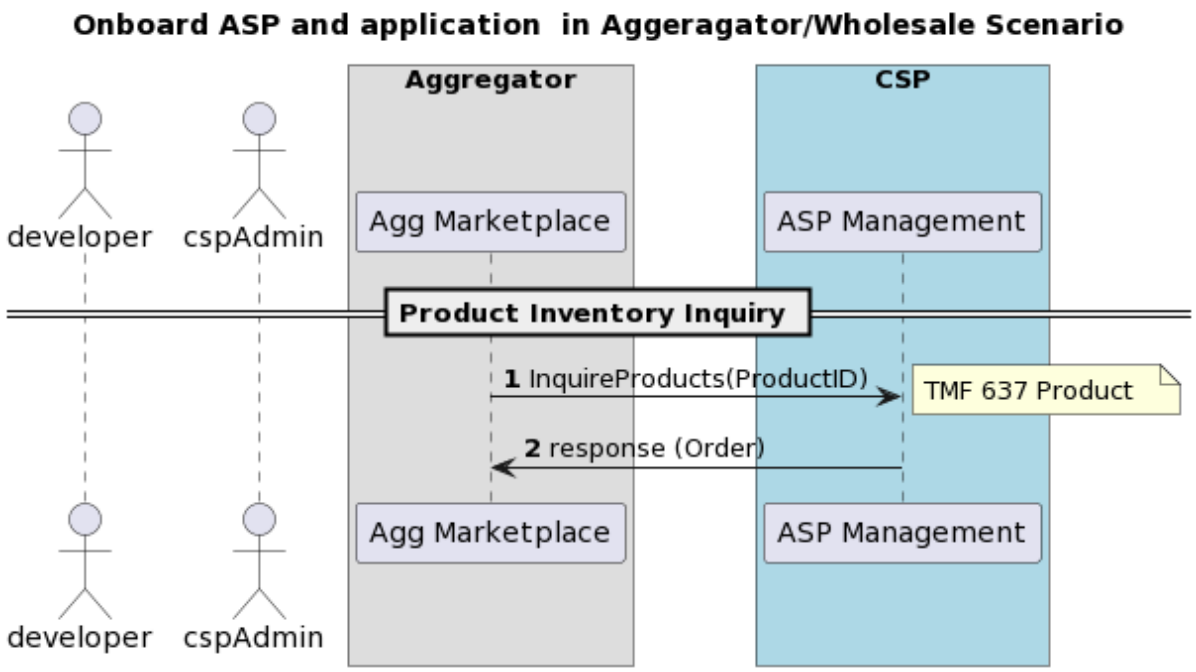
This use case involves the Channel Partner inquiring the CSP about the state of orders that were previously requested to be created and may be still on process by the CSP ; so including submitted, completed or rejected ones.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
2	Read submitted orders	GET /apiProductOrder

UC12: Product Inquiry

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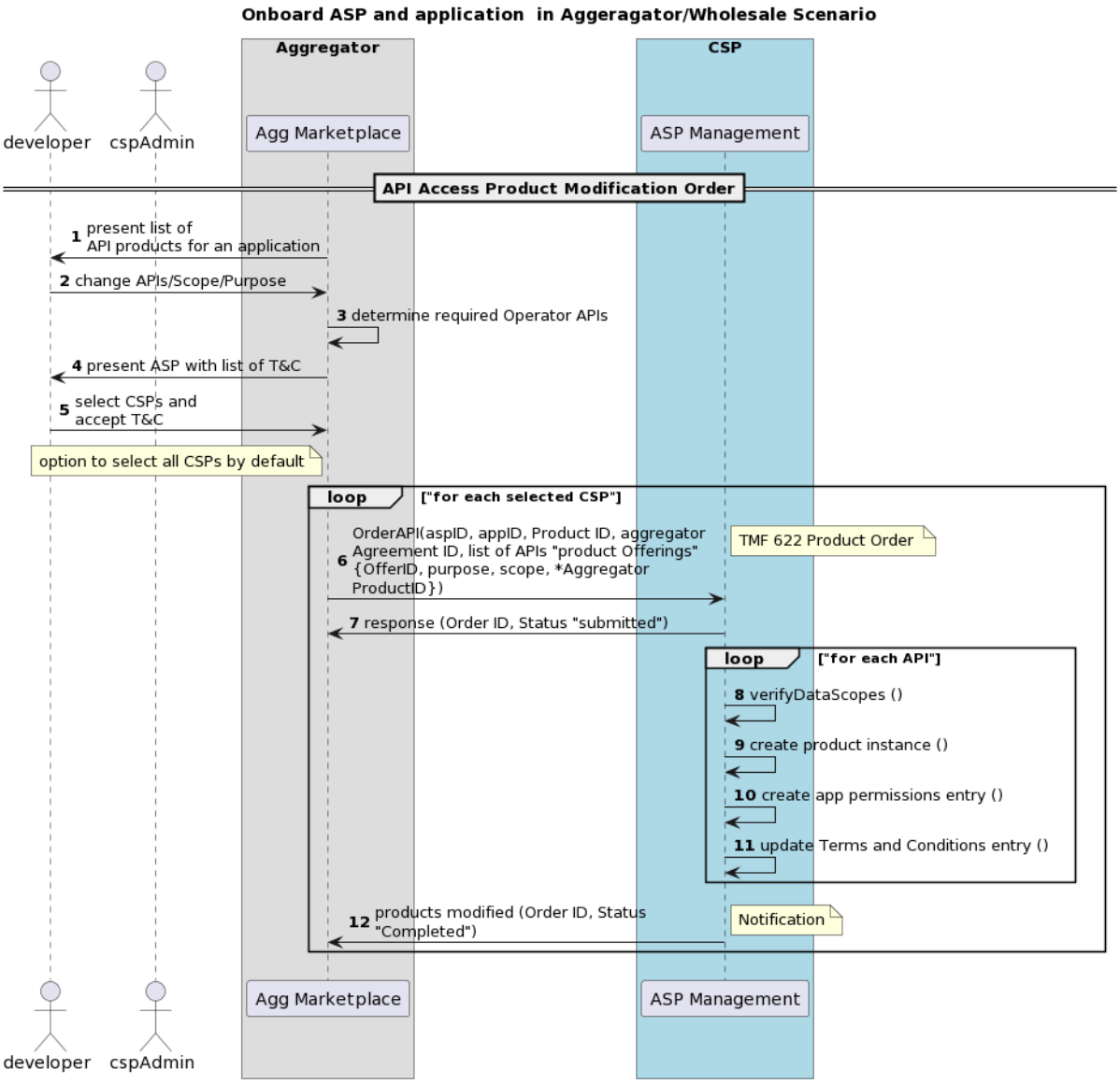
In this use case, the Channel Partner asks the Operator about the status of one of the Products that were previously ordered from one of the ApplicationOwner that the Channel Partner is serving.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
2	Read product inventory	GET /apiProduct

UC13: API Access Product Modification Ordering

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In this use case, the ApplicationOwner wants to modify some of the API/Products which is already using. The change could affect the purpose for the API, or any other configurable characteristic of the ProductSpecification. After receiving the request from the ApplicationOwner, the Channel Partner detects the affected Products and present the ApplicationOwner the Terms and Conditions that must be accepted in order to be able to use the modified Products.

Then, the Channel Partner proceeds to request the modifications to each of the affected Operators. Each Operator has to validate permissions and terms and conditions. Once validations are done, the Operator sends a notification to the Channel Partner and the modification is finished.

The Operate APIs to be used in this scenario are the following:

Step	Description	Operation used
6	Order API product for a change	POST /apiProductOrder

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Note: Many of the resource and payload examples include sample values for Attachment.content. These sample values are not syntactically correct, since any realistic value for content would be far too large and would obscure the readability of the examples.

Managed Entity and Task Resource Models

ApiProductOrder resource

An ApiProductOrder is a specialization of a ProductOrder, which can be used to place an order between a customer and a service provider or between a service provider and a partner and vice versa.

Resource model

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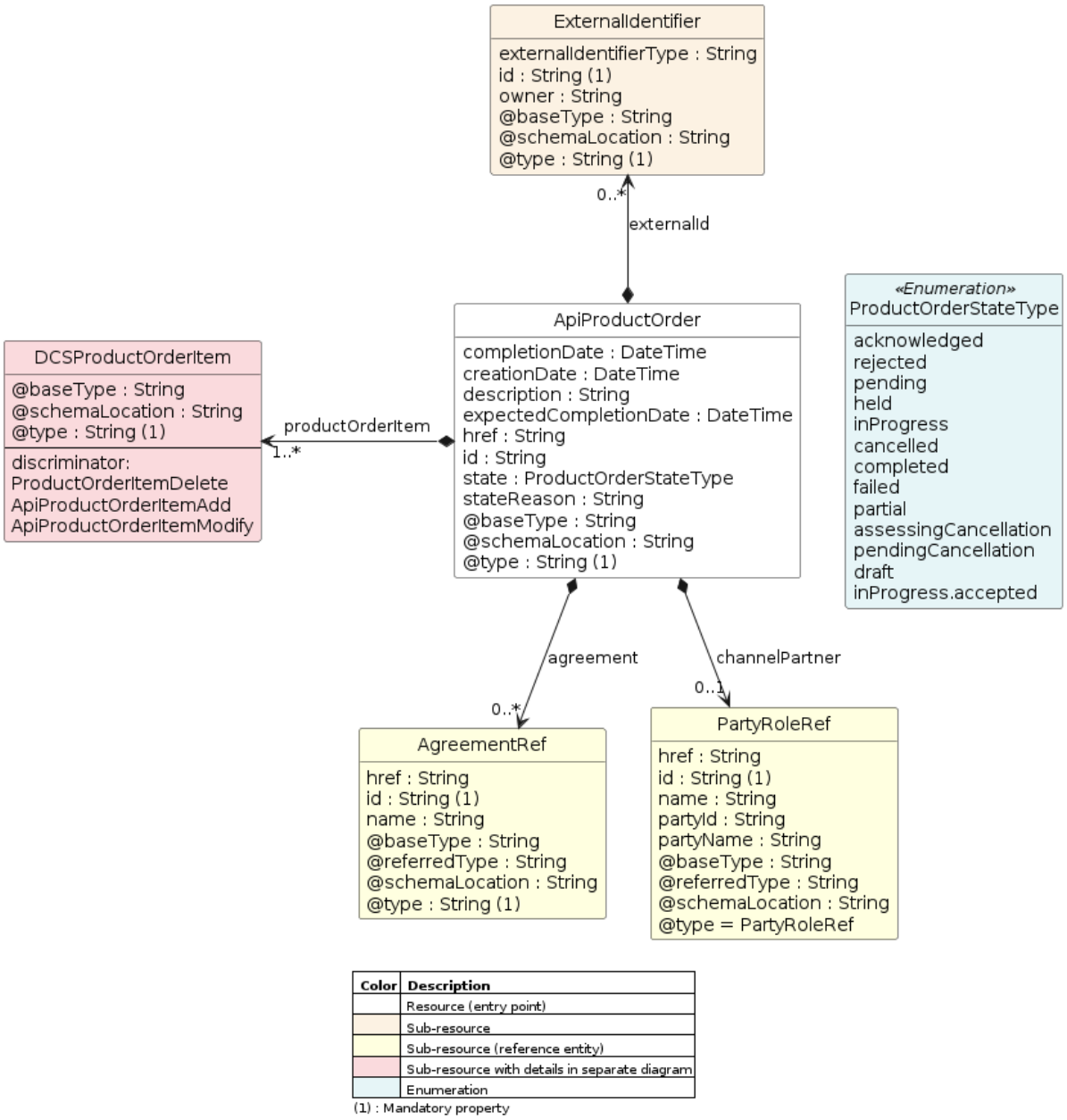


Figure 1 - ApiProductOrder

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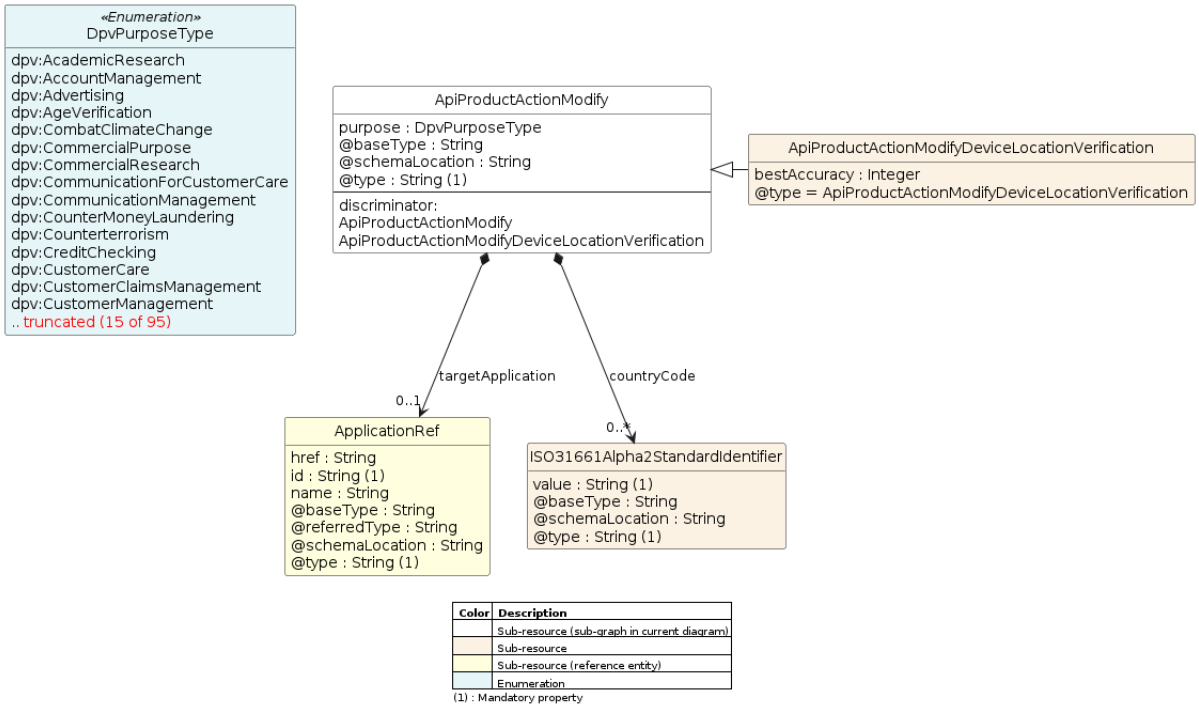


Figure 2 - ApiProductActionModify

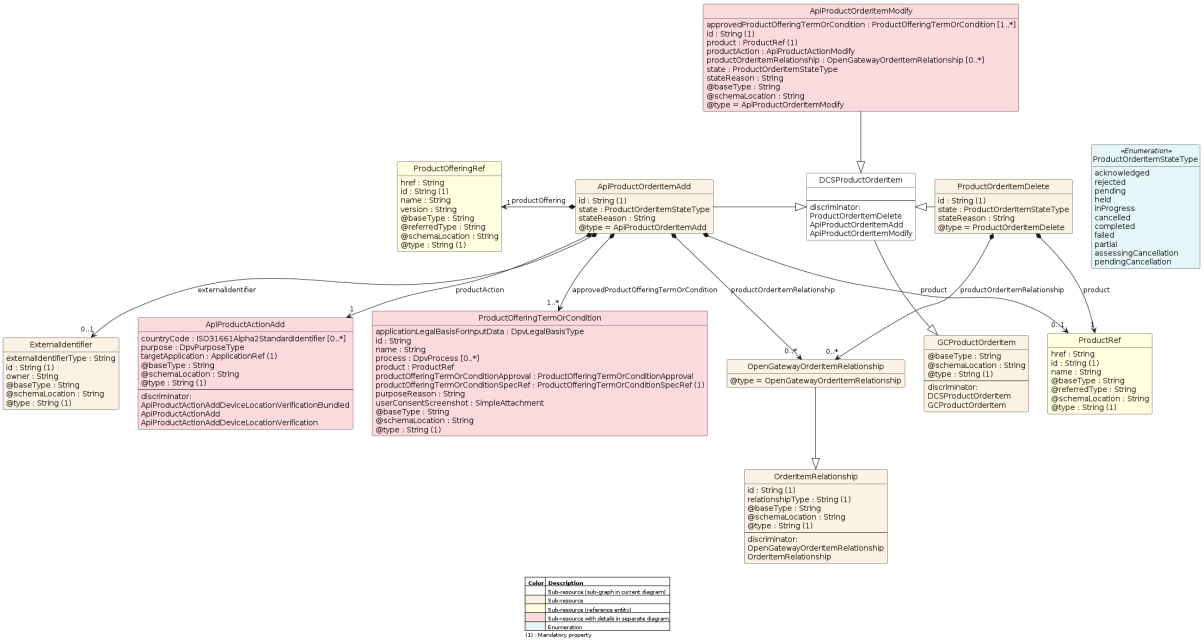


Figure 3 - DCSProductOrderItem

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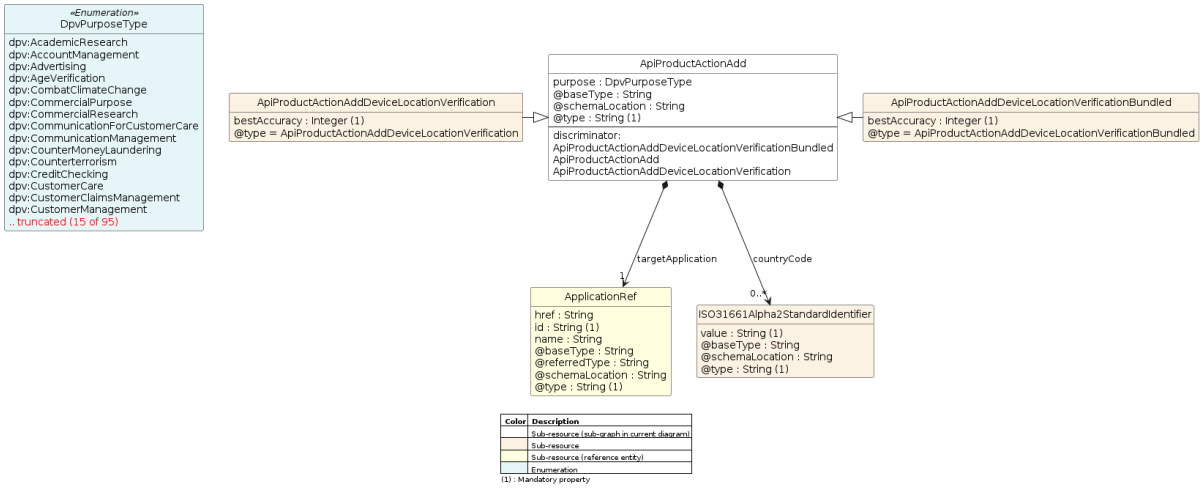


Figure 4 - ApiProductActionAdd

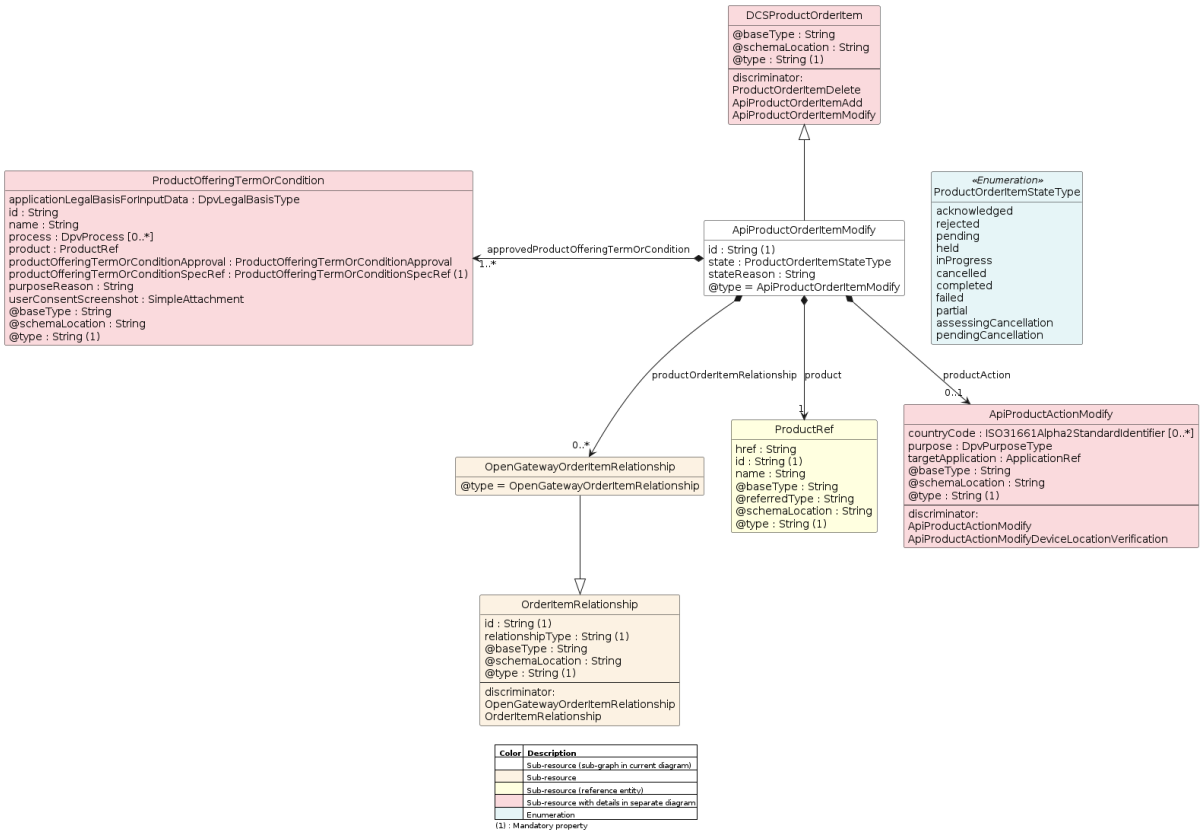


Figure 5 - ApiProductOrderItemModify

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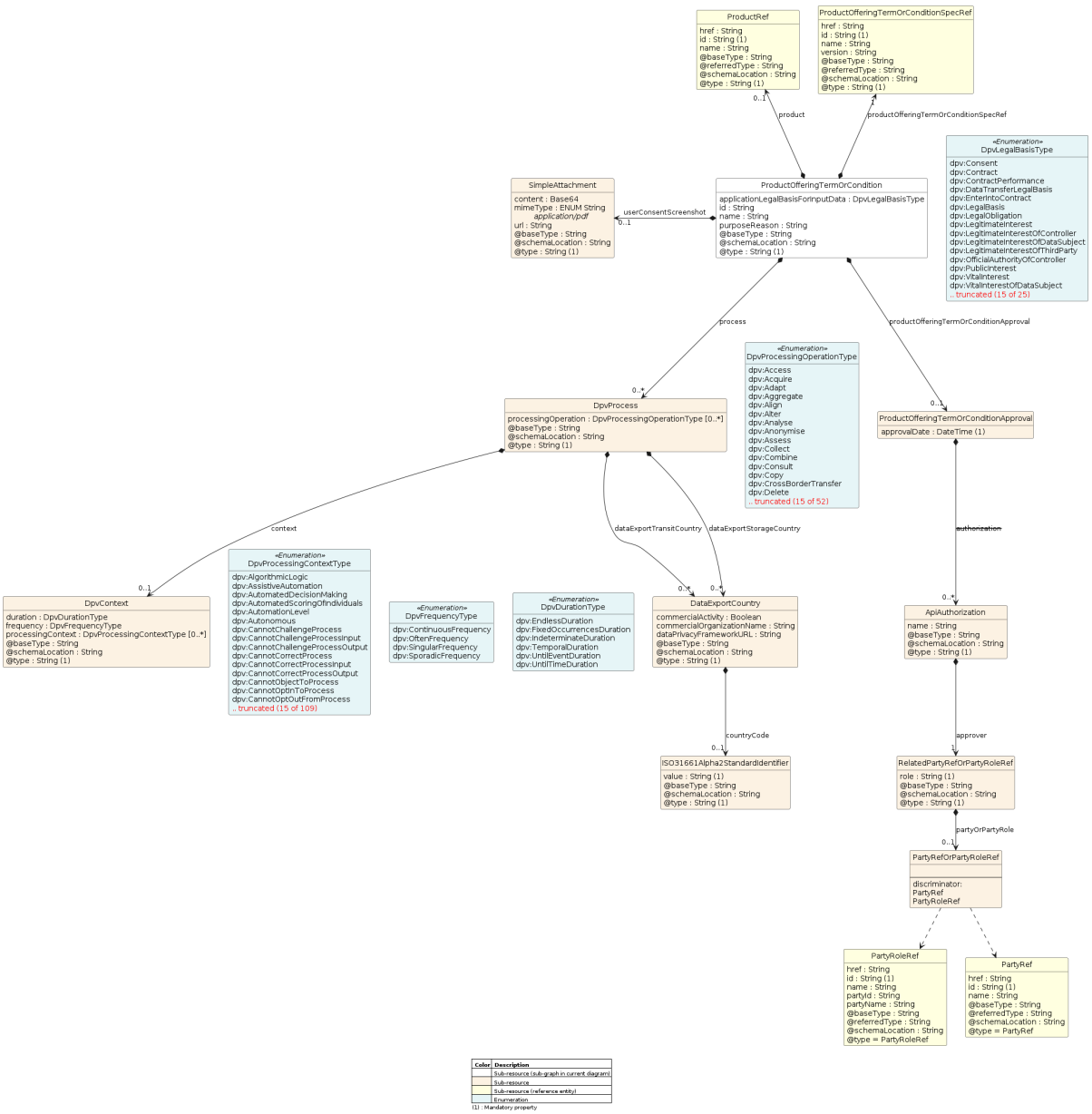


Figure 6 - ProductOfferingTermOrCondition

Field descriptions

ApiProductOrder fields

agreement	An AgreementRef. A reference to an agreement defined in the context of the product order.
channelPartner	A PartyRoleRef. Party role reference. A party role represents the part played by a party in a given context.
completionDate	A DateTime. Date when the ProductOrder was completed.
creationDate	A DateTime. Date and time when the ProductOrder was created.
description	A String. Description of the product order.
expectedCompletionDate	A DateTime. Expected delivery date amended by the provider.

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externalId	An ExternalIdentifier. Can be used to pass the internal identifier of the order in the Channel Partner system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOrderItem	A DCSProductOrderItem. An identified part of the order. A product order is decomposed into one or more order items. DCSProductOrderItem can be instantiated as * ApiProductOrderItemAdd * ApiProductOrderItemModify * ProductOrderItemDelete
state	A ProductOrderStateType. Possible values for the state of the order. ENUMERATED with values: * acknowledged * rejected * pending * held * inProgress * cancelled * completed * failed * partial * assessingCancellation * pendingCancellation * draft * inProgress.accepted
stateReason	A String. A string providing an explanation on the value of the state lifecycle. For instance if the state is rejected, stateReason will provide the reason for rejection.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AgreementRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DCSProductOrderItem sub-resource fields

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	DCSProductOrderItem can be instantiated as * ApiProductOrderItemAdd * ApiProductOrderItemModify * ProductOrderItemDelete

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approvedProductOfferingTermOrCondition	This property is present in subclasses
externalIdentifier	This property is present in subclasses
id	This property is present in subclasses
product	This property is present in subclasses
productAction	This property is present in subclasses
productOffering	This property is present in subclasses
productOrderItemRelationship	This property is present in subclasses
state	This property is present in subclasses
stateReason	This property is present in subclasses

ProductOfferingTermOrCondition sub-resource fields

applicationLegalBasisForInputData	A DpvLegalBasisType. Valid values for Legal Basis based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:Consent * dpv:Contract * dpv:ContractPerformance * dpv:DataTransferLegalBasis * dpv:EnterIntoContract * dpv:LegalBasis * dpv:LegalObligation * dpv:LegitimateInterest * dpv:LegitimateInterestOfController * dpv:LegitimateInterestOfDataSubject * dpv:LegitimateInterestOfThirdParty * dpv:OfficialAuthorityOfController * dpv:PublicInterest * dpv:VitalInterest * dpv:VitalInterestOfDataSubject * dpv:VitalInterestOfNaturalPerson * dpv:InformedConsent * dpv:DataControllerContract * dpv:ImpliedConsent * dpv:DataSubjectContract * dpv:ExplicitlyExpressedConsent * dpv:UninformedConsent * dpv:DataProcessorContract * dpv:ExpressedConsent * dpv:ThirdPartyContract
id	A String. Identifier of the ProductOfferingTermOrCondition.
name	A String. Identifier of the ProductOfferingTermOrCondition.
process	A DpvProcess. Lists all the operations or 'processing' performed on data by the data consumer.
product	A ProductRef. A Product reference.

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productOfferingTermOrConditionApproval	A ProductOfferingTermOrConditionApproval. Approval entity for the ProductOfferingTermOrCondition. Exists when Terms or Conditions are approved.
productOfferingTermOrConditionSpecRef	A ProductOfferingTermOrConditionSpecRef. Reference to the ProductOfferingTermOrCondition specification.
purposeReason	A String. Explain the reason justifying the purpose, indicating specifically which is the reason why the API is employed as part of the application's use case.
userConsentScreenshot	A SimpleAttachment. Complements the description of an element (for instance a product) through video, pictures…
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DpvProcess sub-resource fields

context	A DpvContext. Describes all the context and conditions within which processing takes place.
dataExportStorageCountry	A DataExportCountry. Field to contain all the information related the last data storage site.
dataExportTransitCountry	A DataExportCountry. Field to contain all the traceability of the data until the last storage site.

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processingOperation	A DpvProcessingOperationType. Valid values for the data processing operation based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:Access * dpv:Acquire * dpv:Adapt * dpv:Aggregate * dpv:Align * dpv:Alter * dpv:Analyse * dpv:Anonymise * dpv:Assess * dpv:Collect * dpv:Combine * dpv:Consult * dpv:Copy * dpv:CrossBorderTransfer * dpv:Delete * dpv:Derive * dpv:Destruct * dpv:Disclose * dpv:DiscloseByTransmission * dpv:Display * dpv:Disseminate * dpv:Download * dpv:Erase * dpv:Export * dpv:Filter * dpv:Format * dpv:Generate * dpv:Infer * dpv:MakeAvailable * dpv:Match * dpv:Modify * dpv:Monitor * dpv:Move * dpv:Observe * dpv:Obtain * dpv:Organise * dpv:Profiling * dpv:Pseudonymise * dpv:Query * dpv:Record * dpv:Reformat * dpv:Remove * dpv:Restrict * dpv:Retrieve * dpv:Screen * dpv:Share * dpv:Store * dpv:Structure * dpv:Transfer * dpv:Transform * .. truncated .. 50 of 52
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DpvContext sub-resource fields

duration	A DpvDurationType. Valid values for the data processing duration based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:EndlessDuration * dpv:FixedOccurrencesDuration * dpv:IndeterminateDuration * dpv:TemporalDuration * dpv:UntilEventDuration * dpv:UntilTimeDuration
frequency	A DpvFrequencyType. Valid values for the data processing frequency based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:ContinuousFrequency * dpv:OftenFrequency * dpv:SingularFrequency * dpv:SporadicFrequency

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processingContext	A DpvProcessingContextType. Valid values for the data processing context classes based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AlgorithmicLogic * dpv:AssistiveAutomation * dpv:AutomatedDecisionMaking * dpv:AutomatedScoringOfIndividuals * dpv:AutomationLevel * dpv:Autonomous * dpv:CannotChallengeProcess * dpv:CannotChallengeProcessInput * dpv:CannotChallengeProcessOutput * dpv:CannotCorrectProcess * dpv:CannotCorrectProcessInput * dpv:CannotCorrectProcessOutput * dpv:CannotObjectToProcess * dpv:CannotOptInToProcess * dpv:CannotOptOutFromProcess * dpv:CannotReverseProcessEffects * dpv:CannotReverseProcessInput * dpv:CannotReverseProcessOutput * dpv:CannotWithdrawFromProcess * dpv:ChallengingProcess * dpv:ChallengingProcessInput * dpv:ChallengingProcessOutput * dpv:ConditionalAutomation * dpv:ConsentControl * dpv:CorrectingProcess * dpv:CorrectingProcessInput * dpv:CorrectingProcessOutput * dpv:CompletelyManualProcessing * dpv:DataControllerDataSource * dpv:DataPublishedByDataSubject * dpv:DataSource * dpv:DataSubject * dpv:DataSubjectDataSource * dpv:DataSubjectScale * dpv:DataVolume * dpv:DecisionMaking * dpv:EntityActiveInvolvement * dpv:EntityInvolvement * dpv:EntityNonInvolvement * dpv:EntityNonPermissiveInvolvement * dpv:EntityPassiveInvolvement * dpv:EntityPermissiveInvolvement * dpv:EvaluationOfIndividuals * dpv:EvaluationScoring * dpv:FullAutomation * dpv:GeographicCoverage * dpv:GlobalScale * dpv:HighAutomation * dpv:HugeDataVolume * dpv:HugeScaleOfDataSubjects * .. truncated .. 50 of 109

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DataExportCountry sub-resource fields

commercialActivity	A Boolean. Flag to indicate if commercial activity applies to these data management.
commercialOrganizationName	A String. Organization who is managing the data.
countryCode	An ISO31661Alpha2StandardIdentifier. The corresponding identification of the resource in different standard, regulatory definitions. The standard specification identifier (e.g., ISO 3166-1 Alpha-2) and the corresponding value (e.g., BE) relevant to a particular resource.
dataPrivacyFrameworkURL	A String. URL to the organization or governmental data privacy framework if applies.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ISO31661Alpha2StandardIdentifier sub-resource fields

value	A String. The value of the resource in the ISO 3166-1 Alpha 2 standard.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DataExportCountry sub-resource fields

See [DataExportCountry] for details.

ProductRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingTermOrConditionApproval sub-resource fields

approvalDate	A DateTime. Date of the approval.
~~authorization~~	DEPRECATED An ApiAuthorization. DEPRECATED not possible yet to have the ID of the individual that made the approval.

ApiAuthorization sub-resource fields

approver	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
name	A String. Name of the required authorization.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingTermOrConditionSpecRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the ProductOfferingTermOrCondition specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

SimpleAttachment sub-resource fields

content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types. ENUMERATED with values: * application/pdf
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingTermOrCondition sub-resource fields

See [ProductOfferingTermOrCondition] for details.

ProductRef sub-resource fields

See [ProductRef] for details.

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ProductRef sub-resource fields

See [ProductRef] for details.

ProductRef sub-resource fields

See [ProductRef] for details.

ApiProductActionModify sub-resource fields

countryCode	An ISO31661Alpha2StandardIdentifier. One or several countries where is operated. Must be chosen from possible values described in the API ProductSpecification.

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95
targetApplication	An ApplicationRef. Reference to an Application.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ApiProductActionModify can be instantiated as * ApiProductActionModifyDeviceLocationVerification
bestAccuracy	This property is present in subclasses

ApplicationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiProductActionModifyDeviceLocationVerification sub-resource fields

bestAccuracy	An Integer.
countryCode	An ISO31661Alpha2StandardIdentifier. One or several countries where is operated. Must be chosen from possible values described in the API ProductSpecification.

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95
targetApplication	An ApplicationRef. Reference to an Application.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiProductActionAdd sub-resource fields

countryCode	An ISO31661Alpha2StandardIdentifier. One or several countries where API is operated. Must be chosen from possible values described in the API ProductSpecification.

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95
targetApplication	An ApplicationRef. Reference to an Application.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ApiProductActionAdd can be instantiated as * ApiProductActionAddDeviceLocationVerification * ApiProductActionAddDeviceLocationVerificationBundled
bestAccuracy	This property is present in subclasses

ApiProductActionAddDeviceLocationVerification sub-resource fields

bestAccuracy	An Integer.
countryCode	An ISO31661Alpha2StandardIdentifier. One or several countries where API is operated. Must be chosen from possible values described in the API ProductSpecification.

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95
targetApplication	An ApplicationRef. Reference to an Application.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiProductActionAddDeviceLocationVerificationBundled sub-resource fields

bestAccuracy	An Integer.
countryCode	An ISO31661Alpha2StandardIdentifier. One or several countries where API is operated. Must be chosen from possible values described in the API ProductSpecification.

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95
targetApplication	An ApplicationRef. Reference to an Application.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the product offering.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OpenGatewayOrderItemRelationship sub-resource fields

id	A String. Id of the related Order item (must be in the same Order).
relationshipType	A String. Relationship type in the Open Gateway context. bundles is used when the id in the OrderItemRelationship is the id of a ProductOffering bundled into the productOffering referred in the orderItem. requires is used when the id in the OrderItemRelationship is the id of a ProductOffering required by the productOffering referred in the orderItem. ENUMERATED with values: * bundles * requires
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OrderItemRelationship sub-resource fields

id	A String. Id of the related Order item (must be in the same Order).
relationshipType	A String. Relationship type as relies on, bundles, etc…
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	OrderItemRelationship can be instantiated as * OpenGatewayOrderItemRelationship

OpenGatewayOrderItemRelationship sub-resource fields

See [OpenGatewayOrderItemRelationship] for details.

OpenGatewayOrderItemRelationship sub-resource fields

See [OpenGatewayOrderItemRelationship] for details.

ApiProductOrderItemAdd sub-resource fields

approvedProductOfferingTermOrCondition	A ProductOfferingTermOrCondition. List of approved ProductOfferingTermOrCondition by the ApplicationOwner.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
id	A String. Identifier of the ProductOrder item (generally it is a sequence number 01, 02, 03, …).
product	A ProductRef. A Product reference.
productAction	An ApiProductActionAdd. An ApiProductActionAdd is a specialization of a ProductAction, for GSMA Open GW context for action to add a product. ApiProductActionAdd can be instantiated as * ApiProductActionAddDeviceLocationVerification * ApiProductActionAddDeviceLocationVerificationBundled
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
productOrderItemRelationship	An OpenGatewayOrderItemRelationship. Collection of order items relationships in the Open Gateway context.

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state	A ProductOrderItemStateType. Possible values for the state of the product order item. ENUMERATED with values: * acknowledged * rejected * pending * held * inProgress * cancelled * completed * failed * partial * assessingCancellation * pendingCancellation
stateReason	A String. A string providing an explanation on the value of the state lifecycle. For instance if the state is rejected, stateReason will provide the reason for rejection.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiProductOrderItemModify sub-resource fields

approvedProductOfferingTermOrCondition	A ProductOfferingTermOrCondition. List of approved ProductOfferingTermOrCondition by the ApplicationOwner.
id	A String. Identifier of the ProductOrder item (generally it is a sequence number 01, 02, 03, …).
product	A ProductRef. A Product reference.
productAction	An ApiProductActionModify. An ApiProductActionModify is a specialization of a ProductAction, for GSMA Open GW context for action to modify a product. The values in the schema shall override the values in the Product; it means if the purpose values were [A,B] then if in the modify productAction the purpose values are [A,C] it means the purpose B will be removed and purpose C will be added. ApiProductActionModify can be instantiated as * ApiProductActionModifyDeviceLocationVerification
productOrderItemRelationship	An OpenGatewayOrderItemRelationship. Collection of order items relationships in the Open Gateway context.

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state	A ProductOrderItemStateType. Possible values for the state of the product order item. ENUMERATED with values: * acknowledged * rejected * pending * held * inProgress * cancelled * completed * failed * partial * assessingCancellation * pendingCancellation
stateReason	A String. A string providing an explanation on the value of the state lifecycle. For instance if the state is rejected, stateReason will provide the reason for rejection.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GCProductOrderItem sub-resource fields

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	GCProductOrderItem can be instantiated as * DCSProductOrderItem

ProductOrderItemDelete sub-resource fields

id	A String. Identifier of the ProductOrder item (generally it is a sequence number 01, 02, 03, …).
product	A ProductRef. A Product reference.
productOrderItemRelationship	An OpenGatewayOrderItemRelationship. Collection of order items relationships in the Open Gateway context.

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state	A ProductOrderItemStateType. Possible values for the state of the product order item. ENUMERATED with values: * acknowledged * rejected * pending * held * inProgress * cancelled * completed * failed * partial * assessingCancellation * pendingCancellation
stateReason	A String. A string providing an explanation on the value of the state lifecycle. For instance if the state is rejected, stateReason will provide the reason for rejection.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ApiProductOrder' resource object.


```
{
  "@type": "ApiProductOrder",
  "id": "86387671-0997-461b-99c7-d624ffb0640c",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
  "state": "completed",
  "creationDate": "2025-02-02T12:00:00.00Z",
  "completionDate": "2025-02-02T12:01:00.00Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_2232342",
      "owner": "ChannelPartner"
    }
  ],
  "description": "Camara API order for an Application Owner",
  "productOrderItem": [
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "id": "1",
      "state": "completed",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
        "name": "Location Verification Antifraud CAMARA API Standalone",
```


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```
        "@referredType": "OpenGatewayProductOffering"
      },
      "product": {
        "@type": "ProductRef",
        "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
        "@referredType": "ApiProduct"
      },
      "productAction": {
        "@type": "ApiProductActionAddDeviceLocationVerification",
        "@baseType": "ApiProductAction",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerification.schema.json",
        "purpose": "dpv:FraudPreventionAndDetection",
        "bestAccuracy": 500,
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "product_1234",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-10-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
```


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```
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
      ]
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
```


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```
            "@referredType": "PartyRole"
    }
}
```

ApiProduct resource

ApiProduct is a specialization of a Product being an instance of an atomic ProductOffering or a ProductSpecification that is realized by an API (the Service).

Resource model

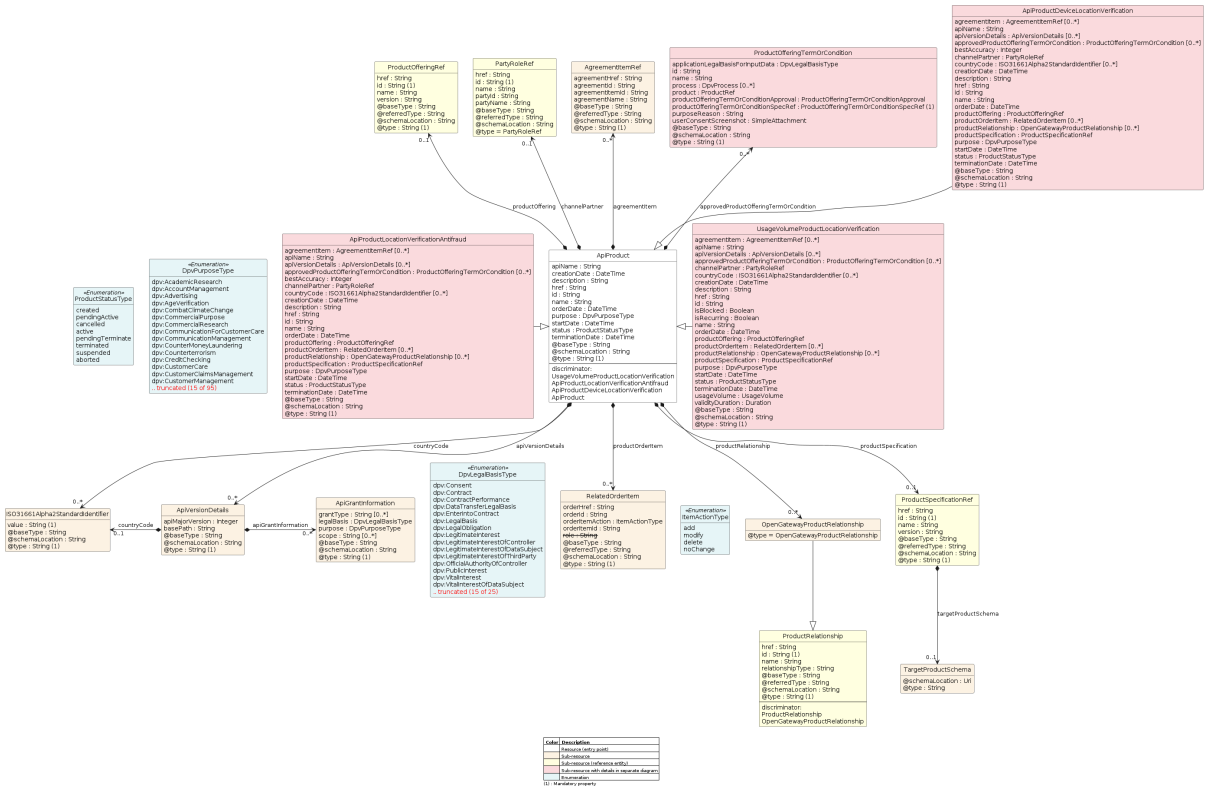


Figure 7 - ApiProduct

See Figure 6 for ProductOfferingTermOrCondition diagrams.

Field descriptions

ApiProduct fields

agreementItem	An AgreementItemRef. Collection of agreement items that involves this Product.
apiName	A String. Standardized name of the API.
apiVersionDetails	An ApiVersionDetails. A set of ApiProduct characteristics grouped together by API major version.
approvedProductOfferingTermOrCondition	A ProductOfferingTermOrCondition. Approved Terms or Conditions of the ProductOffering.

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channelPartner	A PartyRoleRef. Party role reference. A party role represents the part played by a party in a given context.
countryCode	An ISO31661Alpha2StandardIdentifier. One or several countries where API is operated.
creationDate	A DateTime. Date and time when the product was created.
description	A String. Is the description of the product. It could be same description than Product Offering one.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the product. It could be the same as the name of the product offering.
orderDate	A DateTime. Is the date when the product was ordered.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
productOrderItem	A RelatedOrderItem. A reference to the Product Order items that involved this Product.
productRelationship	An OpenGatewayProductRelationship. Collection of product relationships.
productSpecification	A ProductSpecificationRef. ProductSpecification reference. A product Specification represents entities that are orderable from the provider of the catalog.

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95
startDate	A DateTime. Is the date from which the product starts.

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status	A ProductStatusType. Possible values for the status of the product. ENUMERATED with values: * created * pendingActive * cancelled * active * pendingTerminate * terminated * suspended * aborted
terminationDate	A DateTime. Is the date when the product was terminated.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ApiProduct can be instantiated as * ApiProductDeviceLocationVerification * ApiProductLocationVerificationAntifraud * UsageVolumeProductLocationVerification
bestAccuracy	This property is present in subclasses
isBlocked	This property is present in subclasses
isRecurring	This property is present in subclasses
usageVolume	This property is present in subclasses
validityDuration	This property is present in subclasses

AgreementItemRef sub-resource fields

agreementHref	A String. Reference of the related entity.
agreementId	A String. Unique identifier of a related entity.
agreementItemId	A String. Id of an item of a Agreement.
agreementName	A String. Name of the related entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiVersionDetails sub-resource fields

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apiGrantInformation	An ApiGrantInformation. Collection of API grant information that describe the Product. ApiGrantInformation is a structured set of information regarding the grant of access to an API product, detailing its purpose, scope, grant type, and legal basis for usage.
apiMajorVersion	An Integer. Major version of the API standard specification.
basePath	A String. Base path of the API.
countryCode	An ISO31661Alpha2StandardIdentifier. The corresponding identification of the resource in different standard, regulatory definitions. The standard specification identifier (e.g., ISO 3166-1 Alpha-2) and the corresponding value (e.g., BE) relevant to a particular resource.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiGrantInformation sub-resource fields

grantType	A String. One or several grantType of the API, as stated in the API that has been purchased. Examples: "authorization_code" "client_credentials" "urn:ietf:params:oauth:grant-type:token-exchange" "urn:ietf:params:oauth:grant-type:jwt-bearer" "urn:ietf:params:oauth:grant-type:saml2-bearer" "urn:ietf:params:oauth:grant-type:device_code" "urn:openid:params:grant-type:ciba".

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legalBasis	A DpvLegalBasisType. Valid values for Legal Basis based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:Consent * dpv:Contract * dpv:ContractPerformance * dpv:DataTransferLegalBasis * dpv:EnterIntoContract * dpv:LegalBasis * dpv:LegalObligation * dpv:LegitimateInterest * dpv:LegitimateInterestOfController * dpv:LegitimateInterestOfDataSubject * dpv:LegitimateInterestOfThirdParty * dpv:OfficialAuthorityOfController * dpv:PublicInterest * dpv:VitalInterest * dpv:VitalInterestOfDataSubject * dpv:VitalInterestOfNaturalPerson * dpv:InformedConsent * dpv:DataControllerContract * dpv:ImpliedConsent * dpv:DataSubjectContract * dpv:ExplicitlyExpressedConsent * dpv:UninformedConsent * dpv:DataProcessorContract * dpv:ExpressedConsent * dpv:ThirdPartyContract

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95

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scope	A String. One or several scopes of the API, as stated in the API that has been purchased. Examples: "sim-swap:retrieve-date".
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ISO31661Alpha2StandardIdentifier sub-resource fields

See [ISO31661Alpha2StandardIdentifier] for details.

ProductOfferingTermOrCondition sub-resource fields

See [ProductOfferingTermOrCondition] for details.

DpvProcess sub-resource fields

See [DpvProcess] for details.

DpvContext sub-resource fields

See [DpvContext] for details.

DataExportCountry sub-resource fields

See [DataExportCountry] for details.

DataExportCountry sub-resource fields

See [DataExportCountry] for details.

ProductRef sub-resource fields

See [ProductRef] for details.

ProductOfferingTermOrConditionApproval sub-resource fields

See [ProductOfferingTermOrConditionApproval] for details.

ApiAuthorization sub-resource fields

See [ApiAuthorization] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

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PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ProductOfferingTermOrConditionSpecRef sub-resource fields

See [ProductOfferingTermOrConditionSpecRef] for details.

SimpleAttachment sub-resource fields

See [SimpleAttachment] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ISO31661Alpha2StandardIdentifier sub-resource fields

See [ISO31661Alpha2StandardIdentifier] for details.

ProductOfferingRef sub-resource fields

See [ProductOfferingRef] for details.

RelatedOrderItem sub-resource fields

orderHref	A String. Reference of the related entity.
orderId	A String. Unique identifier of a related Order.
orderItemAction	An ItemActionType. Action to be performed on the entity managed by the item. ENUMERATED with values: * add * modify * delete * noChange
orderItemId	A String. Id of an item of a product order.
~~role~~	DEPRECATED A String. Role played by the order.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OpenGatewayProductRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.

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name	A String. Name of the referred entity.
relationshipType	A String. Relationship in the Open Gateway context. bundles is used when the id in the productRelationship is the id of a product bundled into the main product. appliesOn is used when the id in the productRelationship is the id of a product to which the main product applies on (like a usage volume product applies on an API product). ENUMERATED with values: * bundles * appliesOn
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Relationship type as relies on, bundles, etc…
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ProductRelationship can be instantiated as * OpenGatewayProductRelationship

ProductSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
targetProductSchema	A TargetProductSchema. The reference object to the schema and type of target product which is described by product specification.
version	A String. Version of the product specification.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TargetProductSchema sub-resource fields

@schemaLocation	A Uri. This field provides a link to the schema describing the target product.
@type	A String. Class type of the target product.

UsageVolume sub-resource fields

volumeInitialAmount	An Integer. Initial amount of volumeUnit when corresponding Service Balance is created.
volumeUnit	A String. Unit of the volume. If the schema is specialized with an enum, the values must be taken from this list : apiCall, apiNotification.

Json representation sample(s)

We provide below a JSON representation as example of the 'ApiProduct' resource object.


```
{
  "@type": "ApiProduct",
  "id": "456",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/456",
  "name": "SIM Swap  CAMARA API Standalone",
  "creationDate": "2025-03-02T12:02:00.00Z",
  "startDate": "2025-03-02T12:00:03.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
    "id": "32",
    "name": "SIM Swap  CAMARA API Standalone",
    "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
    {
      "@type": "RelatedOrderItem",
      "orderId": "098",
      "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/098",
      "orderItemId": "1",
      "orderItemAction": "add"
    }
  ],
  "productSpecification": {
    "@type": "ProductSpecificationRef",
    "id": "684",
    "href": "https://mycsp.com/tmf-
```


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```
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/684",
    "name": "SIM Swap CAMARA API",
    "version": "1.0.0",
    "@referredType": "ApiProductSpecification"
  },
  "apiName": "sim-swap",
  "apiVersionDetails": [
    {
      "@type": "ApiVersionDetails",
      "apiMajorVersion": 1,
      "basePath": "sim-swap",
      "apiGrantInformation": [
        {
          "@type": "ApiGrantInformation",
          "purpose": "dpv:FraudPreventionAndDetection",
          "scope": [
            "sim-swap:retrieve-date"
          ],
          "grantType": [
            "urn:openid:params:grant-type:ciba",
            "authorization_code"
          ],
          "legalBasis": "dpv:Consent"
        }
      ]
    }
  ],
  "purpose": "dpv:FraudPreventionAndDetection",
  "approvedProductOfferingTermOrCondition": [
    {
      "@type": "ProductOfferingTermOrCondition",
      "productOfferingTermOrConditionSpecRef": {
        "@type": "ProductOfferingTermOrConditionSpecRef",
        "id": "9876",
        "@referredType": "OpenGatewayProductOfferingTermOrConditionSpecification"

      },
      "productOfferingTermOrConditionApproval": {
        "approvalDate": "2023-11-20T00:00:00.000Z"
      },
      "applicationLegalBasisForInputData": "dpv:Consent",
      "process": [
        {
          "@type": "DpvProcess",
          "processingOperation": [
            "dpv:Assess"
          ],
          "context": {
            "@type": "DpvContext",
            "processingContext": [
              "dpv:AutomatedDecisionMaking",
              "dpv:AutomatedScoringOfIndividuals"
            ],
            "duration": "dpv:FixedOccurrencesDuration",
            "frequency": "dpv:SingularFrequency"
          },
          "dataExportTransitCountry": [
            {
              "@type": "DataExportCountry",
              "countryCode": {
                "@type": "ISO31661Alpha2StandardIdentifier",
                "value": "US"
              },
```


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```
              "commercialOrganizationName": "Globy Corporation",
              "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
            }
          ]
        },
        {
          "@type": "DpvProcess",
          "processingOperation": [
            "dpv:Profiling"
          ],
          "context": {
            "@type": "DpvContext",
            "processingContext": [
              "dpv:AutomatedScoringOfIndividuals"
            ],
            "duration": "dpv:EndlessDuration",
            "frequency": "dpv:SingularFrequency"
          },
          "dataExportTransitCountry": [
            {
              "@type": "DataExportCountry",
              "countryCode": {
                "@type": "ISO31661Alpha2StandardIdentifier",
                "value": "US"
              },
              "commercialOrganizationName": "Globy Corporation",
              "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
            }
          ],
          "dataExportStorageCountry": [
            {
              "@type": "DataExportCountry",
              "countryCode": {
                "@type": "ISO31661Alpha2StandardIdentifier",
                "value": "US"
              },
              "commercialOrganizationName": "Globy Corporation",
              "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
            }
          ]
        }
      ],
      "userConsentScreenshot": {
        "@type": "SimpleAttachment",
        "content": "JVBERi...JSVFT0Y=",
        "mimeType": "application/pdf"
      }
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```


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Application resource

Application is a logical resource that can consume API products. It's a specialization of a LogicalResource in the Open Gateway context.

Resource model

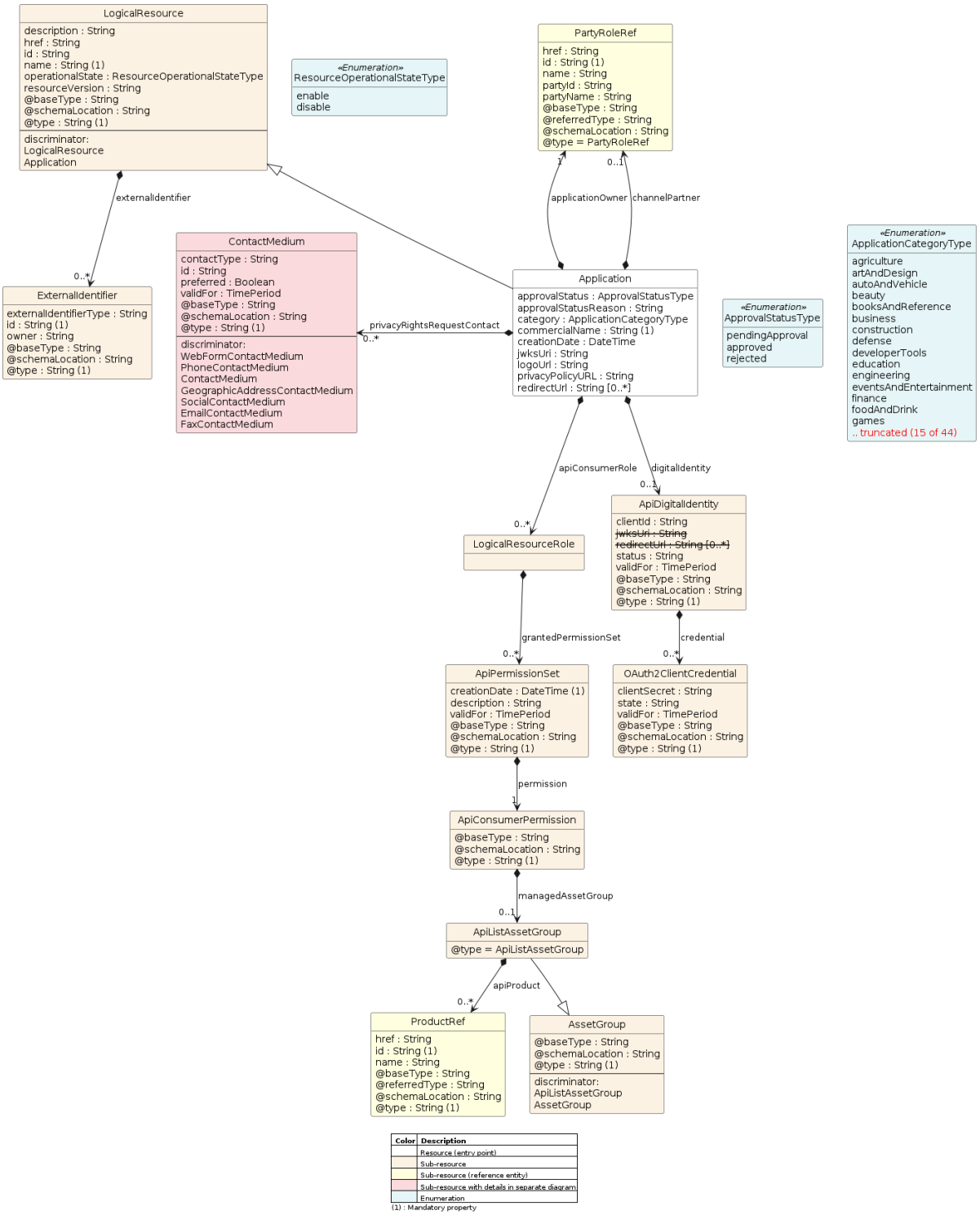


Figure 8 - Application

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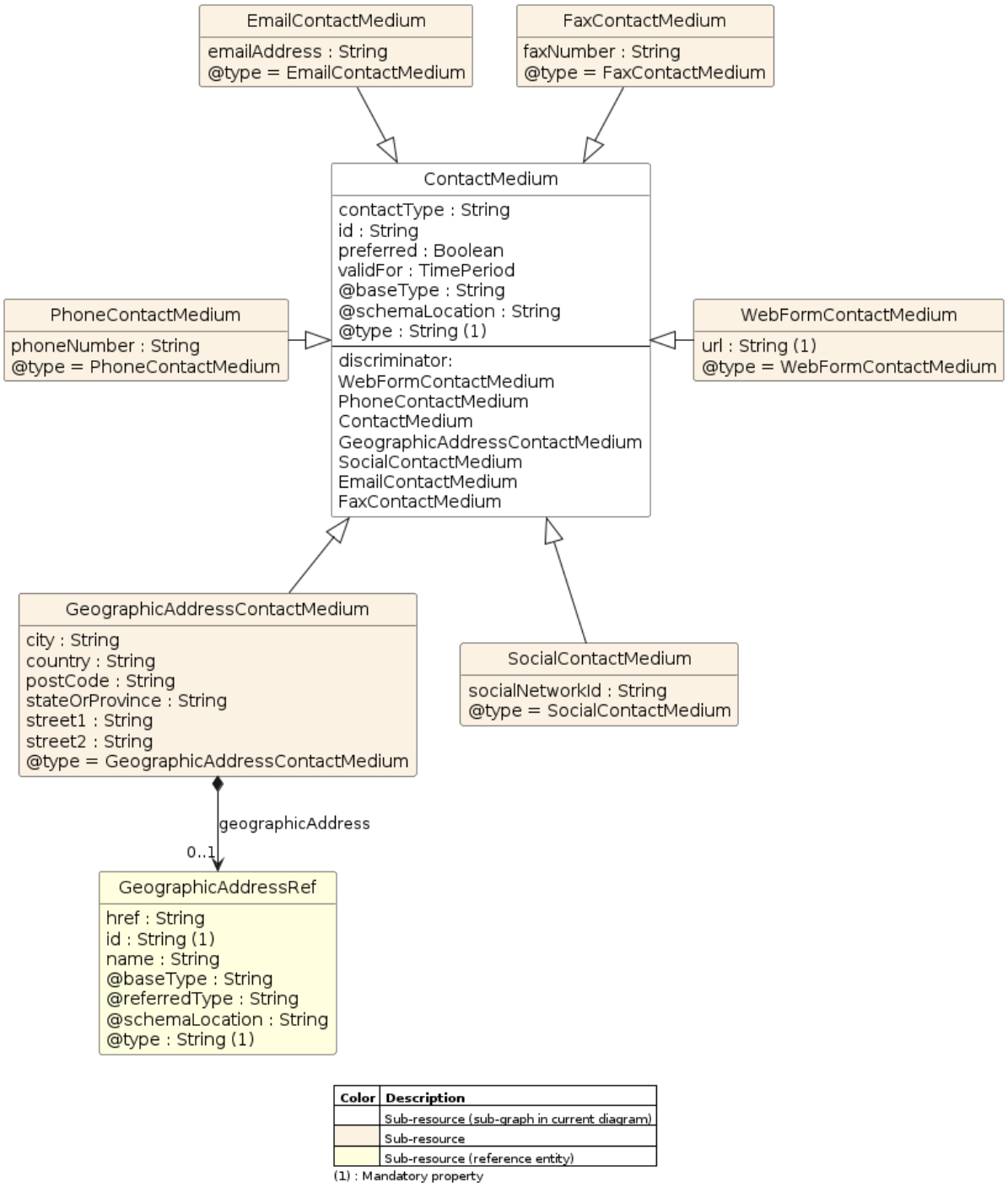


Figure 9 - ContactMedium

Field descriptions

Application fields

apiConsumerRole	A LogicalResourceRole. Each GSMA Service-API plays a role, and has specific permissions for use.
applicationOwner	A PartyRoleRef. Party role reference. A party role represents the part played by a party in a given context.

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approvalStatus	An ApprovalStatusType. Valid values for an ApprovalStatus in the context of an approval process. ENUMERATED with values: * pendingApproval * approved * rejected
approvalStatusReason	A String. The reason why the approval state changed. DEPRECATED in the PATCH (bug to be fixed on next major version).

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category	An ApplicationCategoryType. Valid values for the category of the Application. Please refer to main Application stores in order to retrieve a more precise definition of the categories. ENUMERATED with values: * agriculture * artAndDesign * autoAndVehicle * beauty * booksAndReference * business * construction * defense * developerTools * education * engineering * eventsAndEntertainment * finance * foodAndDrink * games * healthAndFitness * healthcare * information * kids * librariesAndDemo * lifestyle * manufacturing * mapsAndNavigation * media * medical * mining * musicAndAudio * newsAndMagazines * organizations * other * photoAndVideo * productivity * publicService * realEstate * shopping * socialNetworkingAndCommunications * sports * tourism * utilities * trading * transportation * travelAndLocal * water * weather
channelPartner	A PartyRoleRef. Party role reference. A party role represents the part played by a party in a given context.
commercialName	A String. Exact name of the Application as marketed (and known by the end-user).
creationDate	A DateTime. Date and time when the Application was created.

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description	A String. Free-text description of the resource.
digitalIdentity	An ApiDigitalIdentity. ApiDigitalIdentity is a class that allow to describe a digital identity for an individual or a resource or a specific party role. One of these three MUST be provided. If an individual is provided, this identity will be for all her/his partyRole. To avoid confusion it is recommended in this case to not provide partyRoleIdentified. DEPRECATED in the POST and PATCH, this attribute can't be valued by the API consumer in this context.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
jwksUri	A String. Uri pointing to a JSON Web Key (JWK) : data structure that represents a cryptographic key.
logoUrl	A String. Url to retrieve the Application official logo.
name	A String. The name of the resource.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations; values defined by ITU X.731: 'disable': The resource is totally inoperable and unable to provide service to the user(s); 'enable': The resource is partially or fully operable and available for use. ENUMERATED with values: * enable * disable
privacyPolicyURL	A String. Url to Application privacy policy, if applicable.
privacyRightsRequestContact	A ContactMedium. Contact information for the exercise of data protection rights to which the End Users can address. Must be only EmailContactMedium or WebFormContactMedium for this context. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium
redirectUrl	A String. After a user successfully authorizes an application, the authorization server will redirect the user back to the application.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LogicalResourceRole sub-resource fields

grantedPermissionSet	An ApiPermissionSet. The PermissionSet is a set of Permissions granted to a user (party, party role, resource or resource role). The set may be granted explicitly by an authorized user or may be acquired implicitly due to the role that the user is playing.

ApiPermissionSet sub-resource fields

creationDate	A DateTime. Date when the PermissionSet was created and assigned to a user.
description	A String. Text describing the permission set.
permission	An ApiConsumerPermission. An ApiConsumerPermission allows an Application to consume an API.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiConsumerPermission sub-resource fields

managedAssetGroup	An ApiListAssetGroup. Defines a group of API assets to which access can be controlled.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiListAssetGroup sub-resource fields

apiProduct	A ProductRef. The list of products for which permissions are given.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductRef sub-resource fields

See [ProductRef] for details.

AssetGroup sub-resource fields

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	AssetGroup can be instantiated as * ApiListAssetGroup
apiProduct	This property is present in subclasses

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ApiDigitalIdentity sub-resource fields

clientId	A String. OAuth2 Client ID: A public identifier for apps. Even though it is public, it is best that it is not guessable by third parties, so many implementations use something like a 32-character hex string. DEPRECATED in the POST and the PATCH, this attribute can't be valued by the API consumer in this context.
credential	An OAuth2ClientCredential. A list of credential/authentification method that are used for this digital identity.
~~jwksUri~~	DEPRECATED A String. DEPRECATED : moved to Application schema. JSON Web Key (JWK) is a data structure that represents a cryptographic key.
~~redirectUrl~~	DEPRECATED A String. DEPRECATED : moved to Application schema. After a user successfully authorizes an application, the authorization server will redirect the user back to the application.
status	A String. Used to indicate the current lifecycle status of this digital identity (unknown, active, suspended, archived).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

OAuth2ClientCredential sub-resource fields

clientSecret	A String. Secret that permits the Application to authenticate with OAuth2 method.
state	A String. Used to indicate the current lifecycle state of this credential (like active, inactive, expired).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

ContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium

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city	This property is present in subclasses
country	This property is present in subclasses
emailAddress	This property is present in subclasses
faxNumber	This property is present in subclasses
geographicAddress	This property is present in subclasses
phoneNumber	This property is present in subclasses
postCode	This property is present in subclasses
socialNetworkId	This property is present in subclasses
stateOrProvince	This property is present in subclasses
street1	This property is present in subclasses
street2	This property is present in subclasses
url	This property is present in subclasses

GeographicAddressRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EmailContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
emailAddress	A String. Full email address in standard format.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

FaxContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
faxNumber	A String. The fax number of the contact.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressContactMedium sub-resource fields

city	A String. The city.
contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
country	A String. The country.
geographicAddress	A GeographicAddressRef. Reference to a Geographic Address.
id	A String. Identifier for this contact medium.
postCode	A String. Postcode.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
stateOrProvince	A String. State or province.
street1	A String. Describes the street.
street2	A String. Complementary street description.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PhoneContactMedium sub-resource fields

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contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
phoneNumber	A String. The phone number of the contact.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

SocialContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
socialNetworkId	A String. Identifier as a member of a social network.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

WebFormContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
url	A String. Url of the web form.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'Application' resource object.


```
  {
    "@type": "Application",
    "@baseType": "LogicalResource",
    "id": "102",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/102",
    "name": "Evolve West",
    "creationDate": "2024-12-02T14:10:00.00Z",
    "commercialName": "Evolve West",
    "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
    "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
    "description": "Evolve West mobile application",
    "redirectUrl": [
      "https://channelpartner.com/opengateway/application/evolvequest/redirect"
    ],
    "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
    "category": "games",
    "operationalState": "enable",
    "approvalStatus": "approved",
    "digitalIdentity": {
      "@type": "ApiDigitalIdentity",
      "clientId": "evolve_west_channelPartner15",
      "validFor": {
        "startDateTime": "2024-12-02T14:11:00.00Z"
      }
    },
    "apiConsumerRole": [
      {
        "grantedPermissionSet": [
          {
            "@type": "ApiPermissionSet",
            "creationDate": "2025-04-02T15:00:00.00Z",
            "permission": {
              "@type": "ApiConsumerPermission",
              "managedAssetGroup": {
                "@type": "ApiListAssetGroup",
                "@baseType": "AssetGroup",
                "apiProduct": [
                  {
                    "@type": "ProductRef",
                    "id": "123",
                    "@referredType": "ApiProduct"
                  }
                ]
              }
            },
            "validFor": {
              "startDateTime": "2025-04-02T15:00:00.00Z"
```


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```
            }
          }
        ]
      }
    ],
    "applicationOwner": {
      "@type": "PartyRoleRef",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
      "@referredType": "PartyRole"
    },
    "privacyRightsRequestContact": [
      {
        "@type": "WebFormContactMedium",
        "@baseType": "ContactMedium",
        "id": "1",
        "preferred": false,

        "url": "https://contact.evolvewest.com"
      },
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "id": "2",
        "preferred": true,

        "emailAddress": "contact@evolvewest.com"
      }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
          "@referredType": "PartyRole"
    }
  }
```


ApplicationOwner resource

Application Owner represents the role played by the organization that owns an Application that can consume API products. It's a specialization of a PartyRole in the Open Gateway context.

Resource model

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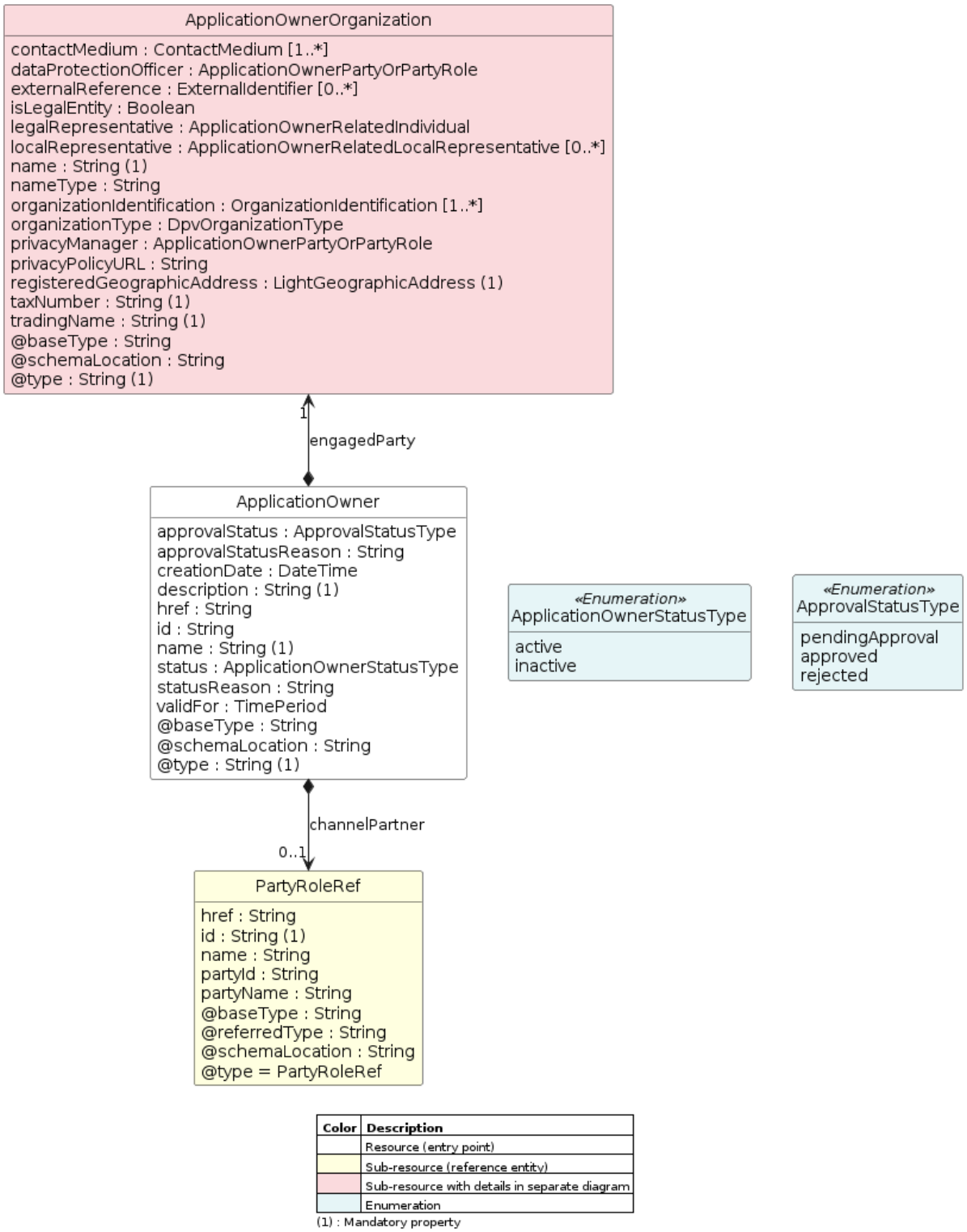


Figure 10 - ApplicationOwner

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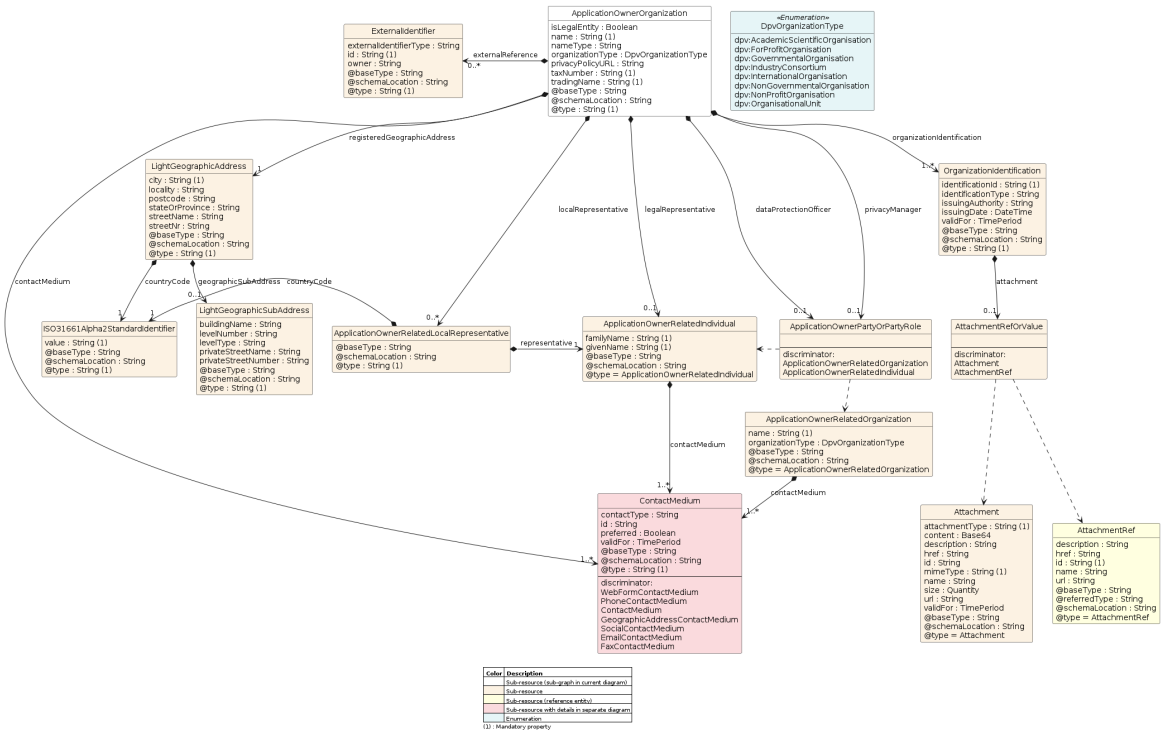


Figure 11 - ApplicationOwnerOrganization

See Figure 9 for ContactMedium diagrams.

Field descriptions

ApplicationOwner fields

approvalStatus	An ApprovalStatusType. Valid values for an ApprovalStatus in the context of an approval process. ENUMERATED with values: * pendingApproval * approved * rejected
approvalStatusReason	A String. The reason why the approval status changed.
channelPartner	A PartyRoleRef. Party role reference. A party role represents the part played by a party in a given context.
creationDate	A DateTime. Date and time when the ApplicationOwner was created.
description	A String. A description of the ApplicationOwner.
engagedParty	An ApplicationOwnerOrganization. Application Organization represents a group of people identified by shared interests or purpose. Examples include business, department and enterprise. Because of the complex nature of many businesses, both organizations and organization units are represented by the same data. It's a specialization of Party in the Open Gateway context.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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name	A String. A word, term, or phrase by which the ApplicationOwner is known and distinguished from other PartyRoles. It's the name of the ApplicationOwner unique entity.
status	An ApplicationOwnerStatusType. Used to track the lifecycle status of the ApplicationOwner in Channel Partner IT. So this attribute is allowed to be contributed by the Channel Partner and patchable. ENUMERATED with values: * active * inactive
statusReason	A String. A string providing an explanation on the value of the Status lifecycle. For instance if the Status is Rejected, statusReason will provide the reason for rejection.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ApplicationOwnerOrganization sub-resource fields

contactMedium	A ContactMedium. List of means for contacting the party, e.g. mobile phone, email address. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium
dataProtectionOfficer	An ApplicationOwnerPartyOrPartyRole. Schema that permits to identify an Individual or an Organisation in the Open Gateway context.
externalReference	An ExternalIdentifier. List of identifiers of the Party in an external system, for example when party information is imported from a commerce system.
isLegalEntity	A Boolean. If value is true, the organization is a legal entity known by a national referential.
legalRepresentative	An ApplicationOwnerRelatedIndividual. ApplicationOwnerRelatedIndividual represents an Individual that has a role within the ApplicationOwner, in the Open Gateway context. Typically an employee of the company.

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localRepresentative	An ApplicationOwnerRelatedLocalRepresentative. Represents a Party representing the Organization within a geographic localization.
name	A String. Organization name (department name for example).
nameType	A String. Type of the name : Co, Inc, Ltd, etc.
organizationIdentification	An OrganizationIdentification. List of official identifiers given to the organization, for example company number in the registry of companies.
organizationType	A DpvOrganizationType. Valid values for a type of organization based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv. ENUMERATED with values: * dpv:AcademicScientificOrganisation * dpv:ForProfitOrganisation * dpv:GovernmentalOrganisation * dpv:IndustryConsortium * dpv:InternationalOrganisation * dpv:NonGovernmentalOrganisation * dpv:NonProfitOrganisation * dpv:OrganisationalUnit
privacyManager	An ApplicationOwnerPartyOrPartyRole. Schema that permits to identify an Individual or an Organisation in the Open Gateway context.
privacyPolicyURL	A String. Url to privacy policy.
registeredGeographicAddress	A LightGeographicAddress. Structured textual way of describing how to find a Property in an urban area (country properties are often defined differently). Note : Address corresponds to SID UrbanPropertyAddress.
taxNumber	A String. Tax number of the Organization in the country.
tradingName	A String. Name that the organization (unit) trades under.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ContactMedium sub-resource fields

See [ContactMedium] for details.

GeographicAddressRef sub-resource fields

See [GeographicAddressRef] for details.

EmailContactMedium sub-resource fields

See [EmailContactMedium] for details.

FaxContactMedium sub-resource fields

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See [FaxContactMedium] for details.

GeographicAddressContactMedium sub-resource fields

See [GeographicAddressContactMedium] for details.

PhoneContactMedium sub-resource fields

See [PhoneContactMedium] for details.

SocialContactMedium sub-resource fields

See [SocialContactMedium] for details.

WebFormContactMedium sub-resource fields

See [WebFormContactMedium] for details.

ApplicationOwnerRelatedIndividual sub-resource fields

contactMedium	A ContactMedium. List of means for contacting the party, e.g. mobile phone, email address. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium
familyName	A String. Contains the non-chosen or inherited name. Also known as last name in the Western context.
givenName	A String. First name of the individual.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApplicationOwnerRelatedOrganization sub-resource fields

contactMedium	A ContactMedium. List of means for contacting the party, e.g. mobile phone, email address. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium

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name	A String. Organization name (department name for example).
organizationType	A DpvOrganizationType. Valid values for a type of organization based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv. ENUMERATED with values: * dpv:AcademicScientificOrganisation * dpv:ForProfitOrganisation * dpv:GovernmentalOrganisation * dpv:IndustryConsortium * dpv:InternationalOrganisation * dpv:NonGovernmentalOrganisation * dpv:NonProfitOrganisation * dpv:OrganisationalUnit
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

ApplicationOwnerRelatedIndividual sub-resource fields

See [ApplicationOwnerRelatedIndividual] for details.

ApplicationOwnerRelatedLocalRepresentative sub-resource fields

countryCode	An ISO31661Alpha2StandardIdentifier. The corresponding identification of the resource in different standard, regulatory definitions. The standard specification identifier (e.g., ISO 3166-1 Alpha-2) and the corresponding value (e.g., BE) relevant to a particular resource.
representative	An ApplicationOwnerRelatedIndividual. ApplicationOwnerRelatedIndividual represents an Individual that has a role within the ApplicationOwner, in the Open Gateway context. Typically an employee of the company.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ISO31661Alpha2StandardIdentifier sub-resource fields

See [ISO31661Alpha2StandardIdentifier] for details.

OrganizationIdentification sub-resource fields

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attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
identificationId	A String. Identifier.
identificationType	A String. Type of identification information used to identify the company in a country or internationally.
issuingAuthority	A String. Authority which has issued the identifier (chamber of commerce…).
issuingDate	A DateTime. Date at which the identifier was issued.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

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description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LightGeographicAddress sub-resource fields

city	A String. City that the address is in.
countryCode	An ISO31661Alpha2StandardIdentifier. The corresponding identification of the resource in different standard, regulatory definitions. The standard specification identifier (e.g., ISO 3166-1 Alpha-2) and the corresponding value (e.g., BE) relevant to a particular resource.
geographicSubAddress	A LightGeographicSubAddress. Representation of a GeographicSubAddress. It is used for addressing within a property in an urban area (country properties are often defined differently). It may refer to a building, a building cluster, or a floor of a multistory building.
locality	A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature, or a suburb, a bounded locality within a city, town or shire principally of urban character.
postcode	A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode).
stateOrProvince	A String. The State or Province that the address is in.
streetName	A String. Name of the street.
streetNr	A String. Number identifying a specific property on a public street.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LightGeographicSubAddress sub-resource fields

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buildingName	A String. Allows for buildings that have well-known names.
levelNumber	A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2.
levelType	A String. Describes level types within a building.
privateStreetName	A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office.
privateStreetNumber	A String. Private streets numbers internal to a private street.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ApplicationOwner' resource object.


```
{
  "@type": "ApplicationOwner",
  "id": "e05b75dc-f843-40d9-8d16-112275a28385",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
  "name": "Funtastic Games application owner role",
  "creationDate": "2024-12-01T13:00:00.00Z",
  "engagedParty": {
    "@type": "ApplicationOwnerOrganization",
    "name": "Funtastic Games SA",
    "tradingName": "Funtastic Games",
    "isLegalEntity": true,
    "organizationType": "dpv:ForProfitOrganisation",
    "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
    "taxNumber": "00.623.904/0001-73",
    "organizationIdentification": [
      {
        "@type": "OrganizationIdentification",
        "identificationType": "CNPJ",
        "identificationId": "00.623.904/0001-73",
        "issuingAuthority": "Receita Federal"
      }
    ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
    "registeredGeographicAddress": {
      "@type": "LightGeographicAddress",
      "streetNr": "123",
```


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```
      "streetName": "Avenida Paulista",
      "locality": "Paraiso",
      "city": "Sao Paulo",
      "stateOrProvince": "Sao Paulo",
      "countryCode": {
        "@type": "ISO31661Alpha2StandardIdentifier",
        "value": "BR"
      },
      "postcode": "01311-000",
      "geographicSubAddress": {
        "@type": "LightGeographicSubAddress",
        "buildingName": "Fogo",
        "levelNumber": "Planta 4"
      }
    },
    "contactMedium": [
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "id": "1",
        "preferred": true,
        "emailAddress": "contact@funtasticgames.com"
      }
    ],
    "legalRepresentative": {
      "@type": "ApplicationOwnerRelatedIndividual",
      "familyName": "OConnor",
      "givenName": "Jane",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
            "id": "1",
          "preferred": true,
          "emailAddress": "jane@funtasticgames.com"
        }
      ]
    },
    "dataProtectionOfficer": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "DPO Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": true,
          "emailAddress": "dpo@funtasticgames.com"
        }
      ]
    },
    "privacyManager": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "Privacy Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": true,
          "emailAddress": "privacy@funtasticgames.com"
```


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```
        }
      ]
    },
    "localRepresentative": [
      {
        "@type": "ApplicationOwnerRelatedLocalRepresentative",
        "countryCode": {
          "@type": "ISO31661Alpha2StandardIdentifier",
          "value": "EU"
        },
        "representative": {
          "@type": "ApplicationOwnerRelatedIndividual",
          "familyName": "Juan",
          "givenName": "Garcia",
          "contactMedium": [
            {
              "@type": "EmailContactMedium",
              "@baseType": "ContactMedium",
              "id": "jgarcia",
              "preferred": true,
                  "emailAddress": "jgarcia@funtasticgames.com"
            }
          ]
        }
      }
    ]
  },
  "status": "inactive",
  "approvalStatus": "pendingApproval",
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```


Monitor resource

Monitoring of resources.

Resource model

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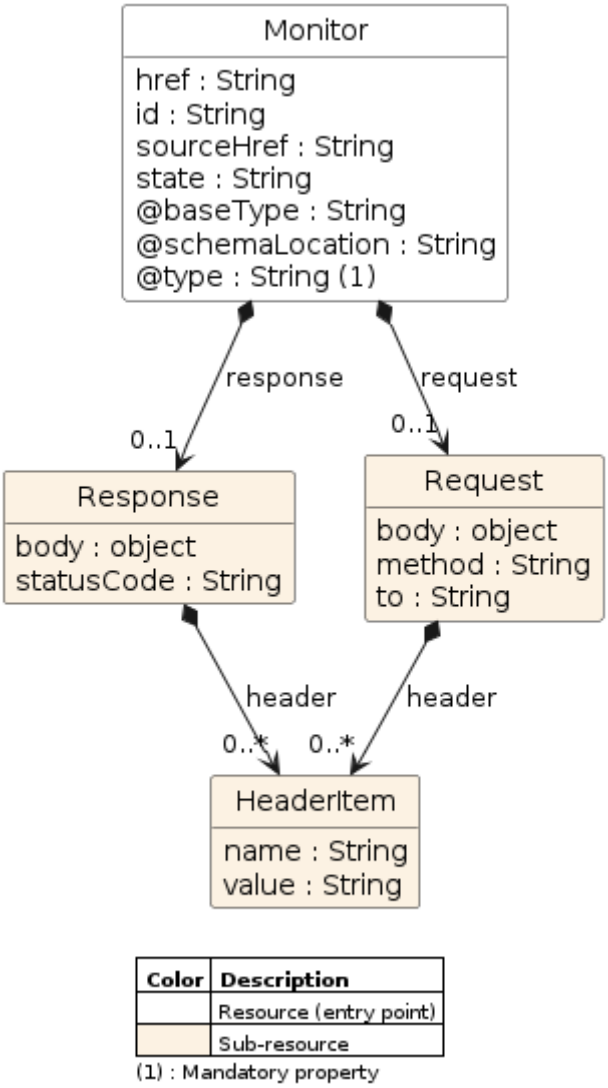


Figure 12 - Monitor

Field descriptions

Monitor fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
request	A Request. A response to a request.
response	A Response. A response to a request.
sourceHref	A String. The monitored resource href.
state	A String. The Monitor state of the resource. inProgress, inError, completed.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

Request sub-resource fields

body	An object. The body of the request. For example for an HTTP request might contain content of a form .
header	A HeaderItem. Items included in the header of the request. For example for an HTTP request might contain requested locale, basic authentication.
method	A String. The protocol of the request, e.g. http.
to	A String. The target of the request, e.g. a URL for an HTTP request.

HeaderItem sub-resource fields

name	A String. The name of the header item, e.g. locale.
value	A String. The value of the header item, e.g. en-us.

Response sub-resource fields

body	An object. The body of the response.
header	A HeaderItem. Items included in the header of the response. For example for an HTTP response might contain negotiated locale.
statusCode	A String. The status of the response. For example for an HTTP response would be codes such as 200, 400, etc.

Json representation sample(s)

We provide below a JSON representation as example of the 'Monitor' resource object.


```
{
  "@type": "Monitor",
  "id": "77795e7f-5213-4b9c-b672-6838a6b76152",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/monitor/77795e7f-5213-4b9c-b672-
6838a6b76152",
  "state": "Completed",
  "request": {
    "method": "POST",
    "to": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application",
    "body": {
      "@type": "Application",
      "@baseType": "LogicalResource",
      "name": "Evolve Quest",
      "commercialName": "Evolve Quest",
      "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
      "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
```


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```
      "description": "Evolve Quest mobile application",
      "redirectUrl": [
        "https://channelpartner.com/opengateway/application/evolvequest/redirect"
      ],
      "jwksUri":
"https://channelpartner.com/opengateway/application/evolvequest/jwk",
      "category": "games",
      "applicationOwner": {
        "@type": "PartyRoleRef",
        "id": "e05b75dc-f843-40d9-8d16-112275a28385",
        "@referredType": "PartyRole"
      },
      "privacyRightsRequestContact": [
        {
          "@type": "WebFormContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": false,

          "url": "https://contact.evolvequest.com"
        },
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "2",
          "preferred": true,

          "emailAddress": "contact@evolvequest.com"
        }
      ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ]
    }
  },
  "response": {
    "statusCode": "201",
    "body": {
      "@type": "Application",
      "@baseType": "LogicalResource",
      "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
      "name": "Evolve Quest",
      "commercialName": "Evolve Quest",
      "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
      "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
      "description": "Evolve Quest mobile application",
      "redirectUrl": [
        "https://channelpartner.com/opengateway/application/evolvequest/redirect"
      ],
      "jwksUri":
"https://channelpartner.com/opengateway/application/evolvequest/jwk",
      "category": "games",
      "operationalState": "enable",
```


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```
      "approvalStatus": "approved",
      "digitalIdentity": {
        "@type": "ApiDigitalIdentity",
        "clientId": "evolve_quest_channelPartner15",
        "validFor": {
          "startDateTime": "2024-12-02T14:01:00.00Z"
        }
      },
      "applicationOwner": {
        "@type": "PartyRoleRef",
        "id": "e05b75dc-f843-40d9-8d16-112275a28385",
          "@referredType": "PartyRole"
      },
      "privacyRightsRequestContact": [
        {
          "@type": "WebFormContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": false,

          "url": "https://contact.evolvequest.com"
        },
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "2",
          "preferred": true,

          "emailAddress": "contact@evolvequest.com"
        }
      ],
      "externalIdentifier": [
        {
          "@type": "ExternalIdentifier",
          "id": "1234",
          "owner": "Developer",
          "externalIdentifierType": "ApplicationIdFromDeveloper"
        }
      ],
      "channelPartner": {
        "@type": "PartyRoleRef",
        "id": "15",
          "@referredType": "PartyRole"
      }
    }
  },
  "sourceHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c"
}
```


ApiProductDeviceLocationVerification resource

An ApiProductDeviceLocationVerification is a specialization of a ApiProduct. It's an example, not a schema to be implemented as is in production.

Resource model

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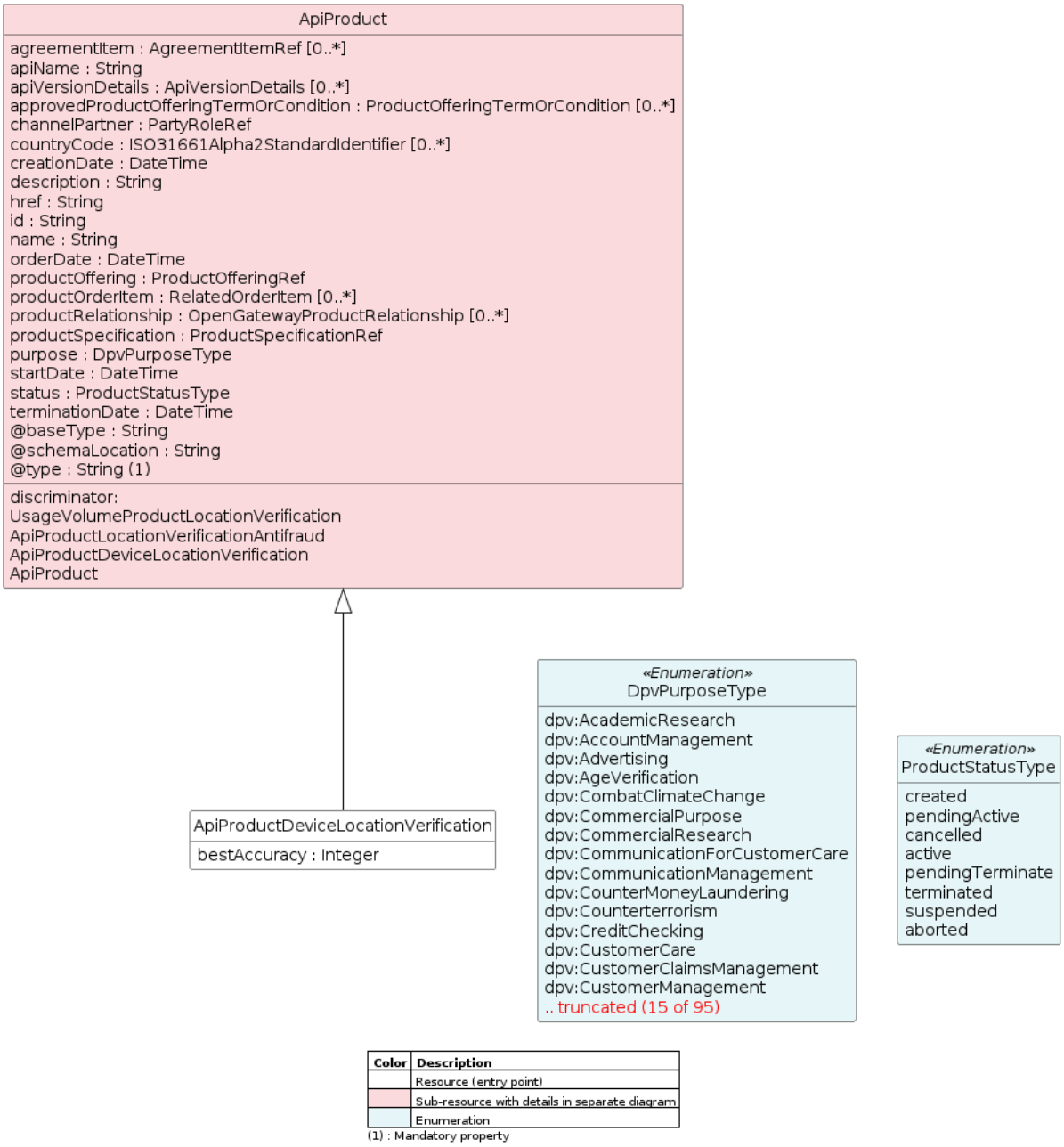


Figure 13 - ApiProductDeviceLocationVerification

See Figure 6 for ProductOfferingTermOrCondition diagrams.

Field descriptions

ApiProductDeviceLocationVerification fields

agreementItem	An AgreementItemRef. Collection of agreement items that involves this Product.
apiName	A String. Standardized name of the API.
apiVersionDetails	An ApiVersionDetails. A set of ApiProduct characteristics grouped together by API major version.

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approvedProductOfferingTermOrCondition	A ProductOfferingTermOrCondition. Approved Terms or Conditions of the ProductOffering.
bestAccuracy	An Integer. The best accuracy is the wished accuracy of the location in the response.
channelPartner	A PartyRoleRef. Party role reference. A party role represents the part played by a party in a given context.
countryCode	An ISO31661Alpha2StandardIdentifier. One or several countries where API is operated.
creationDate	A DateTime. Date and time when the product was created.
description	A String. Is the description of the product. It could be same description than Product Offering one.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the product. It could be the same as the name of the product offering.
orderDate	A DateTime. Is the date when the product was ordered.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
productOrderItem	A RelatedOrderItem. A reference to the Product Order items that involved this Product.
productRelationship	An OpenGatewayProductRelationship. Collection of product relationships.
productSpecification	A ProductSpecificationRef. ProductSpecification reference. A product Specification represents entities that are orderable from the provider of the catalog.

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95
startDate	A DateTime. Is the date from which the product starts.

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status	A ProductStatusType. Possible values for the status of the product. ENUMERATED with values: * created * pendingActive * cancelled * active * pendingTerminate * terminated * suspended * aborted
terminationDate	A DateTime. Is the date when the product was terminated.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AgreementItemRef sub-resource fields

See [AgreementItemRef] for details.

ApiVersionDetails sub-resource fields

See [ApiVersionDetails] for details.

ApiGrantInformation sub-resource fields

See [ApiGrantInformation] for details.

ISO31661Alpha2StandardIdentifier sub-resource fields

See [ISO31661Alpha2StandardIdentifier] for details.

ProductOfferingTermOrCondition sub-resource fields

See [ProductOfferingTermOrCondition] for details.

DpvProcess sub-resource fields

See [DpvProcess] for details.

DpvContext sub-resource fields

See [DpvContext] for details.

DataExportCountry sub-resource fields

See [DataExportCountry] for details.

DataExportCountry sub-resource fields

See [DataExportCountry] for details.

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ProductRef sub-resource fields

See [ProductRef] for details.

ProductOfferingTermOrConditionApproval sub-resource fields

See [ProductOfferingTermOrConditionApproval] for details.

ApiAuthorization sub-resource fields

See [ApiAuthorization] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ProductOfferingTermOrConditionSpecRef sub-resource fields

See [ProductOfferingTermOrConditionSpecRef] for details.

SimpleAttachment sub-resource fields

See [SimpleAttachment] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ISO31661Alpha2StandardIdentifier sub-resource fields

See [ISO31661Alpha2StandardIdentifier] for details.

ProductOfferingRef sub-resource fields

See [ProductOfferingRef] for details.

RelatedOrderItem sub-resource fields

See [RelatedOrderItem] for details.

OpenGatewayProductRelationship sub-resource fields

See [OpenGatewayProductRelationship] for details.

ProductRelationship sub-resource fields

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See [ProductRelationship] for details.

ProductSpecificationRef sub-resource fields

See [ProductSpecificationRef] for details.

TargetProductSchema sub-resource fields

See [TargetProductSchema] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'ApiProductDeviceLocationVerification' resource object.


```
{
  "@type": "ApiProductDeviceLocationVerification",
  "@baseType": "ApiProduct",
  "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Product/ApiProductDeviceLocationVerification.schema.json",
  "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
  "name": "Location Verification Antifraud CAMARA API Standalone",
  "creationDate": "2025-02-02T12:02:00.00Z",
  "startDate": "2025-02-02T12:00:03.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
    "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
    "name": "Location Verification Antifraud CAMARA API Standalone",
    "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
    {
      "@type": "RelatedOrderItem",
      "orderId": "86387671-0997-461b-99c7-d624ffb0640c",
        "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
      "orderItemId": "1",
      "orderItemAction": "add"
    }
  ],
  "productSpecification": {
    "@type": "ProductSpecificationRef",
    "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
9543-0c5e9070ade9",
    "name": "Location Verification Antifraud CAMARA API",
    "version": "1.0.0",
    "@referredType": "ApiProductSpecification"
  },
  "apiName": "location-verification",
  "apiVersionDetails": [
    {
      "@type": "ApiVersionDetails",
```


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```
      "apiMajorVersion": 0,
      "basePath": "location-verification",
      "apiGrantInformation": [
        {
          "@type": "ApiGrantInformation",
          "purpose": "dpv:FraudPreventionAndDetection",
          "scope": [
            "location-verification:verify"
          ],
          "grantType": [
            "urn:openid:params:grant-type:ciba",
            "authorization_code"
          ],
          "legalBasis": "dpv:Consent"
        },
        {
          "@type": "ApiGrantInformation",
          "scope": [
            "location-verification:read"
          ],
          "grantType": [
            "client_credentials"
          ]
        }
      ]
    }
  ],
  "purpose": "dpv:FraudPreventionAndDetection",
  "bestAccuracy": 500,
"approvedProductOfferingTermOrCondition": [
  {
    "@type": "ProductOfferingTermOrCondition",

    "productOfferingTermOrConditionSpecRef": {
      "@type": "ProductOfferingTermOrConditionSpecRef",
      "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
      "@referredType": "OpenGatewayProductOfferingTermOrConditionSpecification"

    },
    "productOfferingTermOrConditionApproval": {
      "approvalDate": "2023-11-20T00:00:00.000Z"
    },
    "applicationLegalBasisForInputData": "dpv:Consent",
    "process": [
      {
        "@type": "DpvProcess",
        "processingOperation": [
          "dpv:Assess"
        ],
        "context": {
          "@type": "DpvContext",
          "processingContext": [
            "dpv:AutomatedDecisionMaking",
            "dpv:AutomatedScoringOfIndividuals"
          ],
          "duration": "dpv:FixedOccurrencesDuration",
          "frequency": "dpv:SingularFrequency"
        },
        "dataExportTransitCountry": [
          {
            "@type": "DataExportCountry",
            "countryCode": {
              "@type": "ISO31661Alpha2StandardIdentifier",
```


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```
              "value": "US"
            },
            "commercialOrganizationName": "Globy Corporation",
            "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
          }
        ]
      },
      {
        "@type": "DpvProcess",
        "processingOperation": [
          "dpv:Profiling"
        ],
        "context": {
          "@type": "DpvContext",
          "processingContext": [
            "dpv:AutomatedScoringOfIndividuals"
          ],
          "duration": "dpv:EndlessDuration",
          "frequency": "dpv:SingularFrequency"
        },
        "dataExportTransitCountry": [
          {
            "@type": "DataExportCountry",
            "countryCode": {
              "@type": "ISO31661Alpha2StandardIdentifier",
              "value": "US"
            },
            "commercialOrganizationName": "Globy Corporation",
            "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
          }
        ],
        "dataExportStorageCountry": [
          {
            "@type": "DataExportCountry",
            "countryCode": {
              "@type": "ISO31661Alpha2StandardIdentifier",
              "value": "US"
            },
            "commercialOrganizationName": "Globy Corporation",
            "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
          }
        ]
      }
    ],
    "userConsentScreenshot": {
      "@type": "SimpleAttachment",
      "content": "JVBERi...JSVFT0Y=",
      "mimeType": "application/pdf"
    }
  }
],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```


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ApiProductLocationVerificationAntifraud resource

An ApiProductLocationVerificationAntifraud is a specialization of a ApiProduct. It's an example, not a schema to be implemented as is in production.

Resource model

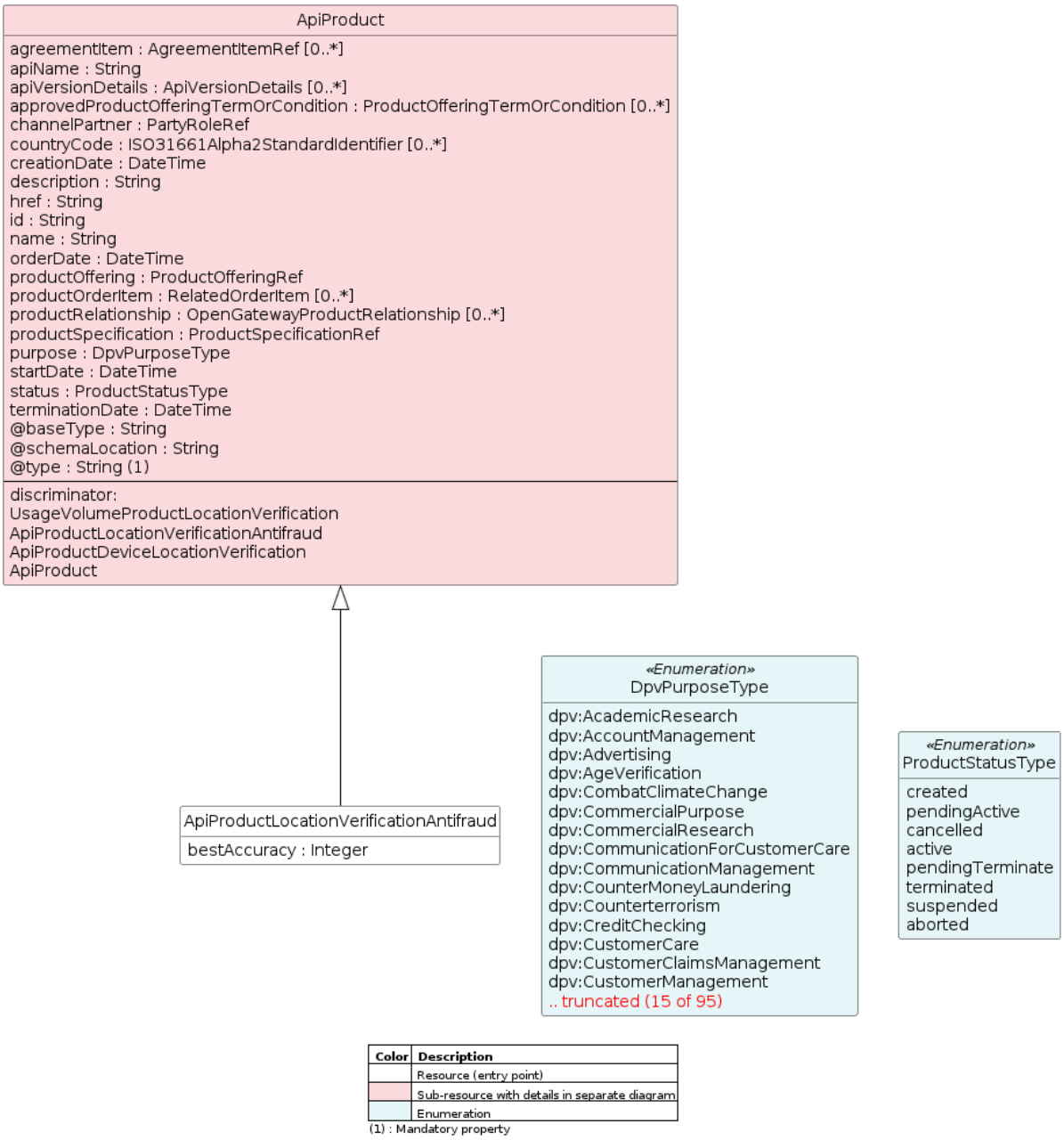


Figure 14 - ApiProductLocationVerificationAntifraud

See Figure 6 for ProductOfferingTermOrCondition diagrams.

Field descriptions

ApiProductLocationVerificationAntifraud fields

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agreementItem	An AgreementItemRef. Collection of agreement items that involves this Product.
apiName	A String. Standardized name of the API.
apiVersionDetails	An ApiVersionDetails. A set of ApiProduct characteristics grouped together by API major version.
approvedProductOfferingTermOrCondition	A ProductOfferingTermOrCondition. Approved Terms or Conditions of the ProductOffering.
bestAccuracy	An Integer. The best accuracy is the wished accuracy of the location in the response.
channelPartner	A PartyRoleRef. Party role reference. A party role represents the part played by a party in a given context.
countryCode	An ISO31661Alpha2StandardIdentifier. One or several countries where API is operated.
creationDate	A DateTime. Date and time when the product was created.
description	A String. Is the description of the product. It could be same description than Product Offering one.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the product. It could be the same as the name of the product offering.
orderDate	A DateTime. Is the date when the product was ordered.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
productOrderItem	A RelatedOrderItem. A reference to the Product Order items that involved this Product.
productRelationship	An OpenGatewayProductRelationship. Collection of product relationships.
productSpecification	A ProductSpecificationRef. ProductSpecification reference. A product Specification represents entities that are orderable from the provider of the catalog.

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95
startDate	A DateTime. Is the date from which the product starts.

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status	A ProductStatusType. Possible values for the status of the product. ENUMERATED with values: * created * pendingActive * cancelled * active * pendingTerminate * terminated * suspended * aborted
terminationDate	A DateTime. Is the date when the product was terminated.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AgreementItemRef sub-resource fields

See [AgreementItemRef] for details.

ApiVersionDetails sub-resource fields

See [ApiVersionDetails] for details.

ApiGrantInformation sub-resource fields

See [ApiGrantInformation] for details.

ISO31661Alpha2StandardIdentifier sub-resource fields

See [ISO31661Alpha2StandardIdentifier] for details.

ProductOfferingTermOrCondition sub-resource fields

See [ProductOfferingTermOrCondition] for details.

DpvProcess sub-resource fields

See [DpvProcess] for details.

DpvContext sub-resource fields

See [DpvContext] for details.

DataExportCountry sub-resource fields

See [DataExportCountry] for details.

DataExportCountry sub-resource fields

See [DataExportCountry] for details.

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ProductRef sub-resource fields

See [ProductRef] for details.

ProductOfferingTermOrConditionApproval sub-resource fields

See [ProductOfferingTermOrConditionApproval] for details.

ApiAuthorization sub-resource fields

See [ApiAuthorization] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ProductOfferingTermOrConditionSpecRef sub-resource fields

See [ProductOfferingTermOrConditionSpecRef] for details.

SimpleAttachment sub-resource fields

See [SimpleAttachment] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ISO31661Alpha2StandardIdentifier sub-resource fields

See [ISO31661Alpha2StandardIdentifier] for details.

ProductOfferingRef sub-resource fields

See [ProductOfferingRef] for details.

RelatedOrderItem sub-resource fields

See [RelatedOrderItem] for details.

OpenGatewayProductRelationship sub-resource fields

See [OpenGatewayProductRelationship] for details.

ProductRelationship sub-resource fields

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See [ProductRelationship] for details.

ProductSpecificationRef sub-resource fields

See [ProductSpecificationRef] for details.

TargetProductSchema sub-resource fields

See [TargetProductSchema] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'ApiProductLocationVerificationAntifraud' resource object.


```
{
  "@type": "ApiProductLocationVerificationAntifraud",
  "@baseType": "ApiProduct",
  "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/ApiProductLocationVerificationAntifraud",
  "id": "3be31371-2755-43f7-a6f6-380234910b9b",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/3be31371-2755-43f7-
a6f6-380234910b9b",
  "creationDate": "2025-01-01T10:02:00.00Z",
  "startDate": "2025-01-01T10:03:00.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
    "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/ab7792f8-6628-4c4b-a557-
699adc26d4ce",
    "name": "Location Verification Antifraud CAMARA API Bundled",
              "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
    {
      "@type": "RelatedOrderItem",
      "orderId": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
      "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
      "orderItemId": "2",
      "orderItemAction": "add"
    }
  ],
  "productSpecification": {
    "@type": "ProductSpecificationRef",
    "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
9543-0c5e9070ade9",
    "name": "Location Verification Antifraud CAMARA API",
    "version": "1.0.0",
    "@referredType": "ApiProductSpecification"
  },
  "apiName": "location-verification",
```


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```
  "apiVersionDetails": [
    {
      "@type": "ApiVersionDetails",
      "apiMajorVersion": 0,
      "basePath": "location-verification",
      "apiGrantInformation": [
        {
          "@type": "ApiGrantInformation",
          "purpose": "dpv:FraudPreventionAndDetection",
          "scope": [
            "location-verification:verify"
          ],
          "grantType": [
            "urn:openid:params:grant-type:ciba",
            "authorization_code"
          ],
          "legalBasis": "dpv:Consent"
        },
        {
          "@type": "ApiGrantInformation",
          "scope": [
            "location-verification:read"
          ],
          "grantType": [
            "client_credentials"
          ]
        }
      ]
    }
  ],
  "purpose": "dpv:FraudPreventionAndDetection",
  "bestAccuracy": 500,
  "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
```


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```
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
    ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
```


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```
}
```

UsageVolumeProductLocationVerification resource

A UsageVolumeProductLocationVerification is a specialization of a ApiProduct. It's an example, not a schema to be implemented as is in production.

Resource model

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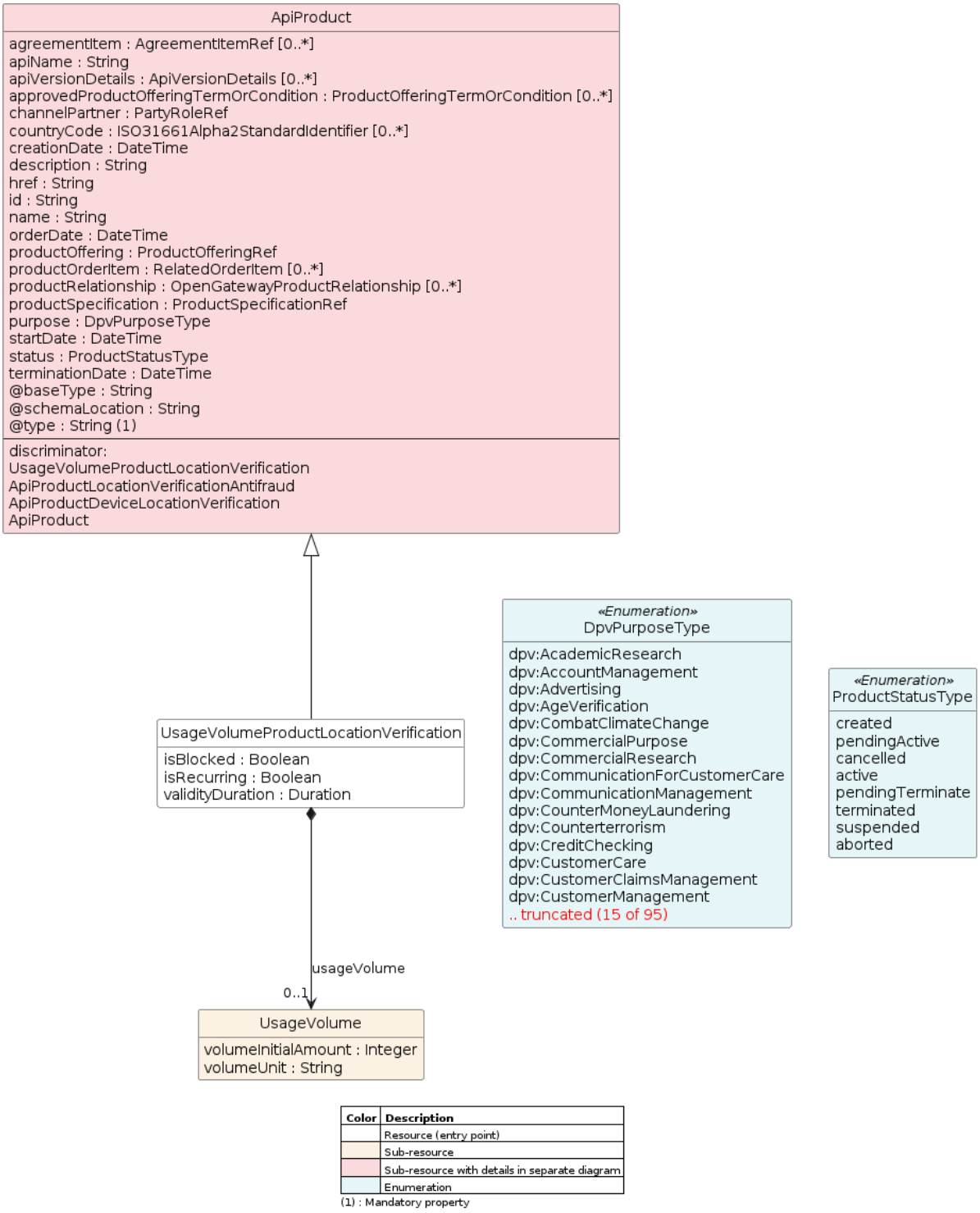


Figure 15 - UsageVolumeProductLocationVerification

See Figure 6 for ProductOfferingTermOrCondition diagrams.

Field descriptions

UsageVolumeProductLocationVerification fields

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agreementItem	An AgreementItemRef. Collection of agreement items that involves this Product.
apiName	A String. Standardized name of the API.
apiVersionDetails	An ApiVersionDetails. A set of ApiProduct characteristics grouped together by API major version.
approvedProductOfferingTermOrCondition	A ProductOfferingTermOrCondition. Approved Terms or Conditions of the ProductOffering.
channelPartner	A PartyRoleRef. Party role reference. A party role represents the part played by a party in a given context.
countryCode	An ISO31661Alpha2StandardIdentifier. One or several countries where API is operated.
creationDate	A DateTime. Date and time when the product was created.
description	A String. Is the description of the product. It could be same description than Product Offering one.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
isBlocked	A Boolean. If true means that the usage of the service can be blocked when a specific value of the Service Balance is reached, for instance when its remaining usage value is 0.
isRecurring	A Boolean. If true means that a new Service Balance will be created after each validity period.
name	A String. Name of the product. It could be the same as the name of the product offering.
orderDate	A DateTime. Is the date when the product was ordered.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
productOrderItem	A RelatedOrderItem. A reference to the Product Order items that involved this Product.
productRelationship	An OpenGatewayProductRelationship. Collection of product relationships.
productSpecification	A ProductSpecificationRef. ProductSpecification reference. A product Specification represents entities that are orderable from the provider of the catalog.

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purpose	A DpvPurposeType. Valid values for the purpose based on W3C Data Privacy Vocabulary v2 https://w3c.github.io/dpv/2.0/dpv/. ENUMERATED with values: * dpv:AcademicResearch * dpv:AccountManagement * dpv:Advertising * dpv:AgeVerification * dpv:CombatClimateChange * dpv:CommercialPurpose * dpv:CommercialResearch * dpv:CommunicationForCustomerCare * dpv:CommunicationManagement * dpv:CounterMoneyLaundering * dpv:Counterterrorism * dpv:CreditChecking * dpv:CustomerCare * dpv:CustomerClaimsManagement * dpv:CustomerManagement * dpv:CustomerOrderManagement * dpv:CustomerRelationshipManagement * dpv:CustomerSolvencyMonitoring * dpv:DataAltruism * dpv:DeliveryOfGoods * dpv:DirectMarketing * dpv:DisputeManagement * dpv:EnforceAccessControl * dpv:EnforceSecurity * dpv:EstablishContractualAgreement * dpv:FraudPreventionAndDetection * dpv:FulfilmentOfContractualObligation * dpv:FulfilmentOfObligation * dpv:HumanResourceManagement * dpv:IdentityAuthentication * dpv:IdentityVerification * dpv:ImproveExistingProductsAndServices * dpv:ImproveHealthcare * dpv:ImproveInternalCRMProcesses * dpv:ImprovePublicServices * dpv:ImproveTransportMobility * dpv:IncreaseServiceRobustness * dpv:InternalResourceOptimisation * dpv:LegalCompliance * dpv:MaintainCreditCheckingDatabase * dpv:MaintainCreditRatingDatabase * dpv:MaintainFraudDatabase * dpv:Marketing * dpv:MemberPartnerManagement * dpv:MisusePreventionAndDetection * dpv:NonCommercialPurpose * dpv:NonCommercialResearch * dpv:OptimisationForConsumer * dpv:OptimisationForController * dpv:OptimiseUserInterface * .. truncated .. 50 of 95
startDate	A DateTime. Is the date from which the product starts.

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status	A ProductStatusType. Possible values for the status of the product. ENUMERATED with values: * created * pendingActive * cancelled * active * pendingTerminate * terminated * suspended * aborted
terminationDate	A DateTime. Is the date when the product was terminated.
usageVolume	A UsageVolume. Usage Volume for Open Gateway context.
validityDuration	A Duration. A time interval in a given unit of time in Open Gateway context.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AgreementItemRef sub-resource fields

See [AgreementItemRef] for details.

ApiVersionDetails sub-resource fields

See [ApiVersionDetails] for details.

ApiGrantInformation sub-resource fields

See [ApiGrantInformation] for details.

ISO31661Alpha2StandardIdentifier sub-resource fields

See [ISO31661Alpha2StandardIdentifier] for details.

ProductOfferingTermOrCondition sub-resource fields

See [ProductOfferingTermOrCondition] for details.

DpvProcess sub-resource fields

See [DpvProcess] for details.

DpvContext sub-resource fields

See [DpvContext] for details.

DataExportCountry sub-resource fields

See [DataExportCountry] for details.

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DataExportCountry sub-resource fields

See [DataExportCountry] for details.

ProductRef sub-resource fields

See [ProductRef] for details.

ProductOfferingTermOrConditionApproval sub-resource fields

See [ProductOfferingTermOrConditionApproval] for details.

ApiAuthorization sub-resource fields

See [ApiAuthorization] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

See [RelatedPartyRefOrPartyRoleRef] for details.

PartyRef sub-resource fields

See [PartyRef] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ProductOfferingTermOrConditionSpecRef sub-resource fields

See [ProductOfferingTermOrConditionSpecRef] for details.

SimpleAttachment sub-resource fields

See [SimpleAttachment] for details.

PartyRoleRef sub-resource fields

See [PartyRoleRef] for details.

ISO31661Alpha2StandardIdentifier sub-resource fields

See [ISO31661Alpha2StandardIdentifier] for details.

ProductOfferingRef sub-resource fields

See [ProductOfferingRef] for details.

RelatedOrderItem sub-resource fields

See [RelatedOrderItem] for details.

OpenGatewayProductRelationship sub-resource fields

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See [OpenGatewayProductRelationship] for details.

ProductRelationship sub-resource fields

See [ProductRelationship] for details.

ProductSpecificationRef sub-resource fields

See [ProductSpecificationRef] for details.

TargetProductSchema sub-resource fields

See [TargetProductSchema] for details.

UsageVolume sub-resource fields

See [UsageVolume] for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'UsageVolumeProductLocationVerification' resource object.


```
{
  "@type": "UsageVolumeProductLocationVerification",
  "@baseType": "ApiProduct",
  "id": "fea0c794-d62d-40bb-a9a4-934336194dca",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/fea0c794-d62d-40bb-
a9a4-934336194dca",
  "creationDate": "2025-01-01T10:02:00.00Z",
  "startDate": "2025-01-01T10:03:00.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
    "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/0a2782b7-81a7-4af9-9e4f-
d51bc690ec2d",
    "name": "Location Verification Verify Usage Volume Recurring Bundled",
    "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
    {
      "@type": "RelatedOrderItem",
      "orderId": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
      "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
      "orderItemId": "3",
      "orderItemAction": "add"
    }
  ],
  "productRelationship": [
    {
      "@type": "OpenGatewayProductRelationship",
      "id": "3be31371-2755-43f7-a6f6-380234910b9b",
      "relationshipType": "appliesOn"
```


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```
      }
    ],
    "apiName": "",
    "usageVolume": {
      "volumeUnit": "apiCall",
      "volumeInitialAmount": 10000
    },
    "isBlocked": false,
    "isRecurring": true,
    "validityDuration": {
      "amount": 1,
      "units": "month"
    },
    "approvedProductOfferingTermOrCondition": [
          {
            "@type": "ProductOfferingTermOrCondition",
            "productOfferingTermOrConditionSpecRef": {
              "@type": "ProductOfferingTermOrConditionSpecRef",
              "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d_terms_1",
                   "@referredType":
 "OpenGatewayProductOfferingTermOrConditionSpecification"

            },
            "productOfferingTermOrConditionApproval": {
              "approvalDate": "2023-11-20T00:00:00.000Z"
            }
          }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
            "@referredType": "PartyRole"
    }
  }
```


Notification Resource Models

10 notifications are defined for this API.

Notifications related to ApiProductOrder:

- Create Event
- Attribute Value Change Event
- State Change Event

Notifications related to Application:

- Create Event
- Attribute Value Change Event
- Approval Status Change Event

Notifications related to ApplicationOwner:

- Create Event

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- Attribute Value Change Event
- Approval Status Change Event

Notifications related to Monitor:

- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

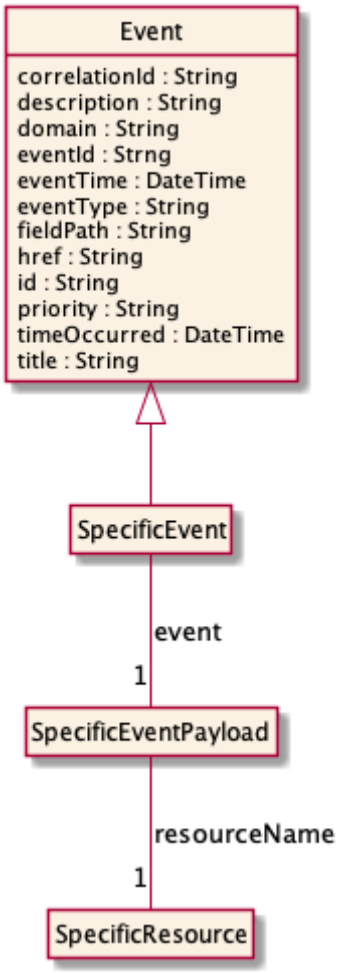


Figure 16 Notification Pattern

ApiProductOrder

Create Event

Message example for an ApiProductOrderCreateEvent event

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```
Content-Type: application/json
{
    "correlationId": "126-54f",
    "description": "Event for a productOrder creation.",
    "domain": "Open Gateway Onboarding",
    "eventId": "125",
    "eventTime": "2025-02-02T12:00:00.00Z",
    "eventType": "ApiProductOrderCreateEvent",
    "priority": "1",
    "timeOccurred": "2025-02-02T12:00:00.00Z",
    "title": "ApiProductOrderCreateEvent",
    "event": {
      "productOrder": {
        "@type": "ApiProductOrder",
        "id": "86387671-0997-461b-99c7-d624ffb0640c",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
        "state": "acknowledged",
        "externalId": [
          {
            "@type": "ExternalIdentifier",
            "externalIdentifierType": "ApplicationOwnerCustomerOrder",
            "id": "order_2232342",
            "owner": "ChannelPartner"
          }
        ],
        "description": "Camara API order for an Application Owner",
        "productOrderItem": [
          {
            "@type": "ApiProductOrderItemAdd",
            "@baseType": "DCSProductOrderItem",
            "id": "1",
            "state": "completed",
            "productOffering": {
              "@type": "ProductOfferingRef",
              "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
              "name": "areocloud_device_location_verification",
              "@referredType": "OpenGatewayProductOffering"
            },
            "product": {
              "@type": "ProductRef",
              "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
              "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
              "@referredType": "ApiProduct"
            },
            "productAction": {
              "@type": "ApiProductActionAddDeviceLocationVerification",
              "@baseType": "ApiProductAction",
              "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerification.schema.json",
              "purpose": "dpv:FraudPreventionAndDetection",
              "bestAccuracy": 500,
              "targetApplication": {
                "@type": "ApplicationRef",
                "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
                "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
                "@referredType": "Application"
```

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```
              }
            },
            "externalIdentifier": {
              "@type": "ExternalIdentifier",
              "externalIdentifierType": "ApplicationOwnerProduct",
              "id": "device_location_1234",
              "owner": "ChannelPartner"
            },
            "approvedProductOfferingTermOrCondition": [
              {
                "@type": "ProductOfferingTermOrCondition",

                "productOfferingTermOrConditionSpecRef": {
                  "@type": "ProductOfferingTermOrConditionSpecRef",
                  "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
                  "name": "device_location_verification_terms",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"
                },
                "productOfferingTermOrConditionApproval": {
                  "approvalDate": "2023-11-20T00:00:00.000Z"
                },
                "applicationLegalBasisForInputData": "dpv:Consent",
                "process": [
                  {
                    "@type": "DpvProcess",
                    "processingOperation": [
                      "dpv:Assess"
                    ],
                    "context": {
                      "@type": "DpvContext",
                      "processingContext": [
                        "dpv:AutomatedDecisionMaking",
                        "dpv:AutomatedScoringOfIndividuals"
                      ],
                      "duration": "dpv:FixedOccurrencesDuration",
                      "frequency": "dpv:SingularFrequency"
                    },
                    "dataExportTransitCountry": [
                      {
                        "@type": "DataExportCountry",
                        "countryCode": {
                          "@type": "ISO31661Alpha2StandardIdentifier",
                          "value": "US"
                        },
                        "commercialOrganizationName": "Globy Corporation",
                        "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                      }
                    ]
                  },
                  {
                    "@type": "DpvProcess",
                    "processingOperation": [
                      "dpv:Profiling"
                    ],
                    "context": {
                      "@type": "DpvContext",
                      "processingContext": [
                        "dpv:AutomatedScoringOfIndividuals"
                      ],
                      "duration": "dpv:EndlessDuration",
                      "frequency": "dpv:SingularFrequency"
```


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```
                    },
                    "dataExportTransitCountry": [
                      {
                        "@type": "DataExportCountry",
                        "countryCode": {
                          "@type": "ISO31661Alpha2StandardIdentifier",
                          "value": "US"
                        },
                        "commercialOrganizationName": "Globy Corporation",
                        "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                      }
                    ],
                    "dataExportStorageCountry": [
                      {
                        "@type": "DataExportCountry",
                        "countryCode": {
                          "@type": "ISO31661Alpha2StandardIdentifier",
                          "value": "US"
                        },
                        "commercialOrganizationName": "Globy Corporation",
                        "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                      }
                    ]
                  }
                ],
                "userConsentScreenshot": {
                  "@type": "SimpleAttachment",
                  "content": "JVBERi...JSVFT0Y=",
                  "mimeType": "application/pdf"
                }
              }
            ]
          }
        ],
        "channelPartner": {
          "@type": "PartyRoleRef",
          "id": "15",
          "@referredType": "PartyRole"
        }
      }
},
"@baseType": "Event",
"@type": "ApiProductOrderCreateEvent"
  }
```


Attribute Value Change Event

Message example for an ApiProductOrderAttributeValueChangeEvent event


```
Content-Type: application/json
{
    "correlationId": "126-54f",
    "description": "Event for a productOrder attribute value change.",
    "domain": "Open Gateway Onboarding",
    "eventId": "125",
    "eventTime": "2025-02-02T12:01:00.00Z",
    "eventType": "ApiProductOrderAttributeValueChangeEvent",
    "priority": "1",
```


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```
    "timeOccurred": "2025-02-02T12:01:00.00Z",
    "title": "ApiProductOrderAttributeValueChangeEvent",
    "event": {
      "productOrder": {
          "@type": "ApiProductOrder",
            "id": "86387671-0997-461b-99c7-d624ffb0640c",
            "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
            "state": "completed",
            "completionDate": "2025-02-02T12:01:00.00Z"
      }
    },
    "@baseType": "Event",
    "@type": "ApiProductOrderAttributeValueChangeEvent"
  }
```


State Change Event

Message example for an ApiProductOrderStateChangeEvent event


```
Content-Type: application/json
{
    "correlationId": "126-54f",
    "description": "Event for a productOrder state change.",
    "domain": "Open Gateway Onboarding",
    "eventId": "125",
    "eventTime": "2025-02-02T12:01:00.00Z",
    "eventType": "ApiProductOrderStateChangeEvent",
    "priority": "1",
    "timeOccurred": "2025-02-02T12:01:00.00Z",
    "title": "ApiProductOrderStateChangeEvent",
    "event": {
      "productOrder": {
        "@type": "ApiProductOrder",
        "id": "86387671-0997-461b-99c7-d624ffb0640c",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
        "state": "completed",
        "completionDate": "2025-02-02T12:01:00.00Z"
      }
    },
    "@baseType": "Event",
    "@type": "ApiProductOrderStateChangeEvent"
  }
```


Application

Create Event

Message example for an ApplicationCreateEvent event


```
Content-Type: application/json
{
  "correlationId": "126-54f",
```


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```
  "description": "Event for a application creation.",
  "domain": "Open Gateway Onboarding",
  "eventId": "125",
  "eventTime": "2024-12-02T14:10:00.00Z",
  "eventType": "ApplicationCreateEvent",
  "priority": "1",
  "timeOccurred": "2024-12-02T14:10:00.00Z",
  "title": "ApplicationCreateEvent",
  "event": {
    "application": {
      "@type": "Application",
      "@baseType": "LogicalResource",
      "id": "102",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/102",
      "name": "Evolve West",
      "creationDate": "2024-12-02T14:10:00.00Z",
      "commercialName": "Evolve West",
      "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
      "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
      "description": "Evolve West mobile application",
      "redirectUrl": [
        "https://channelpartner.com/opengateway/application/evolvequest/redirect"
      ],
      "jwksUri":
"https://channelpartner.com/opengateway/application/evolvequest/jwk",
      "category": "games",
      "operationalState": "disable",
      "approvalStatus": "pendingApproval",
      "digitalIdentity": {
        "@type": "ApiDigitalIdentity",
        "clientId": "evolve_quest_channelPartner15",
        "validFor": {
          "startDateTime": "2024-12-02T14:11:00.00Z"
        }
      },
      "apiConsumerRole": [
        {
          "grantedPermissionSet": [
            {
              "@type": "ApiPermissionSet",
              "creationDate": "2025-04-02T15:00:00.00Z",
              "permission": {
                "@type": "ApiConsumerPermission",
                "managedAssetGroup": {
                  "@type": "ApiListAssetGroup",
                  "@baseType": "AssetGroup",
                  "apiProduct": [
                    {
                      "@type": "ProductRef",
                      "id": "123",
                      "@referredType": "ApiProduct"
                    }
                  ]
                }
              },
              "validFor": {
                "startDateTime": "2025-04-02T15:00:00.00Z"
              }
            }
          ]
```


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```
        }
      ],
      "applicationOwner": {
        "@type": "PartyRoleRef",
        "id": "e05b75dc-f843-40d9-8d16-112275a28385",
        "@referredType": "PartyRole"
      },
      "privacyRightsRequestContact": [
        {
          "@type": "WebFormContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": false,
          "url": "https://contact.evolvequest.com"
        },
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "2",
          "preferred": true,
          "emailAddress": "contact@evolvequest.com"
        }
      ],
      "channelPartner": {
        "@type": "PartyRoleRef",
        "id": "15",
        "@referredType": "PartyRole"
      }
    }
  },
  "@baseType": "Event",
  "@type": "ApplicationCreateEvent"
}
```


Attribute Value Change Event

Message example for an ApplicationAttributeValueChangeEvent event


```
Content-Type: application/json
{
  "correlationId": "126-54f",
  "description": "Event for a application attribute value change.",
  "domain": "Open Gateway Onboarding",
  "eventId": "125",
  "eventTime": "2024-12-02T15:00:00.00Z",
  "eventType": "ApplicationAttributeValueChangeEvent",
  "priority": "1",
  "timeOccurred": "2024-12-02T15:00:00.00Z",
  "title": "ApplicationAttributeValueChangeEvent",
  "event": {
    "application": {
      "@type": "Application",
      "@baseType": "LogicalResource",
      "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
      "description": "Evolve Quest awarded mobile application."
    }
  },
```


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```
  "@baseType": "Event",
  "@type": "ApplicationAttributeValueChangeEvent"
}
```

Approval Status Change Event

Message example for an ApplicationApprovalStatusChangeEvent event

```
Content-Type: application/json
{
  "correlationId": "126-54f",
  "description": "Event for a application ApprovalStatus change.",
  "domain": "Open Gateway Onboarding",
  "eventId": "125",
  "eventTime": "2024-12-02T14:00:00.00Z",
  "eventType": "ApplicationApprovalStatusChangeEvent",
  "priority": "1",
  "timeOccurred": "2024-12-02T14:00:00.00Z",
  "title": "ApplicationApprovalStatusChangeEvent",
  "event": {
    "application": {
      "@type": "Application",
      "@baseType": "LogicalResource",
      "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
      "approvalStatus": "approved"
    }
  },
  "@baseType": "Event",
  "@type": "ApplicationApprovalStatusChangeEvent"
}
```

ApplicationOwner

Create Event

Message example for an ApplicationOwnerCreateEvent event

```
Content-Type: application/json
{
  "correlationId": "126-54f",
  "description": "Event for a applicationOwner creation.",
  "domain": "Open Gateway Onboarding",
  "eventId": "125",
  "eventTime": "2024-12-01T13:00:00.00Z",
  "eventType": "ApplicationOwnerCreateEvent",
  "priority": "1",
  "timeOccurred": "2024-12-01T13:00:00.00Z",
  "title": "ApplicationOwnerCreateEvent",
  "event": {
    "applicationOwner": {
      "@type": "ApplicationOwner",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
      "href": "https://mycsp.com/tmf-
```

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```
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
      "name": "Funtastic Games application owner role",
      "creationDate": "2024-12-01T13:00:00.00Z",

      "engagedParty": {
        "@type": "ApplicationOwnerOrganization",
        "name": "Funtastic Games SA",
        "tradingName": "Funtastic Games",
        "isLegalEntity": true,
        "organizationType": "dpv:ForProfitOrganisation",
        "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
        "taxNumber": "00.623.904/0001-73",
        "organizationIdentification": [
          {
            "@type": "OrganizationIdentification",
            "identificationType": "CNPJ",
            "identificationId": "00.623.904/0001-73",
            "issuingAuthority": "Receita Federal"
          }
        ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
        "registeredGeographicAddress": {
          "@type": "LightGeographicAddress",
          "streetNr": "123",
          "streetName": "Avenida Paulista",
          "locality": "Paraiso",
          "city": "Sao Paulo",
          "stateOrProvince": "Sao Paulo",
          "countryCode": {
            "@type": "ISO31661Alpha2StandardIdentifier",
            "value": "BR"
          },
          "postcode": "01311-000",
          "geographicSubAddress": {
            "@type": "LightGeographicSubAddress",
            "buildingName": "Fogo",
            "levelNumber": "Planta 4"
          }
        },
        "contactMedium": [
          {
            "@type": "EmailContactMedium",
            "@baseType": "ContactMedium",
            "id": "1",
            "preferred": true,
            "emailAddress": "contact@funtasticgames.com"
          }
        ],
        "legalRepresentative": {
          "@type": "ApplicationOwnerRelatedIndividual",
          "familyName": "OConnor",
          "givenName": "Jane",
          "contactMedium": [
            {
              "@type": "EmailContactMedium",
```


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```
            "@baseType": "ContactMedium",
            "id": "1",
            "preferred": true,
            "emailAddress": "jane@funtasticgames.com"
          }
        ]
      },
      "dataProtectionOfficer": {
        "@type": "ApplicationOwnerRelatedOrganization",
        "name": "DPO Office",
        "organizationType": "dpv:OrganisationalUnit",
        "contactMedium": [
          {
            "@type": "EmailContactMedium",
            "@baseType": "ContactMedium",
            "id": "1",
            "preferred": true,
            "emailAddress": "dpo@funtasticgames.com"
          }
        ]
      },
      "privacyManager": {
        "@type": "ApplicationOwnerRelatedOrganization",
        "name": "Privacy Office",
        "organizationType": "dpv:OrganisationalUnit",
        "contactMedium": [
          {
            "@type": "EmailContactMedium",
            "@baseType": "ContactMedium",
            "id": "1",
            "preferred": true,
            "emailAddress": "privacy@funtasticgames.com"
          }
        ]
      },
      "localRepresentative": [
        {
          "@type": "ApplicationOwnerRelatedLocalRepresentative",
          "countryCode": {
            "@type": "ISO31661Alpha2StandardIdentifier",
            "value": "EU"
          },
          "representative": {
            "@type": "ApplicationOwnerRelatedIndividual",
            "familyName": "Juan",
            "givenName": "Garcia",
            "contactMedium": [
              {
                "@type": "EmailContactMedium",
                "@baseType": "ContactMedium",
                "id": "jgarcia",
                "preferred": true,
                "emailAddress": "jgarcia@funtasticgames.com"
              }
            ]
          }
        }
      ]
    },
    "status": "inactive",
    "approvalStatus": "pendingApproval",
    "channelPartner": {
      "@type": "PartyRoleRef",
```


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```
        "id": "15",
        "@referredType": "PartyRole"
      }
    }
  },
  "@baseType": "Event",
  "@type": "ApplicationOwnerCreateEvent"
}
```

Attribute Value Change Event

Message example for an ApplicationOwnerAttributeValueChangeEvent event

```
Content-Type: application/json
{
  "correlationId": "126-54f",
  "description": "Event for a applicationOwner attribute value change.",
  "domain": "Open Gateway Onboarding",
  "eventId": "125",
  "eventTime": "2024-12-01T15:00:00.00Z",
  "eventType": "ApplicationOwnerAttributeValueChangeEvent",
  "priority": "1",
  "timeOccurred": "2024-12-01T15:00:00.00Z",
  "title": "ApplicationOwnerAttributeValueChangeEvent",
  "event": {
    "applicationOwner": {
      "@type": "ApplicationOwner",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
      "description": "Application owner role in the GSMA Open Gateway context."
    }
  },
  "@baseType": "Event",
  "@type": "ApplicationOwnerAttributeValueChangeEvent"
}
```

Approval Status Change Event

Message example for an ApplicationOwnerApprovalStatusChangeEvent event

```
Content-Type: application/json
{
  "correlationId": "126-54f",
  "description": "Event for a applicationOwner approvalStatus change.",
  "domain": "Open Gateway Onboarding",
  "eventId": "125",
  "eventTime": "2024-12-01T13:10:00.00Z",
  "eventType": "ApplicationOwnerApprovalStatusChangeEvent",
  "priority": "1",
  "timeOccurred": "2024-12-01T13:10:00.00Z",
  "title": "ApplicationOwnerApprovalStatusChangeEvent",
  "event": {
    "applicationOwner": {
      "@type": "ApplicationOwner",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
```

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```
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
      "approvalStatus": "approved"
    }
  },
  "@baseType": "Event",
  "@type": "ApplicationOwnerApprovalStatusChangeEvent"
}
```


Monitor

State Change Event

Message example for an MonitorStateChangeEvent event


```
Content-Type: application/json
{
  "correlationId": "987-46i",
  "description": "Event for a monitor state change.",
  "domain": "Open Gateway Onboarding",
  "eventId": "987",
  "eventTime": "2024-12-02T14:00:00.00Z",
  "eventType": "MonitorStateChangeEvent",
  "priority": "1",
  "timeOccurred": "2024-12-02T14:00:00.00Z",
  "title": "MonitorStateChangeEvent",
  "event": {
    "monitor": {
      "@type": "Monitor",
      "id": "77795e7f-5213-4b9c-b672-6838a6b76152",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/monitor/77795e7f-5213-4b9c-b672-
6838a6b76152",
      "state": "Completed"
    }
  },
  "@baseType": "Event",
  "@type": "MonitorStateChangeEvent"
}
```


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API OPERATIONS

Note: Many of the resource and payload examples include sample values for Attachment.content. These sample values are not syntactically correct, since any realistic value for content would be far too large and would obscure the readability of the examples.

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Monitor pattern, as described in TMF630 REST Design Guidelines, SHALL be used when supporting Async requests. A Monitor resource is created for each async operation. For example, in a resource creation operation HTTP POST, the returned 202 Accept response shall have a Link header referencing the monitor created to check the operation status. The consumer can check regularly using the Link in order to check the status of the operation.

```
Link: <https://mycsp.com/tmf-api/productOrderingManagement/v5/monitor/77795e7f-5213-
4b9c-b672-6838a6b76152>;rel=related;title=monitor,
```

(where the 77795e7f-5213-4b9c-b672-6838a6b76152 is the monitor resource identifier assigned for this resource)

Notifications are also described in a subsequent section.

Operations on ApiProductOrder

List or find ApiProductOrder objects

`GET /apiProductOrder?fields=…`

Description

This operation list ApiProductOrder entities. Attribute selection is enabled for all first level

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attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of API Product Order

Request

```
GET /apiProductOrder
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "ApiProductOrder",
    "id": "86387671-0997-461b-99c7-d624ffb0640c",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
    "state": "completed",
    "creationDate": "2025-02-02T12:00:00.00Z",
    "completionDate": "2025-02-02T12:01:00.00Z",
    "externalId": [
      {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerCustomerOrder",
        "id": "order_2232342",
        "owner": "ChannelPartner"
      }
    ],
    "description": "Camara API order for an Application Owner",
    "productOrderItem": [
      {
        "@type": "ApiProductOrderItemAdd",
        "@baseType": "DCSProductOrderItem",
        "id": "1",
        "state": "completed",
        "productOffering": {
          "@type": "ProductOfferingRef",
          "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
          "@referredType": "OpenGatewayProductOffering"
        },
        "product": {
          "@type": "ProductRef",
          "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
          "@referredType": "ApiProduct"
        },
        "productAction": {
          "@type": "ApiProductActionAddDeviceLocationVerification",
          "@baseType": "ApiProductAction",
          "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerification_example.schema.jso
```

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```
n",
          "purpose": "dpv:FraudPreventionAndDetection",
          "bestAccuracy": 500,
          "targetApplication": {
            "@type": "ApplicationRef",
            "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
            "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
            "@referredType": "Application"
          }
        },
        "externalIdentifier": {
          "@type": "ExternalIdentifier",
              "externalIdentifierType": "ApplicationOwnerProduct",
              "id": "device_location_1234",
          "owner": "ChannelPartner"
        },
        "approvedProductOfferingTermOrCondition": [
          {
            "@type": "ProductOfferingTermOrCondition",

            "productOfferingTermOrConditionSpecRef": {
              "@type": "ProductOfferingTermOrConditionSpecRef",
              "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

            },
            "productOfferingTermOrConditionApproval": {
              "approvalDate": "2023-11-20T00:00:00.000Z"
            },
            "applicationLegalBasisForInputData": "dpv:Consent",
            "process": [
              {
                "@type": "DpvProcess",
                "processingOperation": [
                  "dpv:Assess"
                ],
                "context": {
                  "@type": "DpvContext",
                  "processingContext": [
                    "dpv:AutomatedDecisionMaking",
                    "dpv:AutomatedScoringOfIndividuals"
                  ],
                  "duration": "dpv:FixedOccurrencesDuration",
                  "frequency": "dpv:SingularFrequency"
                },
                "dataExportTransitCountry": [
                  {
                    "@type": "DataExportCountry",
                    "countryCode": {
                      "@type": "ISO31661Alpha2StandardIdentifier",
                      "value": "US"
                    },
                    "commercialOrganizationName": "Globy Corporation",
                    "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                  }
                ]
              },
              {
                "@type": "DpvProcess",
```


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```
                "processingOperation": [
                  "dpv:Profiling"
                ],
                "context": {
                  "@type": "DpvContext",
                  "processingContext": [
                    "dpv:AutomatedScoringOfIndividuals"
                  ],
                  "duration": "dpv:EndlessDuration",
                  "frequency": "dpv:SingularFrequency"
                },
                "dataExportTransitCountry": [
                  {
                    "@type": "DataExportCountry",
                    "countryCode": {
                      "@type": "ISO31661Alpha2StandardIdentifier",
                      "value": "US"
                    },
                    "commercialOrganizationName": "Globy Corporation",
                    "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                  }
                ],
                "dataExportStorageCountry": [
                  {
                    "@type": "DataExportCountry",
                    "countryCode": {
                      "@type": "ISO31661Alpha2StandardIdentifier",
                      "value": "US"
                    },
                    "commercialOrganizationName": "Globy Corporation",
                    "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                  }
                ]
              }
            ],
            "userConsentScreenshot": {
              "@type": "SimpleAttachment",
              "content": "JVBERi...JSVFT0Y=",
              "mimeType": "application/pdf"
            }
          }
        ]
      }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
          "@referredType": "PartyRole"
    }
  },
  {
    "@type": "ApiProductOrder",
    "id": "d86065be-f2fd-490b-bb6a-dca24cc7f65a",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/d86065be-f2fd-
490b-bb6a-dca24cc7f65a",
    "state": "completed",
    "creationDate": "2025-03-02T12:00:00.00Z",
    "completionDate": "2025-03-02T12:01:00.00Z",
    "externalId": [
      {
```


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```
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerCustomerOrder",
        "id": "order_2232342",
        "owner": "ChannelPartner"
      }
    ],
    "description": "Camara API Unsubscription for an Application Owner",
    "productOrderItem": [
      {
        "@type": "ProductOrderItemDelete",
        "@baseType": "DCSProductOrderItem",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductOrderItemDelete.schema.json",
        "id": "1",
        "state": "completed",
        "product": {
          "@type": "ProductRef",
          "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
          "@referredType": "ApiProduct"
        }
      }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
          "@referredType": "PartyRole"
    }
  },
  {
  "@type": "ApiProductOrder",
  "id": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
  "state": "completed",
  "creationDate": "2025-01-01T10:00:00.00Z",
  "completionDate": "2025-01-01T10:01:00.00Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_6453",
      "owner": "ChannelPartner"
    }
  ],
  "productOrderItem": [
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "state": "completed",
      "id": "1",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a",
        "@referredType": "OpenGatewayProductOffering"
      },
      "product": {
        "@type": "ProductRef",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/33944716-d86f-4122-
```


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```
827b-307d74e6ca50",
        "id": "33944716-d86f-4122-827b-307d74e6ca50",
        "@referredType": "ApiProduct"
      },
      "productAction": {
        "@type": "ApiProductActionAdd",
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ],
      "productOrderItemRelationship": [
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "2",
         "relationshipType": "bundles"
        },
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "3",
         "relationshipType": "bundles"
        }
      ]
    },
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "state": "completed",
      "id": "2",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
        "@referredType": "OpenGatewayProductOffering"
      },
      "product": {
        "@type": "ProductRef",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/3be31371-2755-43f7-
```


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```
a6f6-380234910b9b",
        "id": "3be31371-2755-43f7-a6f6-380234910b9b",
        "@referredType": "ApiProduct"
      },
      "productAction": {
        "@type": "ApiProductActionAddDeviceLocationVerificationBundled",
        "@baseType": "ApiProductActionAdd",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerificationBundled.schema.json
",
        "purpose": "dpv:FraudPreventionAndDetection",
        "bestAccuracy": 500,
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
```


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```
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
      ]
    },
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "state": "completed",
      "id": "3",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
              "@referredType": "OpenGatewayProductOffering"
```


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```
      },
      "product": {
        "@type": "ProductRef",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/fea0c794-d62d-40bb-
a9a4-934336194dca",
        "id": "fea0c794-d62d-40bb-a9a4-934336194dca",
        "@referredType": "ApiProduct"
      },
      "productAction": {
        "@type": "ApiProductActionAdd",
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ]
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
]
```


Retrieve a list of API Product Order with field selection and filter

Request

```
GET /apiProductOrder?filter=state=completed&fields=state
Content-Type: application/json
```

Response

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```
200
[
  {
    "@type": "ApiProductOrder",
    "id": "86387671-0997-461b-99c7-d624ffb0640c",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
    "state": "completed"
  },
  {
    "@type": "ApiProductOrder",
    "id": "d86065be-f2fd-490b-bb6a-dca24cc7f65a",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/d86065be-f2fd-
490b-bb6a-dca24cc7f65a",
    "state": "completed"
  },
  {
  "@type": "ApiProductOrder",
  "id": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
  "state": "completed"
  }
]
```

Retrieves a ApiProductOrder by ID

`GET /apiProductOrder/{id}?fields=…`

Description

This operation retrieves an ApiProductOrder entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve an API Product Order with add action from its identifier

Request

```
GET /apiProductOrder/86387671-0997-461b-99c7-d624ffb0640c
Content-Type: application/json
```

Response

```
200
{
  "@type": "ApiProductOrder",
  "id": "86387671-0997-461b-99c7-d624ffb0640c",
  "href": "https://mycsp.com/tmf-
```

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```
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
  "state": "completed",
  "creationDate": "2025-02-02T12:00:00.00Z",
  "completionDate": "2025-02-02T12:01:00.00Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_2232342",
      "owner": "ChannelPartner"
    }
  ],
  "description": "Camara API order for an Application Owner",
  "productOrderItem": [
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductOrderItemAdd.schema.json",
      "id": "1",
      "state": "completed",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
        "@referredType": "OpenGatewayProductOffering"
      },
      "product": {
        "@type": "ProductRef",
        "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
        "@referredType": "ApiProduct"
      },
      "productAction": {
        "@type": "ApiProductActionAddDeviceLocationVerification",
        "@baseType": "ApiProductAction",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerification_example.schema.jso
n",
        "purpose": "dpv:FraudPreventionAndDetection",
        "bestAccuracy": 500,
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_1234",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
```


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```
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-10-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
```


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```
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
      ]
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```

Retrieve an API Product Order with delete action from its identifier

Request

```
GET /apiProductOrder/d86065be-f2fd-490b-bb6a-dca24cc7f65a
Content-Type: application/json
```

Response

```
200
{
  "@type": "ApiProductOrder",
  "id": "d86065be-f2fd-490b-bb6a-dca24cc7f65a",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/d86065be-f2fd-
490b-bb6a-dca24cc7f65a",
  "state": "completed",
  "creationDate": "2025-03-02T12:00:00.00Z",
  "completionDate": "2025-03-02T12:01:00.00Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_2232342",
      "owner": "ChannelPartner"
    }
  ],
```

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```
  "description": "Camara API Unsubscription for an Application Owner",
  "productOrderItem": [
    {
      "@type": "ProductOrderItemDelete",
      "@baseType": "DCSProductOrderItem",
      "id": "1",
      "state": "completed",
      "product": {
        "@type": "ProductRef",
        "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
        "@referredType": "ApiProduct"
      }
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```

Retrieve an API Product Order with add action for a bundle ProductOffering from its identifier

Request

```
GET /apiProductOrder/c5c005c5-34f9-4e7d-99df-8a30da246f8b
Content-Type: application/json
```

Response

```
200
{
  "@type": "ApiProductOrder",
  "id": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
  "state": "completed",
  "creationDate": "2025-01-01T10:00:00.00Z",
  "completionDate": "2025-01-01T10:01:00.00Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_6453",
      "owner": "ChannelPartner"
    }
  ],
  "productOrderItem": [
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "state": "completed",
      "id": "1",
```

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```
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a",
        "@referredType": "OpenGatewayProductOffering"
      },
      "product": {
        "@type": "ProductRef",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/33944716-d86f-4122-
827b-307d74e6ca50",
        "id": "33944716-d86f-4122-827b-307d74e6ca50",
        "@referredType": "ApiProduct"
      },
      "productAction": {
        "@type": "ApiProductActionAdd",
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ],
      "productOrderItemRelationship": [
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "2",
         "relationshipType": "bundles"
        },
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "3",
         "relationshipType": "bundles"
        }
      ]
    },
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "state": "completed",
      "id": "2",
```


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```
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
        "@referredType": "OpenGatewayProductOffering"
      },
      "product": {
        "@type": "ProductRef",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/3be31371-2755-43f7-
a6f6-380234910b9b",
        "id": "3be31371-2755-43f7-a6f6-380234910b9b",
        "@referredType": "ApiProduct"
      },
      "productAction": {
        "@type": "ApiProductActionAddDeviceLocationVerificationBundled",
        "@baseType": "ApiProductActionAdd",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerificationBundled.schema.json
",
        "purpose": "dpv:FraudPreventionAndDetection",
        "bestAccuracy": 500,
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
                ],
```


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```
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
      ]
    },
```


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```
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "state": "completed",
      "id": "3",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
        "@referredType": "OpenGatewayProductOffering"
      },
      "product": {
        "@type": "ProductRef",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/fea0c794-d62d-40bb-
a9a4-934336194dca",
        "id": "fea0c794-d62d-40bb-a9a4-934336194dca",
        "@referredType": "ApiProduct"
      },
      "productAction": {
        "@type": "ApiProductActionAdd",
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ],
      "productOrderItemRelationship": [
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "2",
         "relationshipType": "requires"
        }
      ]
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
```


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```
  }
}
```

Creates a ApiProductOrder

`POST /apiProductOrder?fields=…`

Description

This operation creates an ApiProductOrder entity.

Mandatory Attributes

Mandatory Attributes	Rule
productOrderItem	
@type	M

Usage samples

Create an API Product Order for ordering (subscribe) a new ProductOffering

Request

```
POST /apiProductOrder
Content-Type: application/json
{
  "@type": "ApiProductOrder",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_2232342",
      "owner": "ChannelPartner"
    }
  ],
  "description": "Camara API order for an Application Owner",
  "productOrderItem": [
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "id": "1",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
        "name": "Location Verification Antifraud CAMARA API Standalone",
              "@referredType": "OpenGatewayProductOffering"
      },
      "productAction": {
        "@type": "ApiProductActionAddDeviceLocationVerification",
        "@baseType": "ApiProductActionAdd",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerification.schema.json",
        "purpose": "dpv:FraudPreventionAndDetection",
        "bestAccuracy": 500,
        "targetApplication": {
```

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```
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_1234",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
            "name": "device_location_verification_terms",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-10-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
```


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```
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
      ]
    }
  ]
}
```


Response


```
201
{
  "@type": "ApiProductOrder",
  "id": "86387671-0997-461b-99c7-d624ffb0640c",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
  "state": "acknowledged",
  "creationDate": "2025-02-02T12:00:00.00Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_2232342",
      "owner": "ChannelPartner"
    }
```


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```
  ],
  "description": "Camara API order for an Application Owner",
  "productOrderItem": [
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "id": "1",
      "state": "completed",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
        "name": "Location Verification Antifraud CAMARA API Standalone",
              "@referredType": "OpenGatewayProductOffering"
      },
      "productAction": {
        "@type": "ApiProductActionAddDeviceLocationVerification",
        "@baseType": "ApiProductActionAdd",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerification.schema.json",
        "purpose": "dpv:FraudPreventionAndDetection",
        "bestAccuracy": 500,
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
              "externalIdentifierType": "ApplicationOwnerProduct",
              "id": "device_location_1234",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
            "name": "device_location_verification_terms",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-10-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
```


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```
                ],
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
      ]
```


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```
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```

Create an API Product Order for ordering (unsubscribe) the removal of an existing Product

Request

```
POST /apiProductOrder
Content-Type: application/json
{
  "@type": "ApiProductOrder",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_2232342",
      "owner": "ChannelPartner"
    }
  ],
  "description": "Camara API Unsubscription for an Application Owner",
  "productOrderItem": [
    {
      "@type": "ProductOrderItemDelete",
      "@baseType": "DCSProductOrderItem",
      "id": "1",
      "product": {
        "@type": "ProductRef",
        "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
        "@referredType": "ApiProduct"
      }
    }
  ]
}
```

Response

```
201
{
  "@type": "ApiProductOrder",
  "id": "d86065be-f2fd-490b-bb6a-dca24cc7f65a",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/d86065be-f2fd-
490b-bb6a-dca24cc7f65a",
  "state": "acknowledged",
  "creationDate": "2025-03-02T12:00:00.00Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_2232342",
      "owner": "ChannelPartner"
```

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```
    }
  ],
  "description": "Camara API Unsubscription for an Application Owner",
  "productOrderItem": [
    {
      "@type": "ProductOrderItemDelete",
      "@baseType": "DCSProductOrderItem",
      "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ProductOrderItemDelete.schema.json",
      "id": "1",
      "state": "acknowledged",
      "product": {
        "@type": "ProductRef",
        "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
        "@referredType": "ApiProduct"
      }
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```

Create an API Product Order for ordering (subscribe) a new ProductOffering bundle

Request

```
POST /apiProductOrder
Content-Type: application/json
{
  "@type": "ApiProductOrder",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_6453",
      "owner": "ChannelPartner"
    }
  ],
  "productOrderItem": [
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "id": "1",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a",
        "@referredType": "OpenGatewayProductOffering"
      },
      "productAction": {
        "@type": "ApiProductActionAdd",
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "@referredType": "Application"
```

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```
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ],
      "productOrderItemRelationship": [
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "2",
         "relationshipType": "bundles"
        },
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "3",
         "relationshipType": "bundles"
        }
      ]
    },
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "id": "2",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
              "@referredType": "OpenGatewayProductOffering"
      },
      "productAction": {
        "@type": "ApiProductActionAddDeviceLocationVerificationBundled",
        "@baseType": "ApiProductActionAdd",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerificationBundled.schema.json
",
        "purpose": "dpv:FraudPreventionAndDetection",
        "bestAccuracy": 500,
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
```


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```
          "owner": "ChannelPartner"
        },
        "approvedProductOfferingTermOrCondition": [
          {
            "@type": "ProductOfferingTermOrCondition",

            "productOfferingTermOrConditionSpecRef": {
              "@type": "ProductOfferingTermOrConditionSpecRef",
              "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce_terms_1",
                   "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

            },
            "productOfferingTermOrConditionApproval": {
              "approvalDate": "2023-11-20T00:00:00.000Z"
            },
            "applicationLegalBasisForInputData": "dpv:Consent",
            "process": [
              {
                "@type": "DpvProcess",
                "processingOperation": [
                  "dpv:Assess"
                ],
                "context": {
                  "@type": "DpvContext",
                  "processingContext": [
                    "dpv:AutomatedDecisionMaking",
                    "dpv:AutomatedScoringOfIndividuals"
                  ],
                  "duration": "dpv:FixedOccurrencesDuration",
                  "frequency": "dpv:SingularFrequency"
                },
                "dataExportTransitCountry": [
                  {
                    "@type": "DataExportCountry",
                    "countryCode": {
                      "@type": "ISO31661Alpha2StandardIdentifier",
                      "value": "US"
                    },
                    "commercialOrganizationName": "Globy Corporation",
                    "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                  }
                ]
              },
              {
                "@type": "DpvProcess",
                "processingOperation": [
                  "dpv:Profiling"
                ],
                "context": {
                  "@type": "DpvContext",
                  "processingContext": [
                    "dpv:AutomatedScoringOfIndividuals"
                  ],
                  "duration": "dpv:EndlessDuration",
                  "frequency": "dpv:SingularFrequency"
                },
                "dataExportTransitCountry": [
                  {
                    "@type": "DataExportCountry",
                    "countryCode": {
                      "@type": "ISO31661Alpha2StandardIdentifier",
```


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```
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
      ]
    },
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "id": "3",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
        "@referredType": "OpenGatewayProductOffering"
      },
      "productAction": {
        "@type": "ApiProductActionAdd",
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
```


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```
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ]
    }
  ]
}
```

Response

```
201
{
  "@type": "ApiProductOrder",
  "id": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
  "state": "acknowledged",
  "creationDate": "2025-01-01T10:00:00.00Z",
  "externalId": [
    {
      "@type": "ExternalIdentifier",
      "externalIdentifierType": "ApplicationOwnerCustomerOrder",
      "id": "order_6453",
      "owner": "ChannelPartner"
    }
  ],
  "productOrderItem": [
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "state": "acknowledged",
      "id": "1",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a",
        "@referredType": "OpenGatewayProductOffering"
      },
      "productAction": {
        "@type": "ApiProductActionAdd",
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
```

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```
            "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ],
      "productOrderItemRelationship": [
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "2",
         "relationshipType": "bundles"
        },
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "3",
         "relationshipType": "bundles"
        }
      ]
    },
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "state": "acknowledged",
      "id": "2",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
             "@referredType": "OpenGatewayProductOffering"
      },
      "productAction": {
        "@type": "ApiProductActionAddDeviceLocationVerificationBundled",
        "@baseType": "ApiProductActionAdd",
        "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Customer/ApiProductActionAddDeviceLocationVerificationBundled.schema.json
",
        "purpose": "dpv:FraudPreventionAndDetection",
        "bestAccuracy": 500,
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce_terms_1",
```


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```
                "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
```


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```
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
      ]
    },
    {
      "@type": "ApiProductOrderItemAdd",
      "@baseType": "DCSProductOrderItem",
      "state": "acknowledged",
      "id": "3",
      "productOffering": {
        "@type": "ProductOfferingRef",
        "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
              "@referredType": "OpenGatewayProductOffering"
      },
      "productAction": {
        "@type": "ApiProductActionAdd",
        "targetApplication": {
          "@type": "ApplicationRef",
          "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
          "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
          "@referredType": "Application"
        }
      },
      "externalIdentifier": {
        "@type": "ExternalIdentifier",
        "externalIdentifierType": "ApplicationOwnerProduct",
        "id": "device_location_bundle_0987",
        "owner": "ChannelPartner"
      },
      "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d_terms_1",
                 "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ],
      "productOrderItemRelationship": [
```


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```
        {
         "@type": "OpenGatewayOrderItemRelationship",
         "id": "2",
         "relationshipType": "requires"
        }
      ]
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```

Operations on ApiProduct

Retrieves a ApiProduct by ID

`GET /apiProduct/{id}?fields=…`

Description

This operation retrieves an ApiProduct entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve an API Product, location verification standalone

Request

```
GET /apiProduct/8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb
Content-Type: application/json
```

Response

```
200
{
  "@type": "ApiProductDeviceLocationVerification",
  "@baseType": "ApiProduct",
  "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Product/ApiProductDeviceLocationVerification.schema.json",
  "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
  "name": "device_location_verification",
  "description": "Open Gateway Device Location Verify",
  "creationDate": "2025-02-02T12:02:00.00Z",
  "startDate": "2025-02-02T12:00:03.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
```

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```
    "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
    "name": "Location Verification Antifraud CAMARA API Standalone",
    "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
    {
      "@type": "RelatedOrderItem",
      "orderId": "86387671-0997-461b-99c7-d624ffb0640c",
        "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
      "orderItemId": "1",
      "orderItemAction": "add"
    }
  ],
  "productSpecification": {
    "@type": "ProductSpecificationRef",
    "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
9543-0c5e9070ade9",
    "name": "Location Verification Antifraud CAMARA API",
    "version": "1.0.0",
    "@referredType": "ApiProductSpecification"
  },
  "apiName": "location-verification",
  "apiVersionDetails": [
    {
      "@type": "ApiVersionDetails",
      "apiMajorVersion": 0,
      "basePath": "location-verification",
      "apiGrantInformation": [
        {
          "@type": "ApiGrantInformation",
          "purpose": "dpv:FraudPreventionAndDetection",
          "scope": [
            "location-verification:verify"
          ],
          "grantType": [
            "urn:openid:params:grant-type:ciba",
            "authorization_code"
          ],
          "legalBasis": "dpv:Consent"
        },
        {
          "@type": "ApiGrantInformation",
          "scope": [
            "location-verification:read"
          ],
          "grantType": [
            "client_credentials"
          ]
        }
      ]
    }
  ],
  "purpose": "dpv:FraudPreventionAndDetection",
  "bestAccuracy": 500,
"approvedProductOfferingTermOrCondition": [
  {
    "@type": "ProductOfferingTermOrCondition",

    "productOfferingTermOrConditionSpecRef": {
```


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```
      "@type": "ProductOfferingTermOrConditionSpecRef",
      "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
      "name": "device_location_verification_terms",
      "@referredType": "OpenGatewayProductOfferingTermOrConditionSpecification"

    },
    "productOfferingTermOrConditionApproval": {
      "approvalDate": "2023-11-20T00:00:00.000Z"
    },
    "applicationLegalBasisForInputData": "dpv:Consent",
    "process": [
      {
        "@type": "DpvProcess",
        "processingOperation": [
          "dpv:Assess"
        ],
        "context": {
          "@type": "DpvContext",
          "processingContext": [
            "dpv:AutomatedDecisionMaking",
            "dpv:AutomatedScoringOfIndividuals"
          ],
          "duration": "dpv:FixedOccurrencesDuration",
          "frequency": "dpv:SingularFrequency"
        },
        "dataExportTransitCountry": [
          {
            "@type": "DataExportCountry",
            "countryCode": {
              "@type": "ISO31661Alpha2StandardIdentifier",
              "value": "US"
            },
            "commercialOrganizationName": "Globy Corporation",
            "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
          }
        ]
      },
      {
        "@type": "DpvProcess",
        "processingOperation": [
          "dpv:Profiling"
        ],
        "context": {
          "@type": "DpvContext",
          "processingContext": [
            "dpv:AutomatedScoringOfIndividuals"
          ],
          "duration": "dpv:EndlessDuration",
          "frequency": "dpv:SingularFrequency"
        },
        "dataExportTransitCountry": [
          {
            "@type": "DataExportCountry",
            "countryCode": {
              "@type": "ISO31661Alpha2StandardIdentifier",
              "value": "US"
            },
            "commercialOrganizationName": "Globy Corporation",
            "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
          }
        ],
```


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```
        "dataExportStorageCountry": [
          {
            "@type": "DataExportCountry",
            "countryCode": {
              "@type": "ISO31661Alpha2StandardIdentifier",
              "value": "US"
            },
            "commercialOrganizationName": "Globy Corporation",
            "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
          }
        ]
      }
    ],
    "userConsentScreenshot": {
      "@type": "SimpleAttachment",
      "content": "JVBERi...JSVFT0Y=",
      "mimeType": "application/pdf"
    }
  }
],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```

Retrieve an API Product, location verification bundle

Request

```
GET /apiProduct/33944716-d86f-4122-827b-307d74e6ca50
Content-Type: application/json
```

Response

```
200
{
  "@type": "ApiProduct",
  "id": "33944716-d86f-4122-827b-307d74e6ca50",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/33944716-d86f-4122-
827b-307d74e6ca50",
  "creationDate": "2025-01-01T10:02:00.00Z",
  "startDate": "2025-01-01T10:03:00.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/5e9b10e5-d707-4a16-89d1-
687c7f0bf81a",
    "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a",
    "name": "Open Gateway Location Verification bundle offer",
    "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
```

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```
    {
      "@type": "RelatedOrderItem",
      "orderId": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
      "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
      "orderItemId": "1",
      "orderItemAction": "add"
    }
  ],
  "productRelationship": [
    {
      "@type": "OpenGatewayProductRelationship",
      "id": "3be31371-2755-43f7-a6f6-380234910b9b",
      "relationshipType": "bundles"
    },
    {
      "@type": "OpenGatewayProductRelationship",
      "id": "fea0c794-d62d-40bb-a9a4-934336194dca",
      "relationshipType": "bundles"
    }
  ],
  "apiName": "",
  "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a_terms_1",
            "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```


Retrieve an API Product, location verification API bundled

Request

```
GET /apiProduct/3be31371-2755-43f7-a6f6-380234910b9b
Content-Type: application/json
```

Response

```
200
{
  "@type": "ApiProductLocationVerificationAntifraud",
```

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```
  "@baseType": "ApiProduct",
  "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/LocationVerificationAntifraudStandalone",
  "id": "3be31371-2755-43f7-a6f6-380234910b9b",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/3be31371-2755-43f7-
a6f6-380234910b9b",
  "creationDate": "2025-01-01T10:02:00.00Z",
  "startDate": "2025-01-01T10:03:00.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
    "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/ab7792f8-6628-4c4b-a557-
699adc26d4ce",
    "name": "Location Verification Antifraud CAMARA API Bundled",
    "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
    {
      "@type": "RelatedOrderItem",
      "orderId": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
      "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
      "orderItemId": "2",
      "orderItemAction": "add"
    }
  ],
  "productSpecification": {
    "@type": "ProductSpecificationRef",
    "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
9543-0c5e9070ade9",
    "name": "Location Verification Antifraud CAMARA API",
    "version": "1.0.0",
    "@referredType": "ApiProductSpecification"
  },
  "apiName": "location-verification",
  "apiVersionDetails": [
    {
      "@type": "ApiVersionDetails",
      "apiMajorVersion": 0,
      "basePath": "location-verification",
      "apiGrantInformation": [
        {
          "@type": "ApiGrantInformation",
          "purpose": "dpv:FraudPreventionAndDetection",
          "scope": [
            "location-verification:verify"
          ],
          "grantType": [
            "urn:openid:params:grant-type:ciba",
            "authorization_code"
          ],
          "legalBasis": "dpv:Consent"
        },
        {
          "@type": "ApiGrantInformation",
          "scope": [
```


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```
            "location-verification:read"
          ],
          "grantType": [
            "client_credentials"
          ]
        }
      ]
    }
  ],
  "purpose": "dpv:FraudPreventionAndDetection",
  "bestAccuracy": 500,
  "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce_terms_1",
            "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedScoringOfIndividuals"
```


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```
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
    ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```

Retrieve an API Product, location verification usage volume bundled

Request

```
GET /apiProduct/fea0c794-d62d-40bb-a9a4-934336194dca
Content-Type: application/json
```

Response

```
200
{
  "@type": "UsageVolumeProductLocationVerification",
  "@baseType": "ApiProduct",
```

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```
  "id": "fea0c794-d62d-40bb-a9a4-934336194dca",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/fea0c794-d62d-40bb-
a9a4-934336194dca",
  "creationDate": "2025-01-01T10:02:00.00Z",
  "startDate": "2025-01-01T10:03:00.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
    "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/0a2782b7-81a7-4af9-9e4f-
d51bc690ec2d",
    "name": "Location Verification Verify Usage Volume Recurring Bundled",
    "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
    {
      "@type": "RelatedOrderItem",
      "orderId": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
      "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
      "orderItemId": "3",
      "orderItemAction": "add"
    }
  ],
  "productSpecification": {
    "@type": "ProductSpecificationRef",
    "id": "a7b90fdb-00f7-498b-944e-88115d00626b",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/a7b90fdb-00f7-498b-
944e-88115d00626b",
    "name": "Location Verification Verify Usage Volume Recurring",
    "version": "1.0.0",
    "@referredType": "UsageVolumeProductSpecification"
  },
  "productRelationship": [
    {
      "@type": "OpenGatewayProductRelationship",
      "id": "3be31371-2755-43f7-a6f6-380234910b9b",
      "relationshipType": "appliesOn"
    }
  ],
  "apiName": "",
  "usageVolume": {
    "volumeUnit": "apiCall",
    "volumeInitialAmount": 10000
  },
  "isBlocked": false,
  "isRecurring": true,
  "validityDuration": {
    "amount": 1,
    "units": "month"
  },
  "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d_terms_1",
            "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"
```


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```
          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```

List or find ApiProduct objects

`GET /apiProduct?fields=…`

Description

This operation list ApiProduct entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of API Product

Request

```
GET /apiProduct
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "ApiProductDeviceLocationVerification",
    "@baseType": "ApiProduct",
    "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Product/ApiProductDeviceLocationVerification.schema.json",
    "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
    "name": "device_location_verification",
    "description": "Open Gateway Device Location Verify",
    "creationDate": "2025-02-02T12:02:00.00Z",
    "startDate": "2025-02-02T12:00:03.00Z",
    "status": "active",
    "productOffering": {
      "@type": "ProductOfferingRef",
      "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb",
      "name": "Location Verification Antifraud CAMARA API Standalone",
```

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```
      "@referredType": "OpenGatewayProductOffering"
    },
    "productOrderItem": [
      {
        "@type": "RelatedOrderItem",
        "orderId": "86387671-0997-461b-99c7-d624ffb0640c",
        "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/86387671-0997-
461b-99c7-d624ffb0640c",
        "orderItemId": "1",
        "orderItemAction": "add"
      }
    ],
    "productSpecification": {
      "@type": "ProductSpecificationRef",
      "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
      "@referredType": "ApiProductSpecification"
    },
    "apiName": "location-verification",
    "apiVersionDetails": [
      {
        "@type": "ApiVersionDetails",
        "apiMajorVersion": 0,
        "basePath": "location-verification",
        "apiGrantInformation": [
          {
            "@type": "ApiGrantInformation",
            "purpose": "dpv:FraudPreventionAndDetection",
            "scope": [
              "location-verification:verify"
            ],
            "grantType": [
              "urn:openid:params:grant-type:ciba",
              "authorization_code"
            ],
            "legalBasis": "dpv:Consent"
          },
          {
            "@type": "ApiGrantInformation",
            "scope": [
              "location-verification:read"
            ],
            "grantType": [
              "client_credentials"
            ]
          }
        ]
      }
    ],
    "purpose": "dpv:FraudPreventionAndDetection",
    "bestAccuracy": 500,
  "approvedProductOfferingTermOrCondition": [
    {
      "@type": "ProductOfferingTermOrCondition",

      "productOfferingTermOrConditionSpecRef": {
        "@type": "ProductOfferingTermOrConditionSpecRef",
        "id": "2d4ef4d3-08ce-441d-ac76-133b6dad0ccb_terms_1",
        "name": "device_location_verification_terms",
        "@referredType": "OpenGatewayProductOfferingTermOrConditionSpecification"

      },
      "productOfferingTermOrConditionApproval": {
```


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```
          "approvalDate": "2023-11-20T00:00:00.000Z"
        },
        "applicationLegalBasisForInputData": "dpv:Consent",
        "process": [
          {
            "@type": "DpvProcess",
            "processingOperation": [
              "dpv:Assess"
            ],
            "context": {
              "@type": "DpvContext",
              "processingContext": [
                "dpv:AutomatedDecisionMaking",
                "dpv:AutomatedScoringOfIndividuals"
              ],
              "duration": "dpv:FixedOccurrencesDuration",
              "frequency": "dpv:SingularFrequency"
            },
            "dataExportTransitCountry": [
              {
                "@type": "DataExportCountry",
                "countryCode": {
                  "@type": "ISO31661Alpha2StandardIdentifier",
                  "value": "US"
                },
                "commercialOrganizationName": "Globy Corporation",
                "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
              }
            ]
          },
          {
            "@type": "DpvProcess",
            "processingOperation": [
              "dpv:Profiling"
            ],
            "context": {
              "@type": "DpvContext",
              "processingContext": [
                "dpv:AutomatedScoringOfIndividuals"
              ],
              "duration": "dpv:EndlessDuration",
              "frequency": "dpv:SingularFrequency"
            },
            "dataExportTransitCountry": [
              {
                "@type": "DataExportCountry",
                "countryCode": {
                  "@type": "ISO31661Alpha2StandardIdentifier",
                  "value": "US"
                },
                "commercialOrganizationName": "Globy Corporation",
                "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
              }
            ],
            "dataExportStorageCountry": [
              {
                "@type": "DataExportCountry",
                "countryCode": {
                  "@type": "ISO31661Alpha2StandardIdentifier",
                  "value": "US"
                },
```


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```
              "commercialOrganizationName": "Globy Corporation",
              "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
            }
          ]
        }
      ],
      "userConsentScreenshot": {
        "@type": "SimpleAttachment",
        "content": "JVBERi...JSVFT0Y=",
        "mimeType": "application/pdf"
      }
    }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
      "@referredType": "PartyRole"
    }
  },
{
  "@type": "ApiProduct",
  "id": "33944716-d86f-4122-827b-307d74e6ca50",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/33944716-d86f-4122-
827b-307d74e6ca50",
  "creationDate": "2025-01-01T10:02:00.00Z",
  "startDate": "2025-01-01T10:03:00.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/5e9b10e5-d707-4a16-89d1-
687c7f0bf81a",
    "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a",
    "name": "Open Gateway Location Verification bundle offer",
    "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
    {
      "@type": "RelatedOrderItem",
      "orderId": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
      "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
      "orderItemId": "1",
      "orderItemAction": "add"
    }
  ],
  "productRelationship": [
    {
      "@type": "OpenGatewayProductRelationship",
      "id": "3be31371-2755-43f7-a6f6-380234910b9b",
      "relationshipType": "bundles"
    },
    {
      "@type": "OpenGatewayProductRelationship",
      "id": "fea0c794-d62d-40bb-a9a4-934336194dca",
      "relationshipType": "bundles"
    }
  ],
  "apiName": "",
  "approvedProductOfferingTermOrCondition": [
```


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```
        {
          "@type": "ProductOfferingTermOrCondition",
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "5e9b10e5-d707-4a16-89d1-687c7f0bf81a_terms_1",
            "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
      ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
},
{
  "@type": "ApiProductLocationVerificationAntifraud",
  "@baseType": "ApiProduct",
  "@schemaLocation": "https://mycsp.com/tmf-
api/schemas/Tmf/Product/DCS/GSMAOperateAPI/LocationVerificationAntifraudStandalone.s
chema.json#/definitions/LocationVerificationAntifraudStandalone",
  "id": "3be31371-2755-43f7-a6f6-380234910b9b",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/3be31371-2755-43f7-
a6f6-380234910b9b",
  "creationDate": "2025-01-01T10:02:00.00Z",
  "startDate": "2025-01-01T10:03:00.00Z",
  "status": "active",
  "productOffering": {
    "@type": "ProductOfferingRef",
    "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/ab7792f8-6628-4c4b-a557-
699adc26d4ce",
    "name": "Location Verification Antifraud CAMARA API Bundled",
    "@referredType": "OpenGatewayProductOffering"
  },
  "productOrderItem": [
    {
      "@type": "RelatedOrderItem",
      "orderId": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
      "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
      "orderItemId": "2",
      "orderItemAction": "add"
    }
  ],
  "productSpecification": {
    "@type": "ProductSpecificationRef",
    "id": "4b6591ef-5ede-4885-9543-0c5e9070ade9",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/4b6591ef-5ede-4885-
9543-0c5e9070ade9",
    "name": "Location Verification Antifraud CAMARA API",
    "version": "1.0.0",
    "@referredType": "ApiProductSpecification"
  },
```


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```
  "apiName": "location-verification",
  "apiVersionDetails": [
    {
      "@type": "ApiVersionDetails",
      "apiMajorVersion": 0,
      "basePath": "location-verification",
      "apiGrantInformation": [
        {
          "@type": "ApiGrantInformation",
          "purpose": "dpv:FraudPreventionAndDetection",
          "scope": [
            "location-verification:verify"
          ],
          "grantType": [
            "urn:openid:params:grant-type:ciba",
            "authorization_code"
          ],
          "legalBasis": "dpv:Consent"
        },
        {
          "@type": "ApiGrantInformation",
          "scope": [
            "location-verification:read"
          ],
          "grantType": [
            "client_credentials"
          ]
        }
      ]
    }
  ],
  "purpose": "dpv:FraudPreventionAndDetection",
  "bestAccuracy": 500,
  "approvedProductOfferingTermOrCondition": [
        {
          "@type": "ProductOfferingTermOrCondition",

          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "ab7792f8-6628-4c4b-a557-699adc26d4ce_terms_1",
            "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          },
          "applicationLegalBasisForInputData": "dpv:Consent",
          "process": [
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Assess"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedDecisionMaking",
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:FixedOccurrencesDuration",
                "frequency": "dpv:SingularFrequency"
              },
```


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```
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            },
            {
              "@type": "DpvProcess",
              "processingOperation": [
                "dpv:Profiling"
              ],
              "context": {
                "@type": "DpvContext",
                "processingContext": [
                  "dpv:AutomatedScoringOfIndividuals"
                ],
                "duration": "dpv:EndlessDuration",
                "frequency": "dpv:SingularFrequency"
              },
              "dataExportTransitCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ],
              "dataExportStorageCountry": [
                {
                  "@type": "DataExportCountry",
                  "countryCode": {
                    "@type": "ISO31661Alpha2StandardIdentifier",
                    "value": "US"
                  },
                  "commercialOrganizationName": "Globy Corporation",
                  "dataPrivacyFrameworkURL":
"https://www.dataprivacyframework.gov/participant/abcde"
                }
              ]
            }
          ],
          "userConsentScreenshot": {
            "@type": "SimpleAttachment",
            "content": "JVBERi...JSVFT0Y=",
            "mimeType": "application/pdf"
          }
        }
    ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
```


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```
    }
  },
  {
    "@type": "UsageVolumeProductLocationVerification",
    "@baseType": "ApiProduct",
    "id": "fea0c794-d62d-40bb-a9a4-934336194dca",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/fea0c794-d62d-40bb-
a9a4-934336194dca",
    "creationDate": "2025-01-01T10:02:00.00Z",
    "startDate": "2025-01-01T10:03:00.00Z",
    "status": "active",
    "productOffering": {
      "@type": "ProductOfferingRef",
      "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productOffering/0a2782b7-81a7-4af9-9e4f-
d51bc690ec2d",
      "name": "Location Verification Verify Usage Volume Recurring Bundled",
      "@referredType": "OpenGatewayProductOffering"
    },
    "productOrderItem": [
      {
        "@type": "RelatedOrderItem",
        "orderId": "c5c005c5-34f9-4e7d-99df-8a30da246f8b",
        "orderHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProductOrder/c5c005c5-34f9-
4e7d-99df-8a30da246f8b",
        "orderItemId": "3",
        "orderItemAction": "add"
      }
    ],
    "productSpecification": {
      "@type": "ProductSpecificationRef",
      "id": "a7b90fdb-00f7-498b-944e-88115d00626b",
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIProductCatalog/v5/productSpecification/a7b90fdb-00f7-498b-
944e-88115d00626b",
      "name": "Location Verification Verify Usage Volume Recurring",
      "version": "1.0.0",
      "@referredType": "UsageVolumeProductSpecification"
    },
    "productRelationship": [
      {
        "@type": "OpenGatewayProductRelationship",
        "id": "3be31371-2755-43f7-a6f6-380234910b9b",
        "relationshipType": "appliesOn"
      }
    ],
    "apiName": "",
    "usageVolume": {
      "volumeUnit": "apiCall",
      "volumeInitialAmount": 10000
    },
    "isBlocked": false,
    "isRecurring": true,
    "validityDuration": {
      "amount": 1,
      "units": "month"
    },
    "approvedProductOfferingTermOrCondition": [
          {
            "@type": "ProductOfferingTermOrCondition",
```


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```
          "productOfferingTermOrConditionSpecRef": {
            "@type": "ProductOfferingTermOrConditionSpecRef",
            "id": "0a2782b7-81a7-4af9-9e4f-d51bc690ec2d_terms_1",
            "@referredType":
"OpenGatewayProductOfferingTermOrConditionSpecification"

          },
          "productOfferingTermOrConditionApproval": {
            "approvalDate": "2023-11-20T00:00:00.000Z"
          }
        }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
]
```

Retrieve a list of API Product with field selection and filter

Request

```
GET /apiProduct?filter=status=active&fields=status
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "ApiProductDeviceLocationVerification",
    "id": "8bb8bcdb-9fcf-4b4f-8b27-2dda848404fb",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/8bb8bcdb-9fcf-4b4f-
8b27-2dda848404fb",
    "status": "active"
  },
  {
    "@type": "ApiProduct",
    "id": "33944716-d86f-4122-827b-307d74e6ca50",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/33944716-d86f-4122-
827b-307d74e6ca50",
    "status": "active"
  },
  {
    "@type": "ApiProductLocationVerificationAntifraud",
    "id": "3be31371-2755-43f7-a6f6-380234910b9b",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/3be31371-2755-43f7-
a6f6-380234910b9b",
    "status": "active"
  },
  {
    "@type": "UsageVolumeProductLocationVerification",
```

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```
    "id": "fea0c794-d62d-40bb-a9a4-934336194dca",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/apiProduct/fea0c794-d62d-40bb-
a9a4-934336194dca",
    "status": "active"
  }
]
```


Operations on Application

List or find Application objects

`GET /application?fields=…`

Description

This operation list Application entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of Application

Request

```
GET /application
Content-Type: application/json
```

Response


```
200
[
  {
    "@type": "Application",
    "@baseType": "LogicalResource",
    "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
    "name": "Evolve Quest",
    "creationDate": "2024-12-02T14:00:00.00Z",
    "commercialName": "Evolve Quest",
    "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
    "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
    "description": "Evolve Quest mobile application",
    "redirectUrl": [
      "https://channelpartner.com/opengateway/application/evolvequest/redirect"
    ],
    "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
    "category": "games",
    "operationalState": "enable",
    "approvalStatus": "approved",
```


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```
    "digitalIdentity": {
      "@type": "ApiDigitalIdentity",
      "clientId": "evolve_quest_channelPartner15",
      "validFor": {
        "startDateTime": "2024-12-02T14:01:00.00Z"
      }
    },
    "applicationOwner": {
      "@type": "PartyRoleRef",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
      "@referredType": "PartyRole"
    },
    "privacyRightsRequestContact": [
      {
        "@type": "WebFormContactMedium",
        "@baseType": "ContactMedium",
        "id": "1",
        "preferred": false,
        "url": "https://contact.evolvequest.com"
      },
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "id": "2",
        "preferred": true,
        "emailAddress": "contact@evolvequest.com"
      }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
      "@referredType": "PartyRole"
    }
  },
  {
    "@type": "Application",
    "@baseType": "LogicalResource",
    "id": "102",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/102",
    "name": "Evolve West",
    "creationDate": "2024-12-02T14:10:00.00Z",
    "commercialName": "Evolve West",
    "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
    "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
    "description": "Evolve West mobile application",
    "redirectUrl": [
      "https://channelpartner.com/opengateway/application/evolvequest/redirect"
    ],
    "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
    "category": "games",
    "operationalState": "enable",
    "approvalStatus": "approved",
    "digitalIdentity": {
      "@type": "ApiDigitalIdentity",
      "clientId": "evolve_west_channelPartner15",
      "validFor": {
        "startDateTime": "2024-12-02T14:11:00.00Z"
      }
    },
    "apiConsumerRole": [
```


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```
      {
        "grantedPermissionSet": [
          {
            "@type": "ApiPermissionSet",
            "creationDate": "2025-04-02T15:00:00.00Z",
            "permission": {
              "@type": "ApiConsumerPermission",
              "managedAssetGroup": {
                "@type": "ApiListAssetGroup",
                "@baseType": "AssetGroup",
                "apiProduct": [
                  {
                    "@type": "ProductRef",
                    "id": "123",
                    "@referredType": "ApiProduct"
                  }
                ]
              }
            },
            "validFor": {
              "startDateTime": "2025-04-02T15:00:00.00Z"
            }
          }
        ]
      }
    ],
    "applicationOwner": {
      "@type": "PartyRoleRef",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
      "@referredType": "PartyRole"
    },
    "privacyRightsRequestContact": [
      {
        "@type": "WebFormContactMedium",
        "@baseType": "ContactMedium",
        "id": "1",
        "preferred": false,
        "url": "https://contact.evolvewest.com"
      },
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "id": "2",
        "preferred": true,
        "emailAddress": "contact@evolvewest.com"
      }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
          "@referredType": "PartyRole"
    }
  }
]
```


Retrieve a list of Application with field selection and filter

Request

```
GET
```

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```
/application?filter=approvalStatus=approved&fields=approvalStatus,applicationOwner
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "Application",
    "@baseType": "LogicalResource",
    "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
    "approvalStatus": "approved",
    "applicationOwner": {
      "@type": "PartyRoleRef",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385"
    }
  },
  {
    "@type": "Application",
    "@baseType": "LogicalResource",
    "id": "102",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/102",
    "approvalStatus": "approved",
    "applicationOwner": {
      "@type": "PartyRoleRef",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
          "@referredType": "PartyRole"
    }
  }
]
```

Retrieves a Application by ID

`GET /application/{id}?fields=…`

Description

This operation retrieves an Application entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve an Application from its identifier

Request

```
GET /application/101
Content-Type: application/json
```

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Response


```
200
{
  "@type": "Application",
  "@baseType": "LogicalResource",
  "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
  "name": "Evolve Quest",
  "creationDate": "2024-12-02T14:00:00.00Z",
  "commercialName": "Evolve Quest",
  "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
  "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
  "description": "Evolve Quest mobile application",
  "redirectUrl": [
    "https://channelpartner.com/opengateway/application/evolvequest/redirect"
  ],
  "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
  "category": "games",
  "operationalState": "enable",
  "approvalStatus": "approved",
  "digitalIdentity": {
    "@type": "ApiDigitalIdentity",
    "clientId": "evolve_quest_channelPartner15",
    "validFor": {
      "startDateTime": "2024-12-02T14:01:00.00Z"
    }
  },
  "applicationOwner": {
    "@type": "PartyRoleRef",
    "id": "e05b75dc-f843-40d9-8d16-112275a28385",
    "@referredType": "PartyRole"
  },
  "privacyRightsRequestContact": [
    {
      "@type": "WebFormContactMedium",
      "@baseType": "ContactMedium",
      "id": "1",
      "preferred": false,
      "url": "https://contact.evolvequest.com"
    },
    {
      "@type": "EmailContactMedium",
      "@baseType": "ContactMedium",
      "id": "2",
      "preferred": true,
      "emailAddress": "contact@evolvequest.com"
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```


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Retrieve an Application with permissions from its identifier

Request

```
GET /application/102
Content-Type: application/json
```

Response

```
200
  {
    "@type": "Application",
    "@baseType": "LogicalResource",
    "id": "102",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/102",
    "name": "Evolve West",
    "creationDate": "2024-12-02T14:10:00.00Z",
    "commercialName": "Evolve West",
    "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
    "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
    "description": "Evolve West mobile application",
    "redirectUrl": [
      "https://channelpartner.com/opengateway/application/evolvequest/redirect"
    ],
    "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
    "category": "games",
    "operationalState": "enable",
    "approvalStatus": "approved",
    "digitalIdentity": {
      "@type": "ApiDigitalIdentity",
      "clientId": "evolve_west_channelPartner15",
      "validFor": {
        "startDateTime": "2024-12-02T14:11:00.00Z"
      }
    },
    "apiConsumerRole": [
      {
        "grantedPermissionSet": [
          {
            "@type": "ApiPermissionSet",
            "creationDate": "2025-04-02T15:00:00.00Z",
            "permission": {
              "@type": "ApiConsumerPermission",
              "managedAssetGroup": {
                "@type": "ApiListAssetGroup",
                "@baseType": "AssetGroup",
                "apiProduct": [
                  {
                    "@type": "ProductRef",
                    "id": "123",
                    "@referredType": "ApiProduct"
                  }
                ]
              }
            },
            "validFor": {
```

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```
              "startDateTime": "2025-04-02T15:00:00.00Z"
            }
          }
        ]
      }
    ],
    "applicationOwner": {
      "@type": "PartyRoleRef",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
      "@referredType": "PartyRole"
    },
    "privacyRightsRequestContact": [
      {
        "@type": "WebFormContactMedium",
        "@baseType": "ContactMedium",
        "id": "1",
        "preferred": false,
        "url": "https://contact.evolvewest.com"
      },
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "id": "2",
        "preferred": true,
        "emailAddress": "contact@evolvewest.com"
      }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
          "@referredType": "PartyRole"
    }
  }
```

Retrieve an Application without digital identity from its identifier

Request

```
GET /application/103
Content-Type: application/json
```

Response

```
200
{
  "@type": "Application",
  "@baseType": "LogicalResource",
  "id": "103",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/103",
  "name": "Evolve East",
  "creationDate": "2024-12-02T14:20:00.00Z",
  "commercialName": "Evolve East",
  "logoUrl": "https://evolveeast.com/logo.jpg",
  "privacyPolicyURL": "https://evolveeast.com/privacyPolicy.html",
  "description": "Evolve East mobile application",
  "category": "games",
```

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```
    "operationalState": "enable",
    "approvalStatus": "approved",
    "applicationOwner": {
      "@type": "PartyRoleRef",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
            "@referredType": "PartyRole"
    },
    "externalIdentifier": [
      {
        "@type": "ExternalIdentifier",
        "id": "application_123",
        "owner": "ChannelPartner",
        "externalIdentifierType": "Application"
      }
    ],
    "privacyRightsRequestContact": [
      {
        "@type": "WebFormContactMedium",
        "@baseType": "ContactMedium",
        "id": "1",
        "preferred": false,
        "url": "https://contact.evolveeast.com"
      },
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "id": "2",
        "preferred": true,
        "emailAddress": "contact@evolveeast.com"
      }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
            "@referredType": "PartyRole"
    }
  }
```

Creates a Application

POST /application?fields=…

Description

This operation creates an Application entity.

Mandatory Attributes

Mandatory Attributes	Rule
applicationOwner	
commercialName	
description	
name	
@type	M

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Usage samples

Create an Application

Request

```
POST /application
Content-Type: application/json
{
  "@type": "Application",
  "@baseType": "LogicalResource",
  "name": "Evolve Quest",
  "commercialName": "Evolve Quest",
  "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
  "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
  "description": "Evolve Quest mobile application",
  "redirectUrl": [
    "https://channelpartner.com/opengateway/application/evolvequest/redirect"
  ],
  "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
  "category": "games",
  "applicationOwner": {
    "@type": "PartyRoleRef",
    "id": "e05b75dc-f843-40d9-8d16-112275a28385",
    "@referredType": "PartyRole"
  },
  "privacyRightsRequestContact": [
    {
      "@type": "WebFormContactMedium",
      "@baseType": "ContactMedium",
      "id": "1",
      "preferred": false,
      "url": "https://contact.evolvequest.com"
    },
    {
      "@type": "EmailContactMedium",
      "@baseType": "ContactMedium",
      "id": "2",
      "preferred": true,
      "emailAddress": "contact@evolvequest.com"
    }
  ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ]
}
```

Response

```
201
{
```

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```
  "@type": "Application",
  "@baseType": "LogicalResource",
  "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
  "name": "Evolve Quest",
  "creationDate": "2024-12-02T14:00:00.00Z",
  "commercialName": "Evolve Quest",
  "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
  "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
  "description": "Evolve Quest mobile application",
  "redirectUrl": [
    "https://channelpartner.com/opengateway/application/evolvequest/redirect"
  ],
  "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
  "category": "games",
  "operationalState": "enable",
  "approvalStatus": "approved",
  "digitalIdentity": {
    "@type": "ApiDigitalIdentity",
    "clientId": "evolve_quest_channelPartner15"
  },
  "applicationOwner": {
    "@type": "PartyRoleRef",
    "id": "e05b75dc-f843-40d9-8d16-112275a28385",
    "@referredType": "PartyRole"
  },
  "privacyRightsRequestContact": [
    {
      "@type": "WebFormContactMedium",
      "@baseType": "ContactMedium",
      "id": "1",
      "preferred": false,
      "url": "https://contact.evolvequest.com"
    },
    {
      "@type": "EmailContactMedium",
      "@baseType": "ContactMedium",
      "id": "2",
      "preferred": true,
      "emailAddress": "contact@evolvequest.com"
    }
  ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```


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Updates partially a Application

`PATCH /application/{id}?fields=…`

Description

This operation allows partial updates of an Application entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
apiConsumerRole	
applicationOwner	
approvalStatus	
channelPartner	
creationDate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
approvalStatusReason	
category	
commercialName	
description	
digitalIdentity	
externalIdentifier	
href	
id	
jwksUri	
logoUrl	
name	
operationalState	
privacyPolicyURL	
privacyRightsRequestContact	

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Patchable Attributes	Rule
redirectUrl	
resourceVersion	

Usage samples

Update an Application with json content type

Request

```
PATCH /application/101
Content-Type: application/json
{
  "@type": "Application",
  "operationalState": "disable"
}
```

Response

```
200
{
  "@type": "Application",
  "@baseType": "LogicalResource",
  "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
  "name": "Evolve Quest",
  "creationDate": "2024-12-02T14:00:00.00Z",
  "commercialName": "Evolve Quest",
  "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
  "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
  "description": "Evolve Quest mobile application",
  "redirectUrl": [
    "https://channelpartner.com/opengateway/application/evolvequest/redirect"
  ],
  "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
  "category": "games",
  "operationalState": "disable",
  "approvalStatus": "approved",
  "digitalIdentity": {
    "@type": "ApiDigitalIdentity",
    "clientId": "evolve_quest_channelPartner15",
    "validFor": {
      "startDateTime": "2024-12-02T14:01:00.00Z"
    }
  },
  "applicationOwner": {
    "@type": "PartyRoleRef",
    "id": "e05b75dc-f843-40d9-8d16-112275a28385",
    "@referredType": "PartyRole"
  },
  "privacyRightsRequestContact": [
    {
```

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```
      "@type": "WebFormContactMedium",
      "@baseType": "ContactMedium",
      "id": "1",
      "preferred": false,

      "url": "https://contact.evolvequest.com"
    },
    {
      "@type": "EmailContactMedium",
      "@baseType": "ContactMedium",
      "id": "2",
      "preferred": true,

      "emailAddress": "contact@evolvequest.com"
    }
  ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```

Update an Application with merge-patch+json content type

Request

```
PATCH /application/101
Content-Type: application/merge-patch+json
{
  "@type": "Application",
  "operationalState": "disable"
}
```

Response

```
200
{
  "@type": "Application",
  "@baseType": "LogicalResource",
  "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
  "name": "Evolve Quest",
  "creationDate": "2024-12-02T14:00:00.00Z",
  "commercialName": "Evolve Quest",
  "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
```

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```
  "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
  "description": "Evolve Quest mobile application",
  "redirectUrl": [
    "https://channelpartner.com/opengateway/application/evolvequest/redirect"
  ],
  "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
  "category": "games",
  "operationalState": "disable",
  "approvalStatus": "approved",
  "digitalIdentity": {
    "@type": "ApiDigitalIdentity",
    "clientId": "evolve_quest_channelPartner15",
    "validFor": {
      "startDateTime": "2024-12-02T14:01:00.00Z"
    }
  },
  "applicationOwner": {
    "@type": "PartyRoleRef",
    "id": "e05b75dc-f843-40d9-8d16-112275a28385",
    "@referredType": "PartyRole"
  },
  "privacyRightsRequestContact": [
    {
      "@type": "WebFormContactMedium",
      "@baseType": "ContactMedium",
      "id": "1",
      "preferred": false,

      "url": "https://contact.evolvequest.com"
    },
    {
      "@type": "EmailContactMedium",
      "@baseType": "ContactMedium",
      "id": "2",
      "preferred": true,

      "emailAddress": "contact@evolvequest.com"
    }
  ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```

Update an Application with json-patch+json content type

Request

```
PATCH /application/101
```

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```
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/operationalState",
    "value": "disable"
  }
]
```

Response

```
200
{
  "@type": "Application",
  "@baseType": "LogicalResource",
  "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
  "name": "Evolve Quest",
  "creationDate": "2024-12-02T14:00:00.00Z",
  "commercialName": "Evolve Quest",
  "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
  "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
  "description": "Evolve Quest mobile application",
  "redirectUrl": [
    "https://channelpartner.com/opengateway/application/evolvequest/redirect"
  ],
  "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
  "category": "games",
  "operationalState": "disable",
  "approvalStatus": "approved",
  "digitalIdentity": {
    "@type": "ApiDigitalIdentity",
    "clientId": "evolve_quest_channelPartner15",
    "validFor": {
      "startDateTime": "2024-12-02T14:01:00.00Z"
    }
  },
  "applicationOwner": {
    "@type": "PartyRoleRef",
    "id": "e05b75dc-f843-40d9-8d16-112275a28385",
    "@referredType": "PartyRole"
  },
  "privacyRightsRequestContact": [
    {
      "@type": "WebFormContactMedium",
      "@baseType": "ContactMedium",
      "id": "1",
      "preferred": false,

      "url": "https://contact.evolvequest.com"
    },
    {
      "@type": "EmailContactMedium",
      "@baseType": "ContactMedium",
      "id": "2",
      "preferred": true,
```

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```
      "emailAddress": "contact@evolvequest.com"
    }
  ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ],
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
    "@referredType": "PartyRole"
  }
}
```

Update an Application with json-patch-query+json content type

Request

```
PATCH /application/101
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path": "/externalIdentifier?id=application_123",
    "value": {
      "@type": "ExternalIdentifier",
      "id": "application_124",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  }
]
```

Response

```
200
{
  "@type": "Application",
  "@baseType": "LogicalResource",
  "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
  "name": "Evolve Quest",
  "creationDate": "2024-12-02T14:00:00.00Z",
  "commercialName": "Evolve Quest",
  "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
  "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
  "description": "Evolve Quest mobile application",
  "redirectUrl": [
```

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```
      "https://channelpartner.com/opengateway/application/evolvequest/redirect"
    ],
    "jwksUri": "https://channelpartner.com/opengateway/application/evolvequest/jwk",
    "category": "games",
    "operationalState": "disable",
    "approvalStatus": "approved",
    "digitalIdentity": {
      "@type": "ApiDigitalIdentity",
      "clientId": "evolve_quest_channelPartner15",
      "validFor": {
        "startDateTime": "2024-12-02T14:01:00.00Z"
      }
    },
    "applicationOwner": {
      "@type": "PartyRoleRef",
      "id": "e05b75dc-f843-40d9-8d16-112275a28385",
      "@referredType": "PartyRole"
    },
    "privacyRightsRequestContact": [
      {
        "@type": "WebFormContactMedium",
        "@baseType": "ContactMedium",
        "id": "1",
        "preferred": false,

        "url": "https://contact.evolvequest.com"
      },
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "id": "2",
        "preferred": true,

        "emailAddress": "contact@evolvequest.com"
      }
    ],
    "externalIdentifier": [
      {
        "@type": "ExternalIdentifier",
        "id": "application_124",
        "owner": "ChannelPartner",
        "externalIdentifierType": "Application"
      }
    ],
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
      "@referredType": "PartyRole"
    }
  }
```


Operations on ApplicationOwner

List or find ApplicationOwner objects

`GET /applicationOwner?fields=…`

Description

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This operation list ApplicationOwner entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of ApplicationOwner

Request

```
GET /applicationOwner
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "ApplicationOwner",
    "id": "e05b75dc-f843-40d9-8d16-112275a28385",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
    "name": "Funtastic Games application owner role",
  "creationDate": "2024-12-01T13:00:00.00Z",
    "description": "Application owner role in the Open Gateway context",
    "engagedParty": {
      "@type": "ApplicationOwnerOrganization",
      "name": "Funtastic Games SA",
      "tradingName": "Funtastic Games",
      "isLegalEntity": true,
      "organizationType": "dpv:ForProfitOrganisation",
      "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
      "taxNumber": "00.623.904/0001-73",
      "organizationIdentification": [
        {
          "@type": "OrganizationIdentification",
          "identificationType": "CNPJ",
          "identificationId": "00.623.904/0001-73",
          "issuingAuthority": "Receita Federal"
        }
      ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
      "registeredGeographicAddress": {
        "@type": "LightGeographicAddress",
        "streetNr": "123",
        "streetName": "Avenida Paulista",
        "locality": "Paraiso",
        "city": "Sao Paulo",
        "stateOrProvince": "Sao Paulo",
        "countryCode": {
```

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```
          "@type": "ISO31661Alpha2StandardIdentifier",
          "value": "BR"
        },
        "postcode": "01311-000",
        "geographicSubAddress": {
          "@type": "LightGeographicSubAddress",
          "buildingName": "Fogo",
          "levelNumber": "Planta 4"
        }
      },
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
            "id": "1",
          "preferred": true,
          "emailAddress": "contact@funtasticgames.com"
        }
      ],
      "legalRepresentative": {
        "@type": "ApplicationOwnerRelatedIndividual",
        "familyName": "OConnor",
        "givenName": "Jane",
        "contactMedium": [
          {
            "@type": "EmailContactMedium",
            "@baseType": "ContactMedium",
                "id": "1",
            "preferred": true,
              "emailAddress": "jane@funtasticgames.com"
          }
        ]
      },
      "dataProtectionOfficer": {
        "@type": "ApplicationOwnerRelatedOrganization",
        "name": "DPO Office",
        "organizationType": "dpv:OrganisationalUnit",
        "contactMedium": [
          {
            "@type": "EmailContactMedium",
            "@baseType": "ContactMedium",
                "id": "1",
            "preferred": true,
              "emailAddress": "dpo@funtasticgames.com"
          }
        ]
      },
      "privacyManager": {
        "@type": "ApplicationOwnerRelatedOrganization",
        "name": "Privacy Office",
        "organizationType": "dpv:OrganisationalUnit",
        "contactMedium": [
          {
            "@type": "EmailContactMedium",
            "@baseType": "ContactMedium",
                "id": "1",
            "preferred": true,
              "emailAddress": "privacy@funtasticgames.com"
          }
        ]
      },
      "localRepresentative": [
        {
```


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```
          "@type": "ApplicationOwnerRelatedLocalRepresentative",
          "countryCode": {
            "@type": "ISO31661Alpha2StandardIdentifier",
            "value": "EU"
          },
          "representative": {
            "@type": "ApplicationOwnerRelatedIndividual",
            "familyName": "Juan",
            "givenName": "Garcia",
            "contactMedium": [
              {
                "@type": "EmailContactMedium",
                "@baseType": "ContactMedium",
                "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Common/EmailContactMedium.schema.json",
                        "id": "jgarcia",
                "preferred": true,
                      "emailAddress": "jgarcia@funtasticgames.com"
              }
            ]
          }
        }
      ]
    },
    "status": "active",
    "approvalStatus": "pendingApproval",
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
          "@referredType": "PartyRole"
    }
  }]
```


Retrieve a list of ApplicationOwner with field selection and filter

Request

```
GET /applicationOwner?filter=approvalStatus=approved&fields=approvalStatus
Content-Type: application/json
```

Response


```
200
[
  {
    "@type": "ApplicationOwner",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
    "id": "e05b75dc-f843-40d9-8d16-112275a28385",
    "approvalStatus": "approved"
  }
]
```


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Retrieves a ApplicationOwner by ID

`GET /applicationOwner/{id}?fields=…`

Description

This operation retrieves an ApplicationOwner entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve an ApplicationOwner from its identifier

Request

```
GET /applicationOwner/31
Content-Type: application/json
```

Response

```
200
{
  "@type": "ApplicationOwner",
  "id": "e05b75dc-f843-40d9-8d16-112275a28385",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
  "name": "Funtastic Games application owner role",
  "creationDate": "2024-12-01T13:00:00.00Z",
  "description": "Application owner role in the Open Gateway context",
  "engagedParty": {
    "@type": "ApplicationOwnerOrganization",
    "name": "Funtastic Games SA",
    "tradingName": "Funtastic Games",
    "isLegalEntity": true,
    "organizationType": "dpv:ForProfitOrganisation",
    "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
    "taxNumber": "00.623.904/0001-73",
    "organizationIdentification": [
      {
        "@type": "OrganizationIdentification",
        "identificationType": "CNPJ",
        "identificationId": "00.623.904/0001-73",
        "issuingAuthority": "Receita Federal"
      }
    ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
    "registeredGeographicAddress": {
      "@type": "LightGeographicAddress",
```

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```
      "streetNr": "123",
      "streetName": "Avenida Paulista",
      "locality": "Paraiso",
      "city": "Sao Paulo",
      "stateOrProvince": "Sao Paulo",
      "countryCode": {
        "@type": "ISO31661Alpha2StandardIdentifier",
        "value": "BR"
      },
      "postcode": "01311-000",
      "geographicSubAddress": {
        "@type": "LightGeographicSubAddress",
        "buildingName": "Fogo",
        "levelNumber": "Planta 4"
      }
    },
    "contactMedium": [
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
          "id": "1",
        "preferred": true,
        "emailAddress": "contact@funtasticgames.com"
      }
    ],
    "legalRepresentative": {
      "@type": "ApplicationOwnerRelatedIndividual",
      "familyName": "OConnor",
      "givenName": "Jane",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "jane@funtasticgames.com"
        }
      ]
    },
    "dataProtectionOfficer": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "DPO Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "dpo@funtasticgames.com"
        }
      ]
    },
    "privacyManager": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "Privacy Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
```


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```
            "emailAddress": "privacy@funtasticgames.com"
          }
        ]
      },
      "localRepresentative": [
        {
          "@type": "ApplicationOwnerRelatedLocalRepresentative",
          "countryCode": {
            "@type": "ISO31661Alpha2StandardIdentifier",
            "value": "EU"
          },
          "representative": {
            "@type": "ApplicationOwnerRelatedIndividual",
            "familyName": "Juan",
            "givenName": "Garcia",
            "contactMedium": [
              {
                "@type": "EmailContactMedium",
                "@baseType": "ContactMedium",
                "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Common/EmailContactMedium.schema.json",
                        "id": "jgarcia",
                "preferred": true,
                    "emailAddress": "jgarcia@funtasticgames.com"
              }
            ]
          }
        }
      ]
    },
    "status": "active",
    "approvalStatus": "pendingApproval",
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
            "@referredType": "PartyRole"
    }
  }
```

Creates a ApplicationOwner

`POST /applicationOwner?fields=…`

Description

This operation creates an ApplicationOwner entity.

Mandatory Attributes

Mandatory Attributes	Rule
description	
engagedParty	
engagedParty.organizationIdentification.identificationId	
name	

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Mandatory Attributes	Rule
@type	M

Usage samples

Create an ApplicationOwner

Request

```
POST /applicationOwner
Content-Type: application/json
{
  "@type": "ApplicationOwner",
  "name": "Funtastic Games application owner role",
  "description": "Application owner role in the Open Gateway context",
  "engagedParty": {
    "@type": "ApplicationOwnerOrganization",
    "name": "Funtastic Games SA",
    "tradingName": "Funtastic Games",
    "isLegalEntity": true,
    "organizationType": "dpv:ForProfitOrganisation",
    "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
    "taxNumber": "00.623.904/0001-73",
    "organizationIdentification": [
      {
        "@type": "OrganizationIdentification",
        "identificationType": "CNPJ",
        "identificationId": "00.623.904/0001-73",
        "issuingAuthority": "Receita Federal"
      }
    ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
    "registeredGeographicAddress": {
      "@type": "LightGeographicAddress",
      "streetNr": "123",
      "streetName": "Avenida Paulista",
      "locality": "Paraiso",
      "city": "Sao Paulo",
      "stateOrProvince": "Sao Paulo",
      "countryCode": {
        "@type": "ISO31661Alpha2StandardIdentifier",
        "value": "BR"
      },
      "postcode": "01311-000",
      "geographicSubAddress": {
        "@type": "LightGeographicSubAddress",
        "buildingName": "Fogo",
        "levelNumber": "Planta 4"
      }
    },
    "contactMedium": [
      {
```

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```
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "id": "1",
        "preferred": true,
        "emailAddress": "contact@funtasticgames.com"
      }
    ],
    "legalRepresentative": {
      "@type": "ApplicationOwnerRelatedIndividual",
      "familyName": "OConnor",
      "givenName": "Jane",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": true,
          "emailAddress": "jane@funtasticgames.com"
        }
      ]
    },
    "dataProtectionOfficer": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "DPO Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": true,
          "emailAddress": "dpo@funtasticgames.com"
        }
      ]
    },
    "privacyManager": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "Privacy Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": true,
          "emailAddress": "privacy@funtasticgames.com"
        }
      ]
    },
    "localRepresentative": [
      {
        "@type": "ApplicationOwnerRelatedLocalRepresentative",
        "countryCode": {
          "@type": "ISO31661Alpha2StandardIdentifier",
          "value": "EU"
        },
        "representative": {
          "@type": "ApplicationOwnerRelatedIndividual",
          "familyName": "Juan",
          "givenName": "Garcia",
          "contactMedium": [
            {
              "@type": "EmailContactMedium",
```


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```
              "@baseType": "ContactMedium",
              "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Common/EmailContactMedium.schema.json",
              "id": "jgarcia",
              "preferred": true,
                  "emailAddress": "jgarcia@funtasticgames.com"
            }
          ]
        }
      }
    ]
  }
}
```

Response

```
201
{
  "@type": "ApplicationOwner",
  "id": "e05b75dc-f843-40d9-8d16-112275a28385",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
  "name": "Funtastic Games application owner role",
  "creationDate": "2024-12-01T13:00:00.00Z",
  "description": "Application owner role in the Open Gateway context",
  "engagedParty": {
    "@type": "ApplicationOwnerOrganization",
    "name": "Funtastic Games SA",
    "tradingName": "Funtastic Games",
    "isLegalEntity": true,
    "organizationType": "dpv:ForProfitOrganisation",
    "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
    "taxNumber": "00.623.904/0001-73",
    "organizationIdentification": [
      {
        "@type": "OrganizationIdentification",
        "identificationType": "CNPJ",
        "identificationId": "00.623.904/0001-73",
        "issuingAuthority": "Receita Federal"
      }
    ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
    "registeredGeographicAddress": {
      "@type": "LightGeographicAddress",
      "streetNr": "123",
      "streetName": "Avenida Paulista",
      "locality": "Paraiso",
      "city": "Sao Paulo",
      "stateOrProvince": "Sao Paulo",
      "countryCode": {
        "@type": "ISO31661Alpha2StandardIdentifier",
        "value": "BR"
```

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```
      },
      "postcode": "01311-000",
      "geographicSubAddress": {
        "@type": "LightGeographicSubAddress",
        "buildingName": "Fogo",
        "levelNumber": "Planta 4"
      }
    },
    "contactMedium": [
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
          "id": "1",
        "preferred": true,
        "emailAddress": "contact@funtasticgames.com"
      }
    ],
    "legalRepresentative": {
      "@type": "ApplicationOwnerRelatedIndividual",
      "familyName": "OConnor",
      "givenName": "Jane",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": true,
          "emailAddress": "jane@funtasticgames.com"
        }
      ]
    },
    "dataProtectionOfficer": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "DPO Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": true,
          "emailAddress": "dpo@funtasticgames.com"
        }
      ]
    },
    "privacyManager": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "Privacy Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "privacy@funtasticgames.com"
        }
      ]
    },
    "localRepresentative": [
      {
        "@type": "ApplicationOwnerRelatedLocalRepresentative",
        "countryCode": {
```


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```
            "@type": "ISO31661Alpha2StandardIdentifier",
            "value": "EU"
          },
          "representative": {
            "@type": "ApplicationOwnerRelatedIndividual",
            "familyName": "Juan",
            "givenName": "Garcia",
            "contactMedium": [
              {
                "@type": "EmailContactMedium",
                "@baseType": "ContactMedium",
                        "id": "jgarcia",
                "preferred": true,
                    "emailAddress": "jgarcia@funtasticgames.com"
              }
            ]
          }
        }
      ]
    },
    "status": "active",
    "approvalStatus": "pendingApproval",
    "channelPartner": {
      "@type": "PartyRoleRef",
      "id": "15",
            "@referredType": "PartyRole"
    }
  }
```

Updates partially a ApplicationOwner

`PATCH /applicationOwner/{id}?fields=…`

Description

This operation allows partial updates of an ApplicationOwner entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
approvalStatus	
approvalStatusReason	
channelPartner	
creationDate	
href	
id	
validFor	

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Non Patchable Attributes	Rule
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
description	
engagedParty	
name	
status	
statusReason	

Usage samples

Update an ApplicationOwner with json content type

Request

```
PATCH /applicationOwner/31
Content-Type: application/json
{
  "@type": "ApplicationOwner",
  "status": "inactive"
}
```

Response

```
200
{
  "@type": "ApplicationOwner",
  "id": "e05b75dc-f843-40d9-8d16-112275a28385",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
  "name": "Funtastic Games application owner role",
  "creationDate": "2024-12-01T13:00:00.00Z",
  "description": "Application owner role in the Open Gateway context",
  "engagedParty": {
    "@type": "ApplicationOwnerOrganization",
    "name": "Funtastic Games SA",
    "tradingName": "Funtastic Games",
    "isLegalEntity": true,
    "organizationType": "dpv:ForProfitOrganisation",
    "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
    "taxNumber": "00.623.904/0001-73",
    "organizationIdentification": [
      {
        "@type": "OrganizationIdentification",
        "identificationType": "CNPJ",
        "identificationId": "00.623.904/0001-73",
```

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```
        "issuingAuthority": "Receita Federal"
      }
    ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
    "registeredGeographicAddress": {
      "@type": "LightGeographicAddress",
      "streetNr": "123",
      "streetName": "Avenida Paulista",
      "locality": "Paraiso",
      "city": "Sao Paulo",
      "stateOrProvince": "Sao Paulo",
      "countryCode": {
        "@type": "ISO31661Alpha2StandardIdentifier",
        "value": "BR"
      },
      "postcode": "01311-000",
      "geographicSubAddress": {
        "@type": "LightGeographicSubAddress",
        "buildingName": "Fogo",
        "levelNumber": "Planta 4"
      }
    },
    "contactMedium": [
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
          "id": "1",
        "preferred": true,
        "emailAddress": "contact@funtasticgames.com"
      }
    ],
    "legalRepresentative": {
      "@type": "ApplicationOwnerRelatedIndividual",
      "familyName": "OConnor",
      "givenName": "Jane",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "jane@funtasticgames.com"
        }
      ]
    },
    "dataProtectionOfficer": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "DPO Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "dpo@funtasticgames.com"
```


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```
        }
      ]
    },
    "privacyManager": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "Privacy Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "privacy@funtasticgames.com"
        }
      ]
    },
    "localRepresentative": [
      {
        "@type": "ApplicationOwnerRelatedLocalRepresentative",
        "countryCode": {
          "@type": "ISO31661Alpha2StandardIdentifier",
          "value": "EU"
        },
        "representative": {
          "@type": "ApplicationOwnerRelatedIndividual",
          "familyName": "Juan",
          "givenName": "Garcia",
          "contactMedium": [
            {
              "@type": "EmailContactMedium",
              "@baseType": "ContactMedium",
              "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Common/EmailContactMedium.schema.json",
                    "id": "jgarcia",
              "preferred": true,
                  "emailAddress": "jgarcia@funtasticgames.com"
            }
          ]
        }
      }
    ]
  },
  "status": "inactive",
  "approvalStatus": "pendingApproval",
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```

Update an ApplicationOwner with merge-patch+json content type

Request

```
PATCH /applicationOwner/31
Content-Type: application/merge-patch+json
{
  "@type": "ApplicationOwner",
```

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```
  "status": "inactive"
}
```

Response

```
200
{
  "@type": "ApplicationOwner",
  "id": "e05b75dc-f843-40d9-8d16-112275a28385",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
  "name": "Funtastic Games application owner role",
  "creationDate": "2024-12-01T13:00:00.00Z",
  "description": "Application owner role in the Open Gateway context",
  "engagedParty": {
    "@type": "ApplicationOwnerOrganization",
    "name": "Funtastic Games SA",
    "tradingName": "Funtastic Games",
    "isLegalEntity": true,
    "organizationType": "dpv:ForProfitOrganisation",
    "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
    "taxNumber": "00.623.904/0001-73",
    "organizationIdentification": [
      {
        "@type": "OrganizationIdentification",
        "identificationType": "CNPJ",
        "identificationId": "00.623.904/0001-73",
        "issuingAuthority": "Receita Federal"
      }
    ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
    "registeredGeographicAddress": {
      "@type": "LightGeographicAddress",
      "streetNr": "123",
      "streetName": "Avenida Paulista",
      "locality": "Paraiso",
      "city": "Sao Paulo",
      "stateOrProvince": "Sao Paulo",
      "countryCode": {
        "@type": "ISO31661Alpha2StandardIdentifier",
        "value": "BR"
      },
      "postcode": "01311-000",
      "geographicSubAddress": {
        "@type": "LightGeographicSubAddress",
        "buildingName": "Fogo",
        "levelNumber": "Planta 4"
      }
    },
    "contactMedium": [
      {
        "@type": "EmailContactMedium",
```

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```
        "@baseType": "ContactMedium",
          "id": "1",
        "preferred": true,
        "emailAddress": "contact@funtasticgames.com"
      }
    ],
    "legalRepresentative": {
      "@type": "ApplicationOwnerRelatedIndividual",
      "familyName": "OConnor",
      "givenName": "Jane",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "jane@funtasticgames.com"
        }
      ]
    },
    "dataProtectionOfficer": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "DPO Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "dpo@funtasticgames.com"
        }
      ]
    },
    "privacyManager": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "Privacy Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "privacy@funtasticgames.com"
        }
      ]
    },
    "localRepresentative": [
      {
        "@type": "ApplicationOwnerRelatedLocalRepresentative",
        "countryCode": {
          "@type": "ISO31661Alpha2StandardIdentifier",
          "value": "EU"
        },
        "representative": {
          "@type": "ApplicationOwnerRelatedIndividual",
          "familyName": "Juan",
          "givenName": "Garcia",
          "contactMedium": [
            {
              "@type": "EmailContactMedium",
              "@baseType": "ContactMedium",
```


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```
              "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Common/EmailContactMedium.schema.json",
                      "id": "jgarcia",
              "preferred": true,
                  "emailAddress": "jgarcia@funtasticgames.com"
            }
          ]
        }
      }
    ]
  },
  "status": "inactive",
  "approvalStatus": "pendingApproval",
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```

Update an ApplicationOwner with json-patch+json content type

Request

```
PATCH /applicationOwner/31
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/status",
    "value": "inactive"
  }
]
```

Response

```
200
{
  "@type": "ApplicationOwner",
  "id": "e05b75dc-f843-40d9-8d16-112275a28385",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
  "name": "Funtastic Games application owner role",
  "creationDate": "2024-12-01T13:00:00.00Z",
  "description": "Application owner role in the Open Gateway context",
  "engagedParty": {
    "@type": "ApplicationOwnerOrganization",
    "name": "Funtastic Games SA",
    "tradingName": "Funtastic Games",
    "isLegalEntity": true,
    "organizationType": "dpv:ForProfitOrganisation",
    "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
    "taxNumber": "00.623.904/0001-73",
    "organizationIdentification": [
      {
        "@type": "OrganizationIdentification",
```

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```
        "identificationType": "CNPJ",
        "identificationId": "00.623.904/0001-73",
        "issuingAuthority": "Receita Federal"
      }
    ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
    "registeredGeographicAddress": {
      "@type": "LightGeographicAddress",
      "streetNr": "123",
      "streetName": "Avenida Paulista",
      "locality": "Paraiso",
      "city": "Sao Paulo",
      "stateOrProvince": "Sao Paulo",
      "countryCode": {
        "@type": "ISO31661Alpha2StandardIdentifier",
        "value": "BR"
      },
      "postcode": "01311-000",
      "geographicSubAddress": {
        "@type": "LightGeographicSubAddress",
        "buildingName": "Fogo",
        "levelNumber": "Planta 4"
      }
    },
    "contactMedium": [
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
          "id": "1",
        "preferred": true,
        "emailAddress": "contact@funtasticgames.com"
      }
    ],
    "legalRepresentative": {
      "@type": "ApplicationOwnerRelatedIndividual",
      "familyName": "OConnor",
      "givenName": "Jane",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "jane@funtasticgames.com"
        }
      ]
    },
    "dataProtectionOfficer": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "DPO Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
```


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```
          "preferred": true,
          "emailAddress": "dpo@funtasticgames.com"
        }
      ]
    },
    "privacyManager": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "Privacy Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
              "id": "1",
          "preferred": true,
          "emailAddress": "privacy@funtasticgames.com"
        }
      ]
    },
    "localRepresentative": [
      {
        "@type": "ApplicationOwnerRelatedLocalRepresentative",
        "countryCode": {
          "@type": "ISO31661Alpha2StandardIdentifier",
          "value": "EU"
        },
        "representative": {
          "@type": "ApplicationOwnerRelatedIndividual",
          "familyName": "Juan",
          "givenName": "Garcia",
          "contactMedium": [
            {
              "@type": "EmailContactMedium",
              "@baseType": "ContactMedium",
              "@schemaLocation": "https://mycsp.com/tmf-
api/schema/Common/EmailContactMedium.schema.json",
                      "id": "jgarcia",
              "preferred": true,
                  "emailAddress": "jgarcia@funtasticgames.com"
            }
          ]
        }
      }
    ]
  },
  "status": "inactive",
  "approvalStatus": "pendingApproval",
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```

Update an ApplicationOwner with json-patch-query+json content type

Request

```
PATCH /applicationOwner/31
Content-Type: application/json-patch-query+json
```

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```
[
  {
    "op": "replace",
    "path": "/engagedParty.contactMedium?id=1",
    "value": {
      "@type": "EmailContactMedium",
      "id": "1",
      "preferred": true,
      "emailAddress": "newContact@funtasticgames.com"
    }
  }
]
```

Response

```
200
{
  "@type": "ApplicationOwner",
  "id": "e05b75dc-f843-40d9-8d16-112275a28385",
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/applicationOwner/e05b75dc-f843-
40d9-8d16-112275a28385",
  "name": "Funtastic Games application owner role",
  "creationDate": "2024-12-01T13:00:00.00Z",
  "description": "Application owner role in the Open Gateway context",
  "engagedParty": {
    "@type": "ApplicationOwnerOrganization",
    "name": "Funtastic Games SA",
    "tradingName": "Funtastic Games",
    "isLegalEntity": true,
    "organizationType": "dpv:ForProfitOrganisation",
    "privacyPolicyURL": "https://www.funtasticgames.com/privacy",
    "taxNumber": "00.623.904/0001-73",
    "organizationIdentification": [
      {
        "@type": "OrganizationIdentification",
        "identificationType": "CNPJ",
        "identificationId": "00.623.904/0001-73",
        "issuingAuthority": "Receita Federal"
      }
    ],
        "externalReference": [
          {
            "@type": "ExternalIdentifier",
            "id": "funtastic_games_asp",
            "externalIdentifierType": "ApplicationOwnerOrganization",
            "owner": "ChannelPartner"
          }
        ],
    "registeredGeographicAddress": {
      "@type": "LightGeographicAddress",
      "streetNr": "123",
      "streetName": "Avenida Paulista",
      "locality": "Paraiso",
      "city": "Sao Paulo",
      "stateOrProvince": "Sao Paulo",
      "countryCode": {
        "@type": "ISO31661Alpha2StandardIdentifier",
        "value": "BR"
      },
```

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```
      "postcode": "01311-000",
      "geographicSubAddress": {
        "@type": "LightGeographicSubAddress",
        "buildingName": "Fogo",
        "levelNumber": "Planta 4"
      }
    },
    "contactMedium": [
      {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
  "id": "1",
        "preferred": true,
        "emailAddress": "newContact@funtasticgames.com"
        }
    ],
    "legalRepresentative": {
      "@type": "ApplicationOwnerRelatedIndividual",
      "familyName": "OConnor",
      "givenName": "Jane",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
      "id": "1",
          "preferred": true,
          "emailAddress": "jane@funtasticgames.com"
        }
      ]
    },
    "dataProtectionOfficer": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "DPO Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
      "id": "1",
          "preferred": true,
          "emailAddress": "dpo@funtasticgames.com"
        }
      ]
    },
    "privacyManager": {
      "@type": "ApplicationOwnerRelatedOrganization",
      "name": "Privacy Office",
      "organizationType": "dpv:OrganisationalUnit",
      "contactMedium": [
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
      "id": "1",
          "preferred": true,
          "emailAddress": "privacy@funtasticgames.com"
        }
      ]
    },
    "localRepresentative": [
      {
        "@type": "ApplicationOwnerRelatedLocalRepresentative",
        "countryCode": {
          "@type": "ISO31661Alpha2StandardIdentifier",
```


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```
          "value": "EU"
        },
        "representative": {
          "@type": "ApplicationOwnerRelatedIndividual",
          "familyName": "Juan",
          "givenName": "Garcia",
          "contactMedium": [
            {
              "@type": "EmailContactMedium",
              "@baseType": "ContactMedium",
              "id": "jgarcia",
              "preferred": true,
                  "emailAddress": "jgarcia@funtasticgames.com"
            }
          ]
        }
      }
    ]
  },
  "status": "active",
  "approvalStatus": "pendingApproval",
  "channelPartner": {
    "@type": "PartyRoleRef",
    "id": "15",
          "@referredType": "PartyRole"
  }
}
```

Operations on Monitor

Retrieves a Monitor by ID

`GET /monitor/{id}?fields=…`

Description

This operation retrieves a Monitor entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a Monitor from its identifier

Request

```
GET /monitor/987654
Content-Type: application/json
```

Response

```
200
{
  "@type": "Monitor",
  "id": "77795e7f-5213-4b9c-b672-6838a6b76152",
```

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```
  "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/monitor/77795e7f-5213-4b9c-b672-
6838a6b76152",
  "state": "Completed",
  "request": {
    "method": "POST",
    "to": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application",
    "body": {
      "@type": "Application",
      "@baseType": "LogicalResource",
      "name": "Evolve Quest",
      "commercialName": "Evolve Quest",
      "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
      "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
      "description": "Evolve Quest mobile application",
      "redirectUrl": [
        "https://channelpartner.com/opengateway/application/evolvequest/redirect"
      ],
      "jwksUri":
"https://channelpartner.com/opengateway/application/evolvequest/jwk",
      "category": "games",
      "applicationOwner": {
        "@type": "PartyRoleRef",
        "id": "e05b75dc-f843-40d9-8d16-112275a28385",
        "@referredType": "PartyRole"
      },
      "privacyRightsRequestContact": [
        {
          "@type": "WebFormContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": false,

          "url": "https://contact.evolvequest.com"
        },
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "2",
          "preferred": true,

          "emailAddress": "contact@evolvequest.com"
        }
      ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ]
    }
  },
  "response": {
    "statusCode": "201",
    "body": {
      "@type": "Application",
      "@baseType": "LogicalResource",
      "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
```


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```
      "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
      "name": "Evolve Quest",
      "commercialName": "Evolve Quest",
      "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
      "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
      "description": "Evolve Quest mobile application",
      "redirectUrl": [
        "https://channelpartner.com/opengateway/application/evolvequest/redirect"
      ],
      "jwksUri":
"https://channelpartner.com/opengateway/application/evolvequest/jwk",
      "category": "games",
      "operationalState": "enable",
      "approvalStatus": "approved",
      "digitalIdentity": {
        "@type": "ApiDigitalIdentity",
        "clientId": "evolve_quest_channelPartner15",
        "validFor": {
          "startDateTime": "2024-12-02T14:01:00.00Z"
        }
      },
      "applicationOwner": {
        "@type": "PartyRoleRef",
        "id": "e05b75dc-f843-40d9-8d16-112275a28385",
        "@referredType": "PartyRole"
      },
      "privacyRightsRequestContact": [
        {
          "@type": "WebFormContactMedium",
          "@baseType": "ContactMedium",
          "id": "1",
          "preferred": false,

          "url": "https://contact.evolvequest.com"
        },
        {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "id": "2",
          "preferred": true,

          "emailAddress": "contact@evolvequest.com"
        }
      ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ],
      "channelPartner": {
        "@type": "PartyRoleRef",
        "id": "15",
          "@referredType": "PartyRole"
      }
    }
  },
```


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```
  "sourceHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c"
}
```

List or find Monitor objects

`GET /monitor?fields=…`

Description

This operation list Monitor entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of Monitor

Request

```
GET /monitor
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "Monitor",
    "id": "77795e7f-5213-4b9c-b672-6838a6b76152",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/monitor/77795e7f-5213-4b9c-b672-
6838a6b76152",
    "state": "Completed",
    "request": {
      "method": "POST",
      "to": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application",
      "body": {
        "@type": "Application",
        "@baseType": "LogicalResource",
        "name": "Evolve Quest",
        "commercialName": "Evolve Quest",
        "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
        "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
        "description": "Evolve Quest mobile application",
        "redirectUrl": [
          "https://channelpartner.com/opengateway/application/evolvequest/redirect"
        ],
        "jwksUri":
"https://channelpartner.com/opengateway/application/evolvequest/jwk",
        "category": "games",
```

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```
        "applicationOwner": {
          "@type": "PartyRoleRef",
          "id": "e05b75dc-f843-40d9-8d16-112275a28385",
          "@referredType": "PartyRole"
        },
        "privacyRightsRequestContact": [
          {
            "@type": "WebFormContactMedium",
            "@baseType": "ContactMedium",
            "id": "1",
            "preferred": false,

            "url": "https://contact.evolvequest.com"
          },
          {
            "@type": "EmailContactMedium",
            "@baseType": "ContactMedium",
            "id": "2",
            "preferred": true,

            "emailAddress": "contact@evolvequest.com"
          }
        ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ]
      }
    },
    "response": {
      "statusCode": "201",
      "body": {
        "@type": "Application",
        "@baseType": "LogicalResource",
        "id": "f1cba17d-789a-4037-b3e3-2c96e887576c",
        "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c",
        "name": "Evolve Quest",
        "commercialName": "Evolve Quest",
        "logoUrl":
"https://channelpartner.com/opengateway/application/evolvequest/logo.jpg",
        "privacyPolicyURL":
"https://channelpartner.com/opengateway/application/evolvequest/privacyPolicy.html",
        "description": "Evolve Quest mobile application",
        "redirectUrl": [
          "https://channelpartner.com/opengateway/application/evolvequest/redirect"
        ],
        "jwksUri":
"https://channelpartner.com/opengateway/application/evolvequest/jwk",
        "category": "games",
        "operationalState": "enable",
        "approvalStatus": "approved",
        "digitalIdentity": {
          "@type": "ApiDigitalIdentity",
          "clientId": "evolve_quest_channelPartner15",
          "validFor": {
            "startDateTime": "2024-12-02T14:01:00.00Z"
          }
```


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```
        },
        "applicationOwner": {
          "@type": "PartyRoleRef",
          "id": "e05b75dc-f843-40d9-8d16-112275a28385",
          "@referredType": "PartyRole"
        },
        "privacyRightsRequestContact": [
          {
            "@type": "WebFormContactMedium",
            "@baseType": "ContactMedium",
            "id": "1",
            "preferred": false,

            "url": "https://contact.evolvequest.com"
          },
          {
            "@type": "EmailContactMedium",
            "@baseType": "ContactMedium",
            "id": "2",
            "preferred": true,

            "emailAddress": "contact@evolvequest.com"
          }
        ],
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "id": "application_123",
      "owner": "ChannelPartner",
      "externalIdentifierType": "Application"
    }
  ],
        "channelPartner": {
          "@type": "PartyRoleRef",
          "id": "15",
          "@referredType": "PartyRole"
        }
      }
    },
    "sourceHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c"
  }
]
```


Retrieve a list of Monitor with field selection and filter

Request

```
GET /monitor?filter=state=Completed&fields=state,sourceHref
Content-Type: application/json
```

Response

```
200
[
  {
```

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```
    "@type": "Monitor",
    "id": "77795e7f-5213-4b9c-b672-6838a6b76152",
    "href": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/monitor/77795e7f-5213-4b9c-b672-
6838a6b76152",
    "state": "Completed",
    "sourceHref": "https://mycsp.com/tmf-
api/openGatewayOperateAPIOnboardingAndOrdering/v5/application/f1cba17d-789a-4037-
b3e3-2c96e887576c"
  }
]
```


Operations on ApiProductDeviceLocationVerification

See operations on ApiProduct.

Operations on ApiProductLocationVerificationAntifraud

See operations on ApiProduct.

Operations on UsageVolumeProductLocationVerification

See operations on ApiProduct.

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 201 if the request was successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version 5.2.1	October 2025	Olivier Arnaud (Orange) olivier.arnaud@orange.com - Ruben Barrado (Telefonica) ruben.barradogonzalez@telefonica.com	
Version 5.2.0	June 2025	Olivier Arnaud (Orange) olivier.arnaud@orange.com	
Version 5.1.2	January 2025	Olivier Arnaud (Orange) olivier.arnaud@orange.com	
Version 5.1.1 (preview)	November 2024	Olivier Arnaud (Orange) olivier.arnaud@orange.com	
Version 5.1.0 (preview)	August 2024	Olivier Arnaud (Orange) olivier.arnaud@orange.com	
Version 5.0.0 (stable)	February 2024	Olivier Arnaud (Orange) olivier.arnaud@orange.com	

Release description v5.2.1 (vs 5.2.0)

Conformance Profile

additional comment on the optional attribute approvedProductOfferingTermOrCondition.applicationLegalBasisForInputData

Release description v5.2.0 (vs 5.1.2)

Operations

additional parameters added by tooling generation (before, after, sort, filter)

Schemas

Changed schemas and add new schemas to allow ordering of composite ProductOffering (bundles)

Changed ApiProduct schema to allow inventory of composite products (bundles) : added productRelationship to ApiProduct

Added example schemas ApiProductLocationVerificationAntifraud and UsageVolumeProductLocationVerification

Replacement of many [entity]_FVO schemas with [entity] schemas , that has impact on many attributes being optional in the POST operation (optional in the OAS but not in the

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conformance profile, so must be kept mandatory in the implementations)

Added stateReason in ApiProductOrder and all ApiProductOrderItem specialized schemas

Added externalIdentifier in ApiProductOrderItemAdd schema

Added creationDate in ApplicationOwner and Application

Added descriptions

User Guide

Additional description of entities for consistency with TMF936 Product Catalog Operate API

Added description of Monitor

Examples

Examples updated for better consistency with GSMA requirements and TMF936 Product Catalog Operate API (based on same catalog examples, allowing bundles ordering and inventory)

Examples updated for better general precision (dates, ids, …)

Release description v5.1.2 (vs 5.1.1)

Operations

additional parameters added by tooling generation (before, after, sort, filter)

Schemas

approvedProductOfferingTermOrCondition.productOfferingTermOrConditionApproval.authorization no more required in POST operation (changes in OAS and Conformance Profile)

Changes on schemas names representation (_RES, _FVO, _MVO) by tooling generation

Change on schemas name related to example DeviceLocationVerification to remove underscore

Attributes description upgrated (Hub, Application)

Removed unused apiProductOrderItem attribute in required list of ApiProductOrder, same for operationalState in Application_FVO

Typos fixed in descriptions

Examples

Examples updated for better consistency with OAS schemas

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Release description v5.1.1 (vs 5.1.0)

End-point changes

ApiGrantInformation

Changed properties from string to enum:

- purpose
- legalBasis

ApiProductActionAdd, ApiProductActionModify, ApiProduct

Changed properties from string to enum:

- purpose

ApiPurposeType

Addition of DpvPurposeType schema for new v2 W3C Data Privacy Vocabulary list of values. Removal of ApiPurposeType schema.

Application

Changed properties from string to enum:

- category

Updated property:

- ApplicationCategoryType enum updated

ApplicationOwnerOrganization and ApplicationOwnerRelatedOrganization

Changed properties from string to enum:

- organizationType

Updated property:

- organizationType enum from list of https://www.w3.org/community/reports/dpvcg/CG-FINAL-dpv-20221205/#vocab-entities-organisation

OrganizationType

Addition of DpvOrganizationType schema for new v2 W3C Data Privacy Vocabulary list of values. Removal of OrganizationType schema.

DpvXXXType schemas

change from DPV v1 values to DPV v2 values

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DpvProcess

New schema

DpvContext

New schema with new enum DpvProcessingContextType

DataExportCountry

Change : now allOf Extensible

ProductOfferingTermOrCondition

Change on schema modelling, adding process attribute based on DpvProcess schema

userConsentScreenshot (addition of url pointing to file, described in SimpleAttachment schema)

specific schemas for examples

Added new schema to make OAS examples fully valid (regarding their @type values)

- ApiProductActionModifyDeviceLocationVerification
- ApiProductActionAddDeviceLocationVerification
- ApiProductDeviceLocationVerification

Conformance Profile changes

New attributes added and existing attributes updated (based on changes in the OAS)

Some minor fixes (wording, dates)

Examples changes

Some examples bug fixed (ApiProductOrder, ApplicationOwner, Application)

Some examples added according to new attributes in OAS.

User Guide changes

Some minor fixes (wording, dates)

Release description v5.1.0 (vs 5.0.0)

End-point changes

Hub

Adding GET /hub/{id}

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Type changes

ApiDigitalIdentity

Marked as deprecated (as input and patchable property)

Marked as deprecated fields:

- clientId (as input property)
- redirectUrl (moved to Application type)
- jwksUri(moved to Application type)

ApiPermissionSet

Marked as deprecated fields:

- validFor

ApiProduct

Added properties to be aligned with future TMF936 Open Gateway Catalogue API:

- purpose
- countryCode (array of ISO country codes of 2 letters: pattern '[A-Z]{2}')

Application

New properties:

- redirectUrl (moved from ApiDigitalIdentity type)
- jwksUri (moved from ApiDigitalIdentity type)

New GDPR required property :

- privacyRightsRequestContact (of ContactMedium type)

Marked as deprecated fields :

- approvalStatusReason (as patchable property)

ApplicationOwnerOrganization

New Properties:

- localRepresentative of type AplicationOwnerRelatedLocalRepresentative

ApplicationOwnerRelatedOrganization

Changed properties from string to enum:

- organizationType

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Updated property:

- organizationType enum from list of https://www.w3.org/community/reports/dpvcg/CG-FINAL-dpv-20221205/#vocab-entities-organisation

DataExportCountry

New type for GDPR required properties with :

- countryCode (ISO country codes of 2 letters: pattern '[A-Z]{2}')
- commercialActivity (boolean)
- commercialOrganizationName (string)
- dataPrivacyFrameworkURL (url) only for EU-USA data privacy agreement.

ProductOfferingTermOrCondition

New property:

- product reference to ApiProduct (optional)

New GDPR properties (optional):

- applicationLegalBasisForInputData (enum from DpvLegalBasisType)
- dataExportTransitCountry (array of DataExportCountry)
- dataExportStorageCountry (array of DataExportCountry)
- duration (enum from DpvDurationType)
- frequency (enum from DpvFrequencyType)
- processingContext (enum from DpvContextClassesType)
- processingOperation (enum from DpvProcessingOperationType)
- purposeReason (string)
- userConsentScreenshot (file attached, described in SimpleAttachment schema)

ProductOfferingTermOrConditionApproval

Marked as deprecated:

- authorization (it is not possible to get the id of the individual authorizing the order)

RelatedOrderItem

Marked as deprecated:

- role

WebFormContactMedium

New ContactMedium sub-type with property url.

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ProductOrderItem

Bug fixed on discriminator list of GCProductOrderItem and GCProductOrderItem_FVO schemas

Examples changes

Some examples bug fixed (ApiProductOrder, ApplicationOwner, Application)

Some examples added according to new attributes in OAS.

User Guide changes

New diagram for Use Case 1

Some minor fixes (wording, dates)

Extension of strongly typed schemas chapter with examples and schemas

Conformance Profile changes

Added possibility of asynchronous responses with code 202 in addition of synchronous with code 201; in that case Monitor ressource is mandatory in GET operation

Added specific rules on specific legislations (GDPR for Europe, Canada, USA)

New attributes added (based on changes in the OAS)

Some minor fixes (wording, dates)

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